



BACHELOR OF ADVANCED COMPUTER SCIENCE (HONOURS)

COMPUTER SCIENCE HONOURS RESEARCH PROJECT

Multi-Page Visually Rich Document Understanding: A Survey of System Design and an Empirical Attribution of Failure

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October 2026

Abstract

Multi-page visually rich document understanding (MP-VRDU) concerns systems that interpret long documents whose relevant information may be distributed across pages, modalities, and structural relationships, often beyond the practical context and compute budgets of multimodal large language models. This thesis is presented as a compilation of a survey paper, an empirical attribution study, and a final system-level method that translates their findings into a working MP-VRDU system.

First, we systematically survey the field and introduce a taxonomy that organises existing systems along four axes: architecture, multimodal representation, inference-time adaptation, and training strategy. Through this taxonomy, we develop an evidence-management perspective on MP-VRDU, showing how systems differ in what document information they preserve, select, and expose to the reasoner, and how successive approaches increasingly move evidence selection and processing outside the MLLM backbone. Second, we conduct a controlled empirical attribution study to determine how MP-VRDU systems should be designed and deployed, isolating failures in representation, selection, and reasoning. We use this controlled attribution framework to derive concrete guidance for designing and deploying MP-VRDU systems. The results show where additional compute is most useful, when richer representations justify their cost, how retrieval depth should be balanced against downstream context, and how reasoner scale and quantisation trade accuracy against memory. Together, these findings provide a practical basis for allocating limited resources across the pipeline rather than maximising every component independently.

Finally, we translate these findings into a deliberately simple adaptive evidence-management pipeline that combines broad evidence acquisition, selective evidence presentation, and structured reasoning. Using a 4-bit Qwen3-VL-8B backbone on a single 16 GB GPU, the method reaches 54.5% accuracy on MMLongBench-Doc, exceeding the corresponding gold-page baseline by 5.0 points and achieving competitive performance against substantially more specialised systems without task-specific training. Together, these results show that effective MP-VRDU depends not only on stronger multimodal backbones, but on managing evidence deliberately across representation, selection, and reasoning, providing a practical basis for building more accurate, faithful, and deployable long-document understanding systems.

Acknowledgments

First, I would like to express my sincere gratitude to my supervisor, Yihao Ding, for his time, guidance, and support throughout this year. No amount of words can fully describe what an excellent supervisor he has been. Without his insight, experience, and willingness to offer advice whenever needed, this thesis and its constituent papers could never have reached their current quality, nor would the survey paper have been accepted to EMNLP. May your ancestors be proud of you, may your descendants be blessed with great fortune, and may you yourself live a long and prosperous life. I am also grateful to Zihan Xu from the University of Melbourne for helping me run experiments on their HPC system while Kaya was down for maintenance in late July. This allowed me to meet the very tight deadline for the ACL August rolling review for EACL, and the feedback from that review round was invaluable in improving this thesis. I would also like to thank my co-authors, Siwen, Daochang, Wei, Daniel, and Yifan, for their feedback and support on the two papers produced this year. Finally, special thanks go to my parents for their (financial) support, which allowed me to work less and devote more time to my studies, and was a major factor in how this honours year turned out.

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Chapter 1

Introduction and Background

Visually rich document understanding (VRDU) concerns systems that extract or infer information from documents by jointly interpreting their textual, visual, structural, and layout information. Documents such as contracts, financial reports, scientific papers, technical manuals, and regulatory filings are rarely plain text: they combine prose with tables, figures, diagrams, typography, spatial organisation, and other layout cues that contribute to meaning. VRDU therefore differs from conventional natural language processing over plain text because information may be encoded not only in the words themselves, but also in where they appear, how they are grouped, and how they relate to surrounding visual elements. Answering a simple question may therefore require reading small text, interpreting a table or figure, resolving spatial relationships between fields, or combining textual and visual evidence from the same page. Early approaches were typically designed for particular document types and tasks, making them narrow and difficult to generalise [1]. More recent systems increasingly rely on large vision-language models that process textual and visual document content together across heterogeneous document types [2, 3], with document question answering becoming a common formulation for evaluating these capabilities [4].

Although VRDU was historically dominated by single-page research, the past several years have seen rapid growth in multi-page document understanding. This reflects the fact that real-world documents commonly extend across tens or hundreds of pages, creating a document-scale setting in which answer-relevant evidence may be sparse, distributed across distant pages, or expressed through several modalities and structural relationships. Simply providing every page to a model is often impractical: textual and visual tokens accumulate with document length, high-resolution page images are expensive to process, and sufficiently long documents can exceed a model’s usable context budget. Even when the document fits, long-context models can struggle to use evidence buried within large inputs reliably [5]. The system must therefore decide not only how each page should be represented, but also which evidence should be retained or selected and how information from different pages should be combined. This is particularly important when a question depends on several pages, such as comparing values across tables, following references between sections, or combining evidence that is individually insufficient. MP-VRDU consequently introduces document-scale evidence-management and cross-page reasoning problems that do not arise to the same extent in the single-page setting, making it a distinct problem rather than a straightforward extension of VRDU.

This thesis examines MP-VRDU from two angles: a survey organising the field by how systems manage document-scale evidence, and an empirical analysis attributing incorrect answers to distinct failure modes. The survey consolidates a fast-moving literature into a single account of how these systems are built. The empirical analysis then tests competing and largely untested claims about how they should be built. Together, they consolidate the current state of the field and help guide future research.

1.1 Research Aims

This thesis addresses three complementary problems in MP-VRDU: how the field should be organised, how its systems should be designed and deployed, and whether these findings can be translated into a competitive working system.

Aim 1: comprehensively survey techniques and architectures for multi-page document understanding, treating it as a problem distinct from the single-page setting. We set out to organise the current landscape of MP-VRDU through an evidence-management perspective, focusing on the design decisions and techniques that arise specifically from long multi-page documents. Prior surveys remain anchored to the single-page setting or organise the field by narrower distinctions such as OCR dependency [6, 7]; none treats multi-page understanding as a distinct problem or takes evidence management as its organising lens.

Aim 2: determine how MP-VRDU systems should be designed and deployed through controlled empirical analysis. We set out to test competing claims about how these systems should be built, isolating key design choices while holding the remaining pipeline fixed. Existing diagnostic work typically studies individual components or aspects of system behaviour in isolation, and much of it is developed outside the document-understanding domain [8–10]. This leaves limited guidance on how these choices should be balanced within a complete MP-VRDU system.

Aim 3: design a simple MP-VRDU system that translates the thesis’s conceptual and empirical findings into a competitive working method. We set out to further validate the evidence-management perspective from Aim 1 and the empirical findings from Aim 2 by translating them directly into the design of an MP-VRDU system. The goal is to achieve performance comparable to or exceeding state-of-the-art systems using a deliberately simple design, demonstrating that these findings provide a practical basis for building effective document-understanding systems.

1.2 Thesis by Compilation

This thesis is structured around two papers addressing Aims 1 and 2, with Aim 3 developed as an additional contribution in Chapter 4. Chapter 2 surveys the MP-VRDU field, additionally serving as this thesis’s literature review, Chapter 3 presents the empirical attribution study and its design implications, and Chapter 4 synthesises both studies, introducing the system developed for Aim 3, and outlines broader implications and future work. The papers are as follows:

1. **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding.** Lewei Xu, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, Wei Liu. *Main Conference of EMNLP 2026*. Available at <https://doi.org/10.13140/RG.2.2.23979.60965>.
2. **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution.** Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, Wei Liu. *Submitted to EACL, under review*. Available at <https://arxiv.org/abs/2608.07943>.

Additional explanatory material, figures, and tables have been added to Chapters 2 and 3 to improve readability and provide context beyond the conventions and space constraints of the original venues. The empirical attribution paper submitted in early August was an earlier version of the study, limited

to a single primary dataset and lacking cross-family model comparisons, judge-human and judge-judge agreement analyses, and the broader design and deployment analysis. These are also weaknesses pointed out by reviewers, and have since been added to Chapter 3.

1.3 Background

1.3.1 A Brief History of Document Understanding

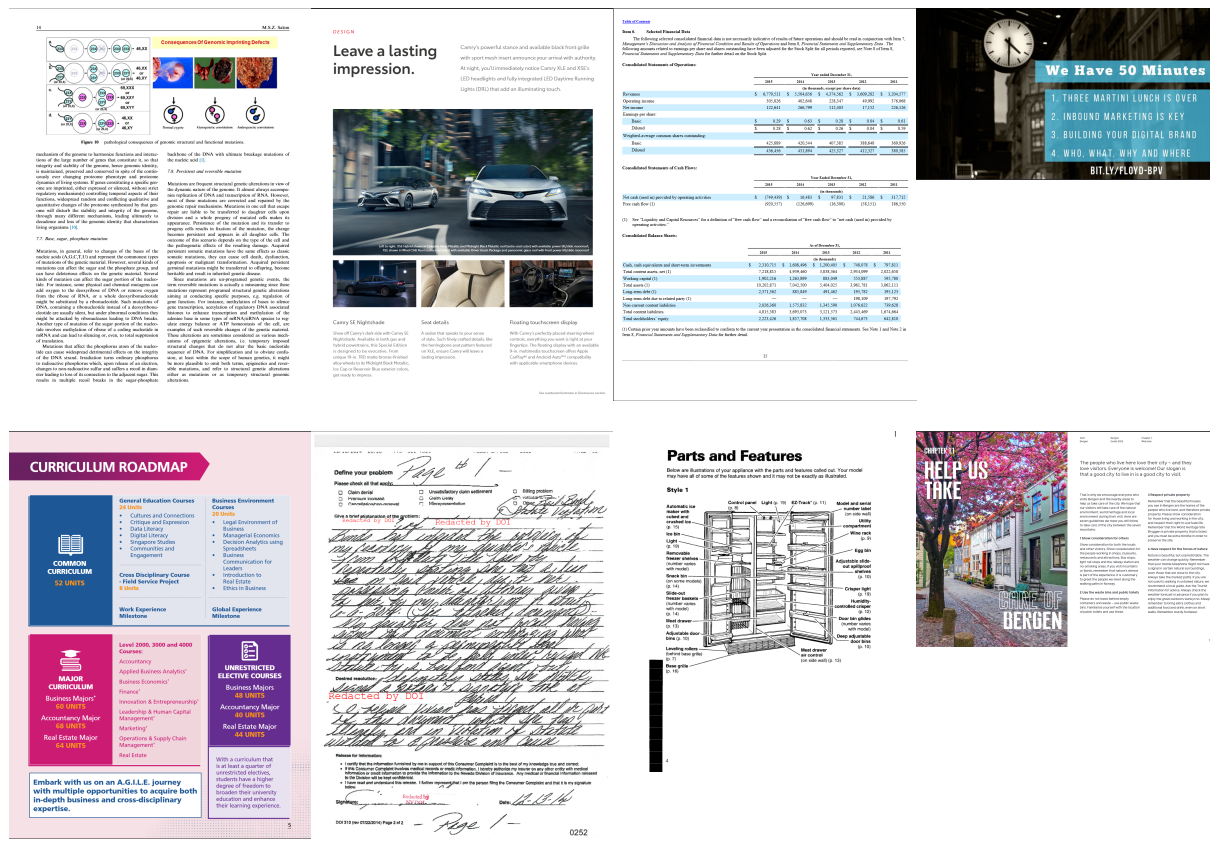


Figure 1.1 Representative pages from MMLongBench-Doc.

Automated document understanding predates the era of instruction-tuned language models and originally addressed a much narrower range of document types than the heterogeneous pages encountered today (Figure 1.1). Early systems were brittle and task-specific, treating document understanding as information extraction over forms and scanned business documents, typically pairing OCR with rule-based or shallow machine-learning classifiers to recover fields such as names, dates, and totals [11, 12]. LayoutLM [1] established in 2020 that layout, the two-dimensional position of text on a page, could be pretrained jointly with textual content rather than attached afterwards, moving the field from task-specific classifiers to a general pretrained backbone. DocVQA [4] reframed the task the following year as natural-language question answering over a document image rather than fixed-field extraction, so that a single model could answer arbitrary questions instead of extracting a predetermined set of fields. Both, however, remained confined to a single page. MP-DocVQA [13] then extended the setting to multiple pages in 2023, pairing a benchmark of multi-page documents with the Hi-VT5 baseline; its questions were still answerable from one page and its documents relatively short, but it marked the point at which

multi-page visually rich document understanding (MP-VRDU) became a distinct problem. Multimodal large language models (MLLMs), emerging over the same period, replaced task-specific encoders with general backbones that consume text, layout, and page images together and answer in natural language [2, 3]. A backbone reads only what fits its context window, however, so scaling to longer documents remained an open problem.

1.3.2 General-Domain MLLMs and MP-VRDU

General-domain MLLMs include both frontier closed-source systems and openly released, open-weight models. Unlike document-specialised models, these backbones are trained for broad multimodal tasks but can nevertheless read rendered pages directly and handle single-page and short multi-page inputs. Hosted products typically ingest both textual and visual document representations jointly, but as documents or collections exceed the available context, direct in-context processing becomes insufficient, motivating retrieval and tool-based access rather than placing the entire document collection in-context. Such techniques are already adopted in current hosted products, with both ChatGPT and Claude using retrieval and agentic tools to access project files and other connected sources [14–16]. These same retrieval and agentic strategies form major directions in MP-VRDU, discussed further in Chapter 2.

An alternative is to pretrain or continue-train models specifically for document understanding, a common approach in SP-VRDU but considerably less prevalent in MP-VRDU, where document-scale pretraining data remain scarce (§2.7.1). Most MP-VRDU systems therefore instead inherit a pretrained general-domain MLLM and concentrate their design choices around how documents are presented to and processed by that backbone. This includes how pages are represented, how evidence is selected and delivered, and how reasoning is organised over the resulting context. The boundary is not absolute, since some methods further train or modify the backbone itself, but the dominant design space lies in the document-specific machinery surrounding it. Chapter 2 examines these choices systematically through architectures, multimodal representation, inference-time adaptation, training, and datasets.

1.3.3 Single-Page VRDU Overview

Although MP-VRDU introduces document-scale problems, much of the machinery used to process each individual page is inherited directly from single-page VRDU. This includes how text, layout, and visual information are represented, as well as the document-oriented encoders and preprocessing techniques used before information is combined across pages. The single-page setting is also comparatively mature: document-specialised systems such as ORCA [17] approach human performance on the DocVQA [4] benchmark, reaching 0.9729 ANLS against a human baseline of 0.9811, with comparable progress on infographics [18], charts [19], and scanned forms. The discussion below therefore follows the organisation of Ding et al.’s survey of MLLM-based VRDU [6], restricted to the page-level techniques that multi-page systems build upon.

MLLM-based VRDU Architectures. At the architectural level, single-page systems divide into two families, shown in Figure 1.2. OCR-dependent frameworks parse the page first, running off-the-shelf OCR or a PDF parser to recover text alongside its bounding-box coordinates. These parsed signals are then encoded, either injected as text into the language model prompt [20] or passed through a vision or multimodal encoder such as LayoutLMv3 that binds text, layout, and image into joint multimodal representations [21, 22], which an adaptor and tokeniser convert into the token sequence the language model consumes. Relying on the parser lets these systems capture structural information without heavy

visual pretraining, but ties their accuracy to the parser's, most visibly on handwritten or low-quality scans where recognition degrades. OCR-free frameworks discard the parser and let a vision encoder read the rendered page directly from pixels, producing visual representations that the language model is trained to interpret through objectives such as text recognition and grounding [3, 23]. Because fine-grained text is legible only at high resolution, these frameworks consume large images and often insert a compression step to reduce the resulting visual token count before the language model [24]; in exchange, they train end to end and handle handwriting and irregular layouts that OCR struggles with.

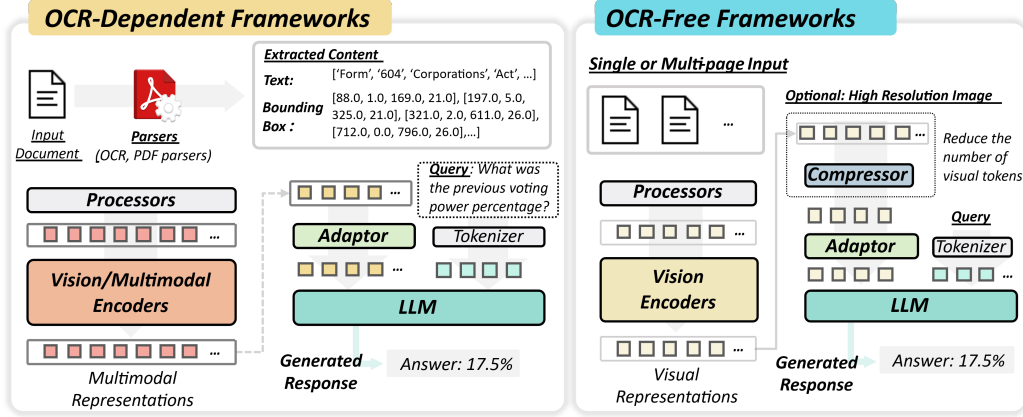


Figure 1.2 OCR-dependent and OCR-free framework architectures for MLLM-based VRDU. Adapted from Ding et al. [6].

Single-page Multimodal Representation. Within both families, a system must still decide how to represent text, layout, and visual content, and how to fuse them (Figure 1.3). Text is recovered as injected OCR strings [20], through joint text-layout-vision encoders [22, 25], or as a pretraining objective for the vision encoder itself [26]. Layout is encoded as positional features attached to text tokens [1, 25], as coordinates verbalised in the prompt [27], or through dedicated layout tokens [28]. Visual content is managed through low-resolution patch embeddings [29], high-resolution shape-adaptive cropping [2, 3], or token compression that cuts the visual sequence by an order of magnitude [24, 30]. Fusion draws these together through dedicated multimodal encoders [21], shared training objectives [3, 26], or prompt-based interleaving of text, layout, and images. Single-page systems make these choices assuming the entire document fits in a single backbone input; the multi-page setting (Chapter 2), breaks that assumption, reframing each choice around which pages or regions the system should attend to in the first place.

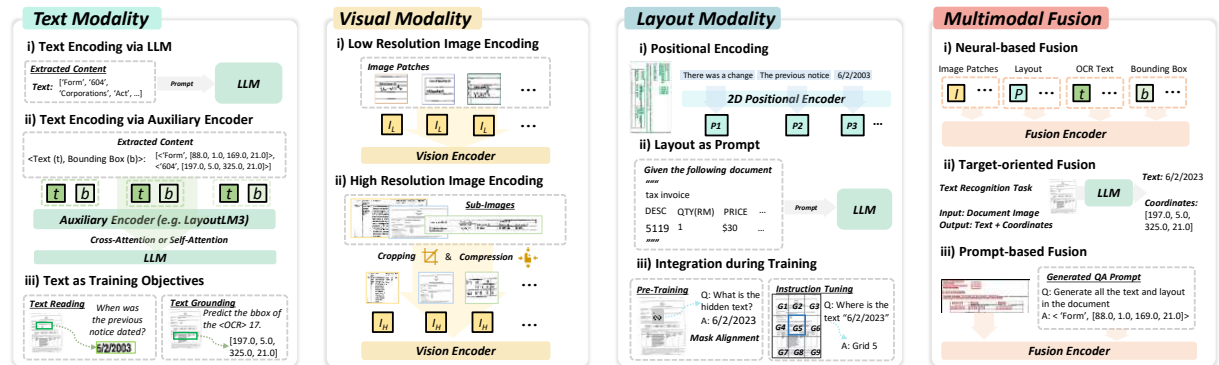


Figure 1.3 Multimodal feature representation and fusion mechanisms. Adapted from Ding et al. [6].

Chapter 2

Survey of Architectures and Techniques

This chapter is adapted from **Managing Evidence at Document Scale: A Survey of Multi-Page Visually Rich Document Understanding** [31], by **Lewei Xu**, Yihao Ding, Zihan Xu, Siwen Luo, Daochang Liu, and Wei Liu, published in the *Main Conference of EMNLP 2026*. The first author was responsible for constructing the surveyed literature corpus, including paper identification and selection; developing the architectural taxonomy used to organise MP-VRDU systems; and preparing the technical content, including the survey analysis, appendix material, and tables. **This chapter additionally serves as the Literature Review of this thesis.**

2.1 Introduction

MP-VRDU has seen rapid growth in recent years, with new architectures, training strategies, and inference methods emerging. Existing work has explored joint multi-page encoding [13], adaptation of pretrained MLLMs [32], retrieval-augmented pipelines that decouple evidence selection from generation [33], and adaptive controllers that gather evidence iteratively at inference time [34], often combined with task-specific training or prompted reasoning loops [35, 36]. However, this body of work has developed along largely parallel lines, with limited cross-family comparison and inconsistent terminology, and few works explicitly distinguish what is genuinely multi-page-distinctive from what is inherited from the single-page setting. As a result, the field lacks a unified view of the design choices that matter most in multi-page settings, motivating the systematic organisation this chapter provides.

Prior surveys have charted VRDU from complementary angles. Subramani et al. [11] survey the OCR-centric deep-learning pipelines of the pre-MLLM era, covering text detection, recognition, and layout analysis as separate stages. Ding et al. [37] and Gbada et al. [38] both treat information extraction over visually rich documents, while a more recent survey by Ding et al. [6] covers the shift to MLLM-based architectures, training strategies, and benchmarks. Closest to the multi-page setting, Gao et al. [7] survey multimodal retrieval-augmented generation as a means of scaling document understanding to the multi-page setting. Several of these include multi-page systems among the work they cover, but none treats multi-page understanding as a distinct problem with its own design axes, addressing it only in passing within a broader single-page or extraction-focused scope.¹

Our contributions are as follows: (1) We define MP-VRDU as a distinct document-understanding problem and organise the field through an evidence-management perspective. (2) We introduce a new architectural taxonomy for MP-VRDU, grouping existing systems into four families according to how they manage document-scale evidence (§2.3). (3) We provide a systematic survey of the architectures, multimodal representations, inference-time and training strategies, datasets, and evaluation practices used in

¹A detailed survey comparison table is provided in Appendix A.1.

2. SURVEY OF ARCHITECTURES AND TECHNIQUES

Model	Venue	OCR	Vision Encoder	LLM Backbone	Mod.	Search	Reasoning	Agentic
Page-to-Document Encoding Models								
Arctic-TILT (2025)	ACL	Yes	U-Net	T5-Large	T, V, L	-	-	-
GRAM (2024)	CVPR	Yes	DocFormerv2	DocFormerv2	T, V, L	-	-	-
Hi-VT5 (2023)	PR	Yes	DiT	T5-Base	T, V, L	-	-	-
RM-T5 (2024)	DAS	Yes	DiT	T5-Base	T, V, L	-	-	-
MLLM-Centric Adaptation Models								
DocR1 (2026)	AAAI	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	-	CoT, HD	-
DocSeeker (2026)	CVPR	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	-	CoT, HD	-
Leopard (2025)	TMLR	No	SigLIP-SO400M	Mult. (La/Mi)	V	-	CoT	-
mPLUG-DocOwl2 (2025)	ACL	No	ViT-L/14	mPLUG-Owl2	V	-	-	-
URaG (2026)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	-	HD	-
Docopilot (2025)	CVPR	No*	InternViT-300M	Mult. (In)	V	-	-	-
Texthawk2 (2024)	preprint	No*	SigLIP-SO400M	Qwen2-7B	V	-	-	-
DocSLM (2026)	CVPR	Yes	SigLIP2	Qwen2.5-1.5B	T, V, L	-	-	-
InstructDoc (2024)	AAAI	Yes	CLIP	FlanT5, BLIP2	T, V, L	-	-	-
LayTokenLLM (2025)	CVPR	Yes	-	Mult. (La/Qw)	T, L	-	-	-
CoR (2026)	preprint	Yes*	Qwen2.5-VL	Qwen2.5-VL-7B	T, V	-	CoT, PE, HD	-
Retrieval-Augmented Pipelines								
AVIR (2026)	MMAAsia	No	Pix2Struct	Qwen2.5-VL-3B	V	D	-	-
DFVC (2026)	MDPI	No	VisRAG, ColQwen	Qwen2-VL-2B	V	D	-	-
DREAM (2025)	MM	No	InternVL2 ViT	InternVL2-4B	V	J	HD	-
M3DocRAG (2024)	preprint	No	ColPali, ColQwen	Mult. (Qw/In/Id)	V	D	-	-
MM-R5 (2025)	AAAI	No	Qwen2.5-VL ViT	Qwen2.5-VL-7B	V	D	CoT	-
MoLoRAG (2025)	EMNLP	No	ColPali	Mult. (Qw/DS/Lv)	V	D, Nav	-	1A
Self-Attention Scoring (2024)	ICDAR	No	Pix2Struct	Pix2Struct	V	D	-	-
SLEUTH (2025)	CVPR	No	ColPali, Qwen3-VL ViT	Mult. (Qw/GL/Ge)	V	D	HD	4A
SV-RAG (2025)	ICLR	No	InternVL2, Phi-3-vision	Mult. (In/Pa/Ph)	V	D	-	-
VDocRAG (2025)	CVPR	No*	Phi3V-4.2B	Phi3V-4.2B	V	D	-	-
CREAM (2024)	ACM	Yes	Pix2Struct	LLaMA2-7B	T, V	J	HD	-
KGP (2024)	AAAI	Yes	-	Mult.	T, L	S, IR, Nav	-	1A
MHier-RAG (2026)	preprint	Yes	Qwen-VL-Plus	Mult. (Qw/DS/GPT)	T, V, L	J	CoT, HD	-
MLDocRAG (2026)	preprint	Yes	Qwen2.5-VL	Mult. (Qw)	T, V, L	D	CoT, HD	-
MultiDocFusion (2026)	ACL	Yes	-	Mult. (Qw/La/Mi)	T, V, L	D	HD	-
M2RAG (2025)	NeurIPS	Yes*	VisRAG-Ret	Mult. (Qw)	T, V	J	-	3A
MDocAgent (2025)	preprint	Yes*	ColPali, ColQwen2	Mult. (La/Qw)	T, V	J	-	5A
PDF-WuKong (2026)	IJCV	Yes*	IXC2-VL-4KHD	IXC2-VL-4KHD	T, V	J	-	-
RAG-DocVQA (2025)	ICDAR	Yes*	DiT, Pix2Struct	Qwen2.5-VL-7B	T, V, L	D	-	-
VisDoMRAG (2025)	NAACL	Yes*	ColPali / ColQwen2	Mult. (Qw/Ge/GPT)	T, V	J	CoT	3A, SC
Adaptive-Trajectory Pipelines								
Doc-V* (2026)	ACL	No	Qwen2.5-VL	Qwen2.5-VL-7B	V	D, IR, QR, Nav	ReAct, HD	1A
DocReact (2025)	ACL	No	ColPali, VisRAG	GPT-4o	V	D, IR, QR	ReAct, QD	1A
MACT (2025)	preprint	No	Multiple	Mult. (Qw/MM/In)	V	D	CoT, PE	4A, SC, SV
VRAG-RL (2025)	NeurIPS	No	Qwen2.5-VL ViT	Qwen2.5-VL-3B/7B	V	D, IR, QR	ReAct, HD	1A
DocLens (2026)	ACL	Yes	-	Mult. (GPT/Ge/Cl)	T, V, L	J	CoT	4A, SA
DMap (2026)	WWW	Yes*	ColPali	Mult. (GPT/Qw)	T, V, L	J	HD	2A, SR
DocAgent (2025)	EMNLP	Yes*	-	Mult. (GPT/Ge/Cl)	T, V, L	J, IR, Nav	ReAct, HD	3A
DocDancer (2026)	preprint	Yes*	Qwen3-VL-235B	Qwen3-4B/30B	T, V, L	J, IR, QR	ReAct	1A
MM-Doc-R1 (2026)	ACL	Yes*	Qwen2.5-VL ViT	Qwen3-4B/8B	T, V	S, IR, QR, Nav	ReAct, HD	3A
SimpleDoc (2025)	EMNLP	Yes*	ColQwen-2.5	Mult. (Qw)	T, V	J, IR, QR	-	1A, SR, SV
ViDoRAG (2025)	EMNLP	Yes*	NV-Embed-V2, ColQwen2	Mult. (Qw/La/GPT)	T, V	J, IR	ReAct, HD	3A, SR, SV

Table 2.1 Survey of MP-VRDU models grouped by architecture. T = text; V = visual; L = layout. CoT = Chain of Thought; ReAct = Reason + Act; PE = Plan-Execute; QD = Question Decomposition; HD = Hierarchical / Coarse-to-Fine. S = Sparse; D = Dense; J = Joint; IR = Iterative Retrieval; QR = Query Reformulation; Nav = Navigation. nA = n agents; SR = Self-Refine; SV = Self-Verify; SC = Self-Consistency; SA = Sampling-Adjudication. OCR column: Yes* = not OCR-dependent; No* = OCR-derived supervision explicitly used during training. *Mult.* denotes a system evaluated on multiple backbones: Qw = Qwen; Ge = Gemini; GL = GLM; Cl = Claude; DS = DeepSeek; In = InternVL; Id = Idefics; La = LLaMA; Lv = LLaVA; Mi = Mistral; MM = MiMo; Ph = Phi; Pa = PaliGemma. ‘-’ indicates not reported.

MP-VRDU (§2.4-§2.7). (4) We identify the principal open challenges that define the current frontier of the field with broader implications and future directions (deferred to Chapter 4).

2.2 Methodology and Scope

This chapter covers MP-VRDU systems published from 2023 (starting from MP-DocVQA’s Hi-VT5 [13]) across major venues such as CVPR, ACL, EMNLP, and ICLR. Candidates were identified through keyword search over multi-page terms (“long document”, “cross-page reasoning”, “page-level retrieval”) combined with task and method terms (“document visual question answering”, “document

RAG”, “long-context vision-language model”), supplemented by forward-citation sweeps anchored on established benchmarks such as MP-DocVQA, MMLongBench-Doc, and LongDocURL. We then manually screened for substantive multi-page contributions such as novel architectures, retrieval mechanisms, or training strategies. The dataset catalogue covers datasets used by the surveyed systems, plus a small number of relevant but underused ones. Rather than a systematic scoping review aiming at exhaustive retrieval, we prioritise representative coverage of the design space, since MP-VRDU is evolving rapidly and its terminology remains in flux. Full surveyed corpus details, search terms and inclusion criteria are available in Appendix A.2.

2.3 Multi-Page Architectural Overview

The defining problem in MP-VRDU is evidence management, and how a system manages cross-page evidence is fixed by its architectural design, which determines **how evidence is selected, preserved, and exposed for reasoning**. We therefore organise the field by architecture rather than by the OCR dependency used in prior surveys [6], into four families. *Page-to-document encoding models* jointly encode pages with cross-page representations (§2.3.1). *MLLM-centric adaptation models* adapt MLLMs to multi-page inputs within context and token budgets (§2.3.2). *Retrieval-augmented pipelines* retrieve relevant evidence before generation (§2.3.3). *Adaptive-trajectory pipelines* iteratively inspect evidence using an LLM controller (§2.3.4). Reported performance across the four families on principal MP-VRDU benchmarks is provided in Appendix A.3.1 (Table A.3).

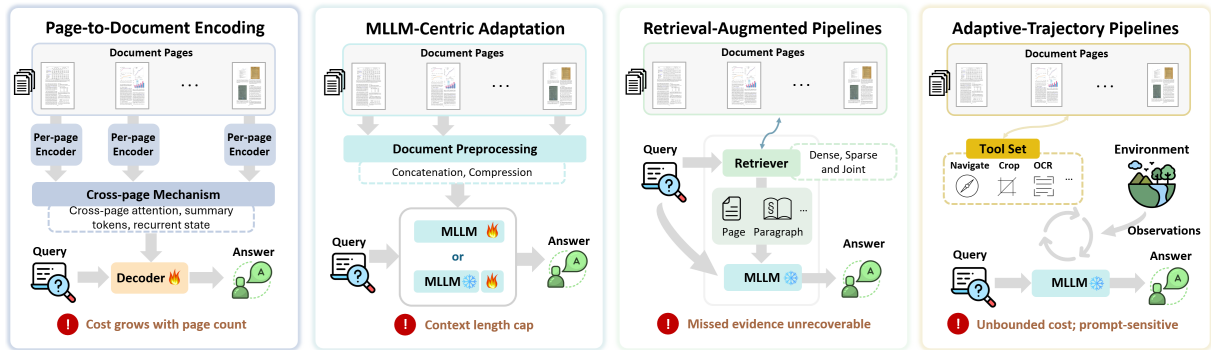


Figure 2.1 Illustration of MP-VRDU Architectures. Snowflake indicates typically frozen; Fire indicates typically trained; Both indicates even split.

2.3.1 Page-to-Document Encoding Models

Page-to-document encoding models process each page through a shared layout-aware document encoder such as LayoutLMv3 [21] that fuses OCR text, layout, and visual features at the page level, then connect per-page representations through a document-level exchange mechanism within a single forward pass. The mechanism may take the form of per-page summary tokens [13], document-global tokens with local-global attention [40], recurrent memory states passed across pages [41], or sparse cross-page attention over long OCR sequences [39]. By preserving page identity, local layout, and reading order in the architecture itself, this design provides a strong inductive bias for multi-page structure and is well suited to documents with regular templates such as multi-page forms and business reports [13, 39]. The principal limitation is intrinsic to the design: every page is still encoded jointly in a single forward pass, and the per-page compression step that enables document-level exchange discards fine-grained evidence

as inputs scale. As the context windows of pretrained MLLMs have widened, the motivation for a dedicated cross-page encoder has weakened, and subsequent families absorb cross-page information into the backbone or select evidence before reading.

2.3.2 MLLM-Centric Adaptation Models

MLLM-centric adaptation models remove the dedicated cross-page encoder entirely and process the document inside a pretrained MLLM backbone whose context window absorbs cross-page information directly, with adaptation ranging from training only lightweight modules to fine-tuning the backbone itself. At the lightweight end, *projectors* that map per-modality features into the backbone’s token space [29] and *adapters* that insert small trainable modules into a frozen backbone for parameter-efficient fine-tuning [28] keep most weights fixed, while visual-token compression further constrains token growth on long inputs to keep multi-page documents within the context budget [24]. At the heavier end, the MLLM backbone itself is unfrozen for continued pretraining or instruction tuning on document corpora, trading parameter efficiency for stronger document-specific grounding [32], with long-context tuning further improving performance on documents of moderate length [3, 30]. The family inherits the foundation MLLM’s instruction-following and reasoning, avoids the engineering overhead of external retrieval or orchestration, and remains parameter-efficient to train. The cost of this design is that performance degrades on longer documents [43]: as most models in this category are OCR-free, reliable text recovery depends on high-resolution input that aggressive token compression readily compromises, and document structure such as section hierarchy and figure-caption linkage must be recovered implicitly from pixels rather than supplied through dedicated tokens [32, 45]. Combined with the attention-dilution problem [5], these limitations motivate subsequent families that move evidence selection out of the backbone altogether.

2.3.3 Retrieval-Augmented Pipelines

Retrieval-augmented pipelines decouple evidence selection from answer generation: a retriever first selects relevant evidence units, such as textual chunks, pages, or graph nodes, and the generator then produces an answer conditioned on this compact subset [33, 55]. This separation allows the pipeline to scale to multi-page inputs that exceed what MLLM-centric and page-to-document encoding models can hold in a single context, since inference cost depends on the retrieved content rather than on document length. The retriever may be a sparse lexical scorer, a dense embedding model, or a structure-aware index over a parsed representation of the document (§2.5.1), and the generator is typically a pretrained MLLM. The family spans single-pass retriever-generator systems [33] to multi-component pipelines that compose multiple LLM-instantiated “agents” in specialised roles such as modality-specific reading branches [36] or modality-fusion synthesis [61], but in either case the control flow is set at design time. Modular decomposition lets the retriever and generator be swapped independently, the retrieved evidence provides a partial audit trail for the answer, and the same interface extends to multi-document corpora without architectural change. Because the flow is fixed, evidence missed or poorly ranked at retrieval is unrecoverable at generation, questions that require aggregating multiple weakly relevant pages frequently fail [51], and upstream errors in chunking, OCR, or modality alignment propagate directly into the answer.

2.3.4 Adaptive-Trajectory Pipelines

Adaptive-trajectory pipelines instantiate the agent paradigm characterised by Plaat et al. [75] and Wang et al. [76], in which one or more LLM controllers reason, act on an external environment, and condition

each subsequent action on observed outcomes rather than executing a plan fixed in advance [34, 65]. Unlike the fixed-role agents of §2.3.3, the controller’s trajectory is not predetermined: it selects each evidence-gathering action at inference time, with actions surfacing as explicit calls to external retrievers, parsers, or OCR engines [67, 74], or as structured responses the framework interprets as a re-retrieval or as navigation over a pre-built document representation [73]. A controller might issue a refined query when the first retrieval returned irrelevant pages, request a higher-resolution view of a page surfaced by an earlier pass, or follow a cross-reference from one section to another. ReAct [77] is the most common controller realisation (§2.5.2), though other variants also appear. Because the trajectory is constructed rather than fixed, compute scales with question difficulty rather than document length, and the full action sequence becomes an inspectable audit trail. The cost of that flexibility is that inference cost is difficult to bound in advance, controllers are sensitive to prompt phrasing and may commit early to wrong evidence and continue confidently down the wrong trajectory without surfacing the error, stochastic generation complicates reproducibility, and the system still inherits the recall ceiling of every retriever, parser, or scoring module it consults.

2.4 MP-VRDU Multimodal Representation

Every evidence-management decision is enacted over a specific modality, and its cost depends on how that modality is represented. While Section 2.3 categorises MP-VRDU systems by where cross-page evidence is processed, this section analyses the problem from a modality perspective. MP-VRDU systems must locate sparse evidence across long multimodal contexts, forcing each modality to trade *page coverage* (how much of the document the system can attend to) against *reading fidelity* (how accurately each attended region is interpreted), since long-context models attend poorly to evidence buried in the middle of long inputs [5]. The single-page mechanisms inherited from prior work, including OCR-text injection, per-image patch encoding, and per-token layout signals, are reviewed in Section 1.3.3; here we concentrate on the multi-page pressures that text (§2.4.1), visual (§2.4.2), and cross-page structural (§2.4.3) evidence introduce.

2.4.1 Text Modality

Text in MP-VRDU presents four bottlenecks. **Volume** across pages routinely exceeds what the MLLM can process in full. This is addressed either by compressing all pages into a fixed budget through page summaries, memory tokens, or long-context encoders [24, 39, 41], or by indexing chunks and passing only selected text to the reader [33, 56, 62]. The first approach fixes compression before the question is seen, so evidence absent from the compressed representation cannot be recovered later in the pipeline; the second makes evidence missed at retrieval unrecoverable at the reading stage, and broadening the retrieved set merely shifts the cost from omission to a related failure mode known as *attention dilution*, in which the model spreads its attention across too many candidate passages and loses focus on the answer-bearing ones [5]. **Continuity** breaks when paragraphs, tables, and section context span page boundaries. This is preserved by recurrent state propagation that carries page context forward across the document [41], structure-aware chunking that groups text under recovered section headings rather than along sliding-window boundaries [60], and adjacent-page expansion at retrieval time that pulls in neighbouring pages when a candidate is surfaced [48]. Each restores local cross-page context at the cost of broader query relevance, since the additional context is selected by proximity rather than by question. **Noise** accumulates as OCR and parsing errors compound across pages: a recognition error rate that is tolerable on a single page multiplies over a long document, and because extracted text is

the retrieval index in most pipelines, these errors degrade retrieval as well as reading, surfacing wrong pages or burying the right ones. It is dampened upstream through layout-aware parsing pipelines that reduce recognition error, or downstream by treating low-confidence tokens as soft cues the reader can discount rather than as ground truth [45], yet how noise propagates through multi-page retrieval indexes remains largely uncharacterised. **Heterogeneity** in text density varies sharply across pages within a single document, yet receives the least attention: most systems apply fixed token budgets and fixed top- k uniformly across pages and queries, over-compressing dense pages while wasting capacity on sparse ones. Adaptive selection that scales the returned set with the per-query score distribution [47] only begins to address this variability, and no current system varies its per-page text budget within a single document.

2.4.2 Visual Modality

Visual evidence preserves signals that text extraction loses, but its cost structure introduces three bottlenecks of its own. **Resolution** cannot be maintained at full document length, since token cost scales with page count multiplied by per-page resolution, forcing a choice between compression and selective high-fidelity reading. Token-reduction modules and adaptive sub-image budgets [24, 30, 43] keep every page available in context but leave fine-grained content, including small text, table cells, and chart annotations, vulnerable to downsampling. Inspecting only a retrieved subset at high resolution [69, 73] preserves detail at the cost of inheriting the selector’s recall ceiling: pages not surfaced by the retriever are never read, regardless of the relevance of their content. **Multi-signal competition** arises because the same visual budget must support text-bearing pixels, non-textual evidence such as charts, figures, and handwriting, and layout cues within a single page. Visual RAG systems, which retrieve over rendered page images rather than over extracted text, encode pages directly so all signals share one representation [33, 55]; the page becomes a single evidence unit that cannot be decomposed at retrieval time. Element-level cropping of tables, figures, and charts [69] preserves fine structure at the cost of detector dependence and separation from the surrounding explanatory text on which interpretation often depends. A third route delegates text recovery to an OCR pipeline so the visual budget can prioritise non-textual evidence [45], which avoids the budget competition at the cost of inheriting OCR fragility on handwriting and complex layouts. **Heterogeneity** in visual complexity varies sharply across pages within a single document, yet receives the least attention: most systems apply uniform resolution and sub-image budgets regardless of whether a page is a sparse cover sheet or a dense table-laden appendix, and evidence-guided allocation that varies the visual budget by content [42] only begins to address per-page and per-query variability.

2.4.3 Cross-Page Structure Modality

Structure turns isolated page observations into document evidence. Per-token layout remains relevant at the multi-page scale, with specialised tokens encoding spatial geometry within each page [28], but the MP-distinctive difficulty is relationships whose scope crosses pages. **Hierarchy and reading order** is the foundational problem: sections, lists, columns, and captions routinely span page boundaries, and the same evidence may serve different reasoning paths depending on which parent section it sits under. Explicit parsers expose this structure as outlines, trees, or hierarchical indices that retrieval can navigate [34, 60], while OCR-free backbones attempt implicit recovery from page images through layout-aware pretraining objectives [24, 30]. The unresolved gap is robustness: parser errors in heading detection or table-boundary segmentation propagate directly into every downstream retrieval and navigation step, and implicit recovery depends on high-resolution input that the compression discussed in §2.4.2 may have already removed. **Cross-page references** demand resolving pointers such as “see Figure 3”

or “as shown in Section 2” to their referents on other pages, rather than ranking pages by surface similarity. Graph-based methods represent these references explicitly as edges between document elements, through knowledge graphs over passages and structural nodes [57] or page graphs with semantic-logical edge scoring [51], improving navigability at the cost of edge quality bounded by extraction heuristics or LLM judgement at construction time. **Spanning elements** are the structural form of the text-continuity problem from §2.4.1: multi-page tables, section-spanning arguments, and theorem-proof or footnote-reference linkages require evidence units larger than a page but smaller than the document. Hierarchical retrieval that routes through section-level units before drilling into chunks [58] and cross-page retrieval that preserves parent context [48] address this partially, but the boundary between useful context and irrelevant neighbours remains hard to set without per-query supervision.

2.5 Inference-Time Adaptation Strategies

Inference-time adaptation improves MP-VRDU performance through changes to the inference pipeline rather than to model parameters: a frozen LLM or vision-language backbone is paired with off-the-shelf retrievers, structural parsers, prompted reasoning loops, or multi-agent orchestrations, with no gradient updates required. These techniques are predominantly used in the retrieval-augmented and adaptive-trajectory frontier, though prompting strategies such as chain-of-thought reasoning also apply to MLLM-centric and page-to-document models that operate over the assembled document directly. We organise these strategies by the evidence-management decision they implement: selecting evidence through retrieval and navigation (§2.5.1), reasoning over assembled evidence (§2.5.2), and iteratively acquiring evidence or distributing acquisition across multiple controllers (§2.5.3).

2.5.1 Retrieval and Navigation

Retrieval narrows long documents to compact evidence units for downstream reasoning. Two broad families have emerged, distinguished by whether passages are scored independently against the query or navigated through explicit document structure.

Similarity-based retrieval. Similarity-based methods score each passage against the query independently and return the top-ranked subset. Sparse lexical signals such as BM25 persist as baselines or auxiliary channels, while dense retrieval in MP-VRDU has consolidated around vision-language late-interaction encoders such as ColPali [78], which extends the late-interaction paradigm of ColBERT [79] to page images by producing token-level query vectors and patch-level page embeddings, then scoring a query-page pair by taking, for each query token, the maximum cosine similarity over all page patches and summing the result (Figure 2.2). This *MaxSim* aggregation preserves fine-grained alignment between specific query terms and specific page regions, critical when a single question token may need to align with one figure caption, table cell, or paragraph buried in a long document. Granularity varies from chunks to whole pages, with adaptive selectors adjusting the per-query retrieval count when evidence density is uneven across the document [47]. A second reranking stage commonly refines an initial dense recall through LLM-based scoring over candidate passages or page summaries [56, 73], optionally under persistent working memory that carries cross-query context across iterations [73]. Independent scoring handles paraphrase and cross-modal queries well, but cannot follow cross-page references or multi-hop dependencies that no single passage carries.

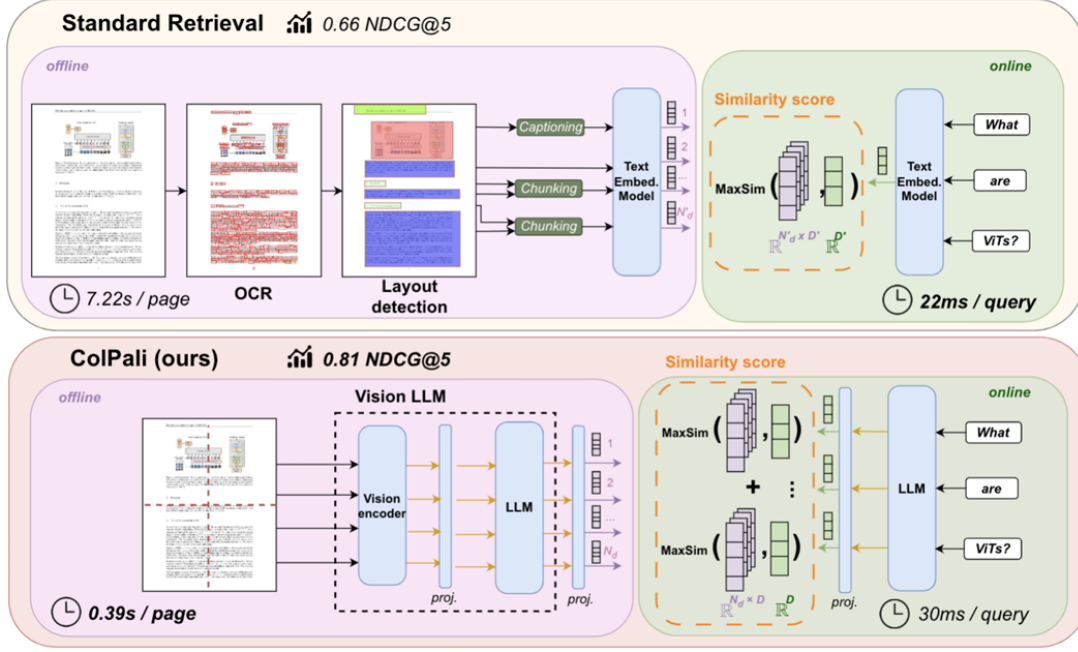


Figure 2.2 ColPali late-interaction retrieval for visually rich document pages. Query-token embeddings are matched against patch-level page embeddings using MaxSim , producing a relevance score that preserves fine-grained alignment between query terms and local page regions.

Relation-aware retrieval. Relation-aware methods retrieve along explicit document structure rather than scoring passages independently. Tree-based variants extract section hierarchies and outlines and group passages along semantic boundaries so retrieved chunks preserve parent context, with some routing queries through dual in-page and cross-page indices to recover both local and global evidence [34, 58, 60]. Graph-based variants treat headings, entities, and cross-references as graph nodes with typed edges, so controllers or retrievers can traverse from one passage to related ones rather than relying on similarity alone; extensions add logical relevance scoring or query-conditioned graph construction to bias traversal toward inferentially connected nodes [51, 57, 59]. Relation-aware methods recover references that similarity scoring misses, at the cost of construction overhead, dependence on parser quality, and traversal cost that grows with the multi-hop count required to answer the question.

2.5.2 Reasoning Strategies

Reasoning strategies act over a single fixed evidence buffer assembled prior to the reasoning call. The buffer may be the full document representation, a retrieved subset, or the output observed at one step of an agentic loop; the unifying property is that the reasoning step itself does not acquire new evidence, which makes these strategies independent of architectural family. They range from purely prompted interventions to strategies requiring changes to decoding, sampling, or control flow, but none updates model parameters. We cover the mainstream reasoning paradigms below.

Sequential Reasoning. A single reasoning thread proceeds linearly, optionally revising itself as it goes. *Chain-of-Thought* (CoT) prompting [80] elicits intermediate reasoning before the final answer, which in MP-VRDU binds evidence from different pages step by step, mitigates attention dilution over long inputs, externalises the reasoning trace for verification, and decomposes multi-hop questions into single-hop steps [35, 43]. Because it operates only over the assembled buffer, its contribution lies in how

faithfully that evidence is used rather than in recovering evidence the buffer omits: missing evidence cannot be reasoned back into existence. *Self-Reflection* extends the single thread with iterative revision, having the model audit or refine its own answer against the buffer, optionally conditioned on feedback from a verifier or critic step [73, 74, 81]. This reduces single-pass error at the cost of additional rounds and a self-evaluation signal whose reliability bounds the gains.

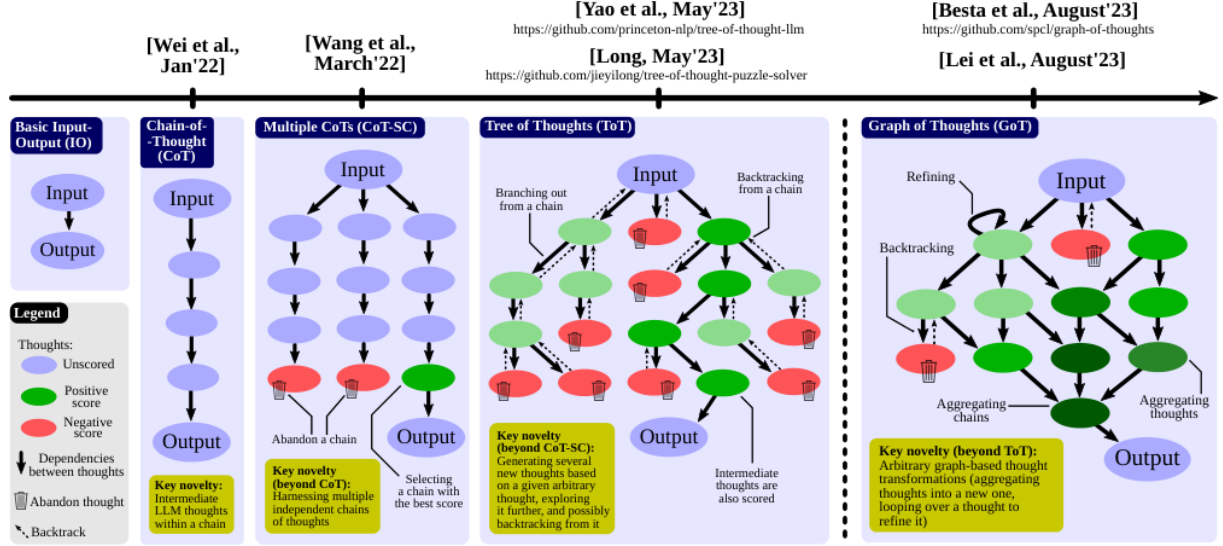


Figure 2.3 Evolution of reasoning topologies used in prompting schemes. Extracted from Besta et al. [82]. Only CoT is used pervasively in MP-VRDU; this diagram is for illustration purposes only.

Parallel Exploration. Several reasoning paths are produced independently and combined into one answer. *Self-Consistency* [83] samples multiple chains and selects the majority answer, trading inference cost for robustness against single-chain failures, with sampling-adjudication variants replacing the majority vote with an LLM judge that compares reasoning paths and selects the most consistent conclusion [69], and parallel-pipeline variants enforcing consistency across modality-specific chains rather than across samples of one chain [64]. *Tree-of-Thoughts* (ToT) [84] instead explores reasoning branches with backtracking, but has seen limited uptake in MP-VRDU, where evidence sparsity rather than search-space branching is the dominant difficulty. These patterns are independent of architectural family: they can wrap a single retrieval-augmented forward pass or operate inside one step of an adaptive-trajectory loop, in which case the loop and the within-step pattern compound their effects on robustness and cost.

2.5.3 Agentic Methods

Agentic methods distribute the MP-VRDU workflow across one or more LLM-instantiated controllers, either iterating evidence acquisition under inference-time control or coordinating across specialised roles. We note that *agentic* is used ambiguously in the MP-VRDU literature, covering both systems whose controller selects actions at inference time, following the framing by Plaet et al. and Wang et al., and systems composed of multiple LLM-instantiated agents in fixed roles; we treat both senses here, following the architectural distinction in §2.3.3–§2.3.4.

ReAct and Tool-Augmented Reasoning. ReAct prompting [77] augments the action space with language thoughts, interleaving thought-action emissions in a single stream whose observations feed the

next step, letting a single controller acquire evidence during inference rather than reason over a fixed buffer. The action space takes two forms in MP-VRDU. Document-side tool calls invoke external retrievers, parsers, or OCR engines, with each round concentrating on a chosen page or region [34, 66], lifting the single-step recall ceiling of retrieval-augmented pipelines at the cost of tool latency and compounding error from early wrong actions. Structured navigation instead operates over a pre-built document representation, with the controller selecting which node, section, or page to inspect next [65], which trades dependence on external tool fidelity for dependence on the quality of the representation built at index time. Adaptive-trajectory systems that lack the thought-action schema, such as those built on reflection-style iterative refinement [73], share the iterative control regime without being ReAct in the strict sense.

Agentic Decomposition. Agentic decomposition distributes the workflow across multiple controllers whose interaction pattern, rather than a single monolithic controller, drives evidence gathering and answer synthesis. Each controller operates on a narrower input and clearer objective than a single backbone tasked with the entire workflow, mitigating the cognitive overload that arises when one model must locate, read, verify, and answer over a long multimodal document [67]. Two patterns recur. *Modality-specialised* decomposition assigns separate controllers to text and visual evidence and adds a synthesis controller that reconciles their outputs, preserving single-modality reading fidelity at the cost of a reconciliation step that lacks access to the original document [36, 61]. *Role-specialised* decomposition assigns controllers to procedural stages such as planning, execution, verification, and synthesis, with explicit routing between stages when a downstream controller detects an error [34, 67]. Separating judgement from generation reduces the cognitive blind spots of self-correction, but introduces inter-controller compatibility costs and a recall ceiling bounded by upstream stages.

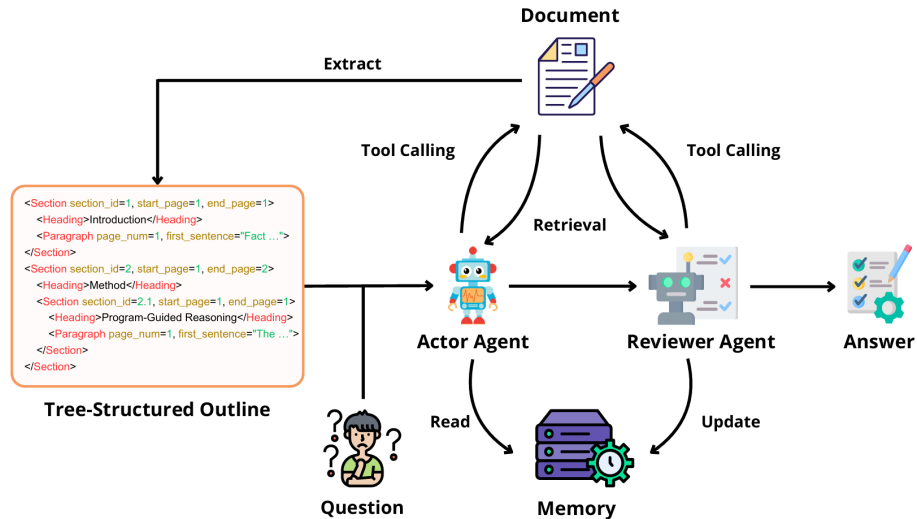


Figure 2.4 Architecture of DocAgent. Included for illustrative purposes. Extracted from Yang et al. [34]. “We first construct the document outline as a compact XML tree structure. An actor agent processes the outline and question as input, leveraging retrieval tools to extract relevant content for answering the question. Next, an evaluator agent reviews the initial response. If a correction is necessary, the actor agent’s memory is updated through reflection, facilitating knowledge transfer across tasks.”

2.6 Training Strategies

VRDU training follows a stepwise pipeline established in the single-page setting [6]: self-supervised pretraining instils document reading through objectives such as masked information modelling and cross-modality alignment, instruction tuning aligns outputs with user queries, and supervised fine-tuning adapts the model to target tasks. MP-VRDU inherits this pipeline but concentrates its effort at the later stages, since document-scale pretraining corpora remain scarce and most systems adopt backbones already pretrained on single-page data, with few exceptions targeting cross-page objectives directly [24, 32]. Fine-tuning instead carries the weight, scaled by LVLM-synthesised supervision (§2.7), with reinforcement learning increasingly layered above it. Training therefore internalises the evidence-management decisions of §2.5 into model parameters, supplying capabilities that prompting cannot: cross-page evidence selection (§2.6.1), document-scale reasoning over assembled evidence (§2.6.2), and reliable agentic acquisition (§2.6.3). Two cross-cutting techniques recur across all three (§2.6.4).

2.6.1 Trained Retrieval Components

Off-the-shelf encoders such as ColPali and ColQwen [78] score each page independently against the query, so retrieved results are dominated by local query-page similarity rather than by document-level context. Retriever training has therefore explored three complementary directions. *Modality-alignment* methods learn visual document embeddings that remain aligned with OCR-derived text, enabling page images to be retrieved without the storage and latency cost of the late-interaction patch-level scoring discussed in §2.5.1 [55]. *Cross-page-fusion* methods enrich each page representation with neighbouring-page context, typically through a lightweight head trained with parameter-efficient methods such as LoRA [85], a tuning approach that adds small trainable matrices to frozen weights; the result makes evidence distributed across adjacent or related pages retrievable as a unit rather than as a set of independently-ranked passages [48]. *Logical-relevance* methods train graph-based scorers to rank pages not only by embedding similarity but also by their role in cross-page reasoning paths, turning retrieval into a learned multi-hop walk rather than a single-step similarity match [51]. The three directions are not interchangeable: modality-alignment improves retrieval throughput, cross-page-fusion improves recall on adjacent evidence, and logical-relevance improves recall on inferentially-connected evidence.

2.6.2 Trained Reasoning over Assembled Evidence

Prompted reasoning leaves quality sensitive to prompt design and base-model behaviour, motivating approaches that train the reasoning component directly so the model produces structured reasoning trajectories on its own rather than relying on a prompt schema imposed at inference time. Three patterns recur. *Trace-imitation* variants supervise the backbone on demonstration chains wrapped in fixed schema tags such as <think>, <evidence>, and <answer>, optionally extended across procedural stages such as plan, execute, integrate, and verify; staged training pipelines further refine through Direct Preference Optimisation (DPO), a method that trains on pairs ranked by preference rather than scalar rewards to prefer faithful chains over hallucinated shortcuts [46]. *Reward-driven* variants optimise the reasoning policy directly with Group Relative Policy Optimisation (GRPO), a reinforcement-learning method that compares grouped rollouts to compute relative rewards; the composite reward typically combines answer accuracy, evidence-page recall, and format compliance, and the policy is often warm-started from a trace-imitation stage rather than learned from scratch [35]. The two are therefore sequential rather than alternative, with reward-driven training refining what imitation alone cannot supply. *Uncertainty-*

calibration variants take a different route, training an aggregation head that selects across candidate outputs by calibrated uncertainty rather than internalising the reasoning chain itself, which yields a learned self-consistency signal that scales to long documents under tight parameter budgets and avoids the cost of sampling multiple chains at test time [45].

2.6.3 Trained Agentic Methods

Prompt-only agentic control is unreliable on smaller open-weight backbones, which cannot consistently follow ReAct schemas, select among tool calls, or terminate trajectories reliably. Two patterns recur. *Trajectory-distillation* variants clone agentic trajectories from a stronger teacher model through supervised fine-tuning on think-action-observation turns, with teacher rollouts collected in closed-loop interaction and filtered for format validity, answer correctness, and evidence coverage; this trains the smaller model to match the trajectory shape of the larger teacher rather than learning the underlying policy from scratch, with some variants additionally conditioning tool selection on a learned representation of the agent’s current goal so that action choices reflect the trajectory’s evolving intent rather than only its most recent observation [71]. *Reinforcement-learned* variants optimise the agentic policy directly against trajectory-level rewards that combine answer correctness, evidence recall, and tool-use shape, often pairing a supervised warm-start with GRPO over a constrained action space such as retrieval and page-fetch calls; learned termination is a recurring concern, with entropy-based stopping letting the trained model commit to an answer once its prediction confidence stabilises rather than running for a fixed number of rounds [65]. Multi-agent variants extend this further by allocating sample budgets per controller across procedural roles such as planner, executor, judge, and answerer rather than running every controller at a fixed budget, spending more compute on the role that the question’s difficulty most stresses [67]. The surveyed evidence does not cleanly separate distillation from reinforcement learning, since reported comparisons confound the training objective with backbone, reward design, and trajectory budget.

2.6.4 Cross-Cutting Training Techniques

Two techniques recur across retriever, reasoner, and orchestration policy training, independent of which component is being optimised. **Parameter-efficient tuning** is the dominant adaptation strategy: full backbone fine-tuning is prohibitive at long-document context lengths and risks erasing pretrained capabilities, so most systems freeze the backbone and train lightweight modules through LoRA matrices, projection heads, or task-specific adapters [28, 48, 56, 71]. The adapted module varies by target: layout-token modules inject structural information [28], projection heads align frozen retrievers with downstream readers [48], and LoRA matrices adapt reasoning or trajectory behaviour without touching backbone weights [56, 71]; the choice between parameter-efficient and full fine-tuning is more often a compute-budget decision than a target-component one. **Multi-stage curriculum training** addresses a different concern: several systems converge on curricula that start on single-page or local objectives before introducing document-scale supervision, exploiting the fact that text reading and layout understanding transfer from single-page data while cross-page integration must be learned separately [24, 45]. Later stages typically add streaming, abstention, or evidence-grounding objectives once foundational reading capabilities are established, with the longest schedules layering five or more stages from image-OCR alignment through to streaming-abstention calibration. Their recurrence across otherwise dissimilar systems indicates that cross-page integration does not emerge from single-page supervision alone.

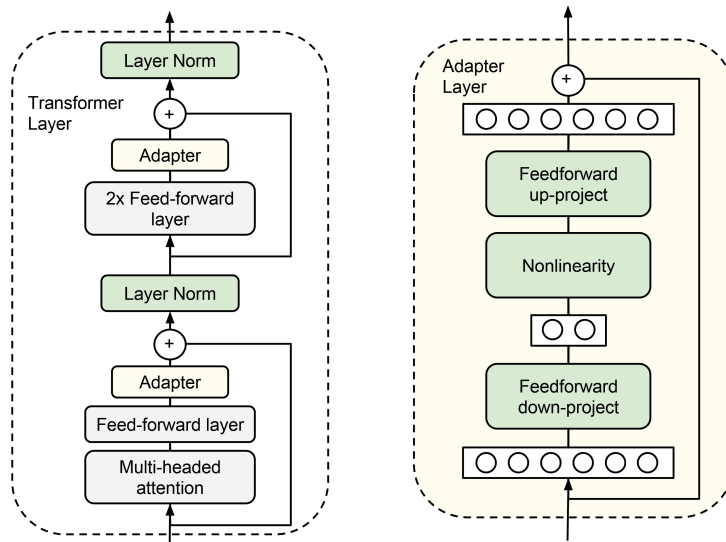


Figure 2.5 Adapter-based parameter-efficient tuning. Included for illustrative purposes. Two adapter modules are inserted into each transformer layer (left); each is a small bottleneck that down-projects to a low dimension, applies a nonlinearity, up-projects back, and adds the result to its input through a residual connection (right). During training the backbone weights are frozen and only the adapter modules and layer norms are updated. Adapted from Houlsby et al. [86].

2.7 Datasets

Multi-page documents impose a structural constraint absent from the single-page setting: per-page annotation does not scale to the page counts typical of MP-VRDU, and multi-page corpora are scarce because long PDFs are harder to source, license, and parse than single pages. This constraint shapes the three roles a dataset can play, pretraining (PT), supervised fine-tuning (SFT), and benchmarking (BM), differently from SP-VRDU, and several datasets occupy more than one role across different systems.

2.7.1 Pretraining

Pretraining is the exception rather than the norm in MP-VRDU. Only a handful of page-to-document and MLLM-centric systems pretrain or continue-train a backbone at all; retrieval-augmented and agentic pipelines inherit pretrained vision-language backbones and skip the stage entirely. Two factors explain its scarcity. Document-scale pretraining is computationally expensive, and genuinely multi-page corpora are few, so most systems that pretrain fall back on single-page archives or repurpose single-page data. Early work drew on single-page OCR archives such as IIT-CDIP [112] and OCR-IDL [113], which supply text-recognition supervision but no cross-page structure. Genuinely multi-page corpora have appeared only recently: DocStruct4M [3] mixes single- and multi-page samples with structure-aware parsing supervision, while the purely multi-page MP-DocStruct1M [24], Doc-750K [32], and Paper-PDF [62] support continued pretraining over genuine multi-page inputs. Where multi-page pretraining does occur, its objective has shifted from text recognition toward structure-aware supervision, but the corpus as a whole shows that most cross-page capability is instilled at fine-tuning rather than pretraining.

Dataset	Pages	Type	Venue	Domain	# Docs	Avg Pages	# Q/A	Multi/S Hop
DocVQA (2021)	SP	PT+BM	WACV	Industry	12767	–	50,000	S
FeTaQA (2022)	SP	PT+BM	TACL	Wikipedia tables	10330	–	10,330	M
InfographicVQA (2022)	SP	PT+BM	WACV	Infographics	5,485	–	30,035	M+S
PDF-VQA Task A (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	81,085	S
PDF-VQA Task B (2023)	SP	PT+BM	ECML-PKDD	Biomedical	–	–	53,872	S
TabFact (2020)	SP	PT+BM	ICLR	Wikipedia tables	16573	–	118,275	M+S
TextVQA (2019)	SP	PT+BM	CVPR	Scene-text images	28408	–	45,336	S
VisualMRC (2021)	SP	PT+BM	AAAI	Webpages	10197	–	30,562	M+S
WikiTableQuestions (2015)	SP	PT+BM	ACL	Wikipedia tables	2108	–	22,033	M
ChartQA (2022)	SP	BM	ACL	Charts	20882	–	32,719	M+S
DocGenome (2024)	MP	PT+BM	NeurIPS	Academic Paper	500000	13	3000	M+S
DuReaderVis (2022)	MP	PT+BM	ACL	Cross-domain	158000	–	15,000	S
Kleister-Charity (2021)	MP	PT+BM	ICDAR	Financial	2778	22.19	21,612	S
Kleister-NDA (2021)	MP	PT+BM	ICDAR	Legal	540	5.98	2,160	S
M-LongDoc (2025)	MP	PT+BM	EMNLP	Cross-domain	180	210.8	851	S
MMVQA (2024)	MP	PT+BM	IJCAI	Biomedical	3146	9.6	262,928	M+S
MP-DocVQA (2023)	MP	PT+BM	PR	Industry	5928	8.27	46,176	S
MultiHiertt (2022)	MP	PT+BM	ACL	Financial reports	2513	~2.5	10,440	M
PaperPDF (2026)	MP	PT+BM	preprint	Academic Paper	–	–	1,110,000	M+S
PDF-VQA Task C (2023)	MP	PT+BM	ECML-PKDD	Biomedical	1147	~10	5,653	M
QASPER (2021)	MP	PT+BM	NAACL	Academic Paper	1585	~8	5,049	M
SlideVQA (2023)	MP	PT+BM	AAAI	Presentations	2619	20	14,484	M+S
MP-DocStruct1M (2025)	MP	PT	ACL	Cross-domain	–	–	1,113,259	S
OpenDocVQA (2025)	MP	SFT+BM	CVPR	Cross-domain	–	–	–	M+S
CoR-Dataset (2026)	MP	SFT	preprint	Cross-domain	–	–	26,088	M+S
Doc-750K (2025)	MP	SFT	CVPR	Academic Paper	–	–	758,000	M+S
MHDocVQA (2025)	MP	SFT	CVPR	Cross-domain	–	9.5	38,020	M
MP-DocReason51K (2025)	MP	SFT	ACL	Cross-domain	–	–	51,754	M+S
BRIDGE (2026)	MP	BM	preprint	Academic Paper	262	–	11,857	M
DocBench (2024)	MP	BM	preprint	Cross-domain	229	–	1,102	S
DUDE (2023)	MP	BM	ICCV	Cross-domain	5019	5.72	41,541	M+S
FetaTab (2024)	MP	BM	NeurIPS	Wikipedia Articles	871	~11	1,016	S
LongDocURL (2025)	MP	BM	ACL	Cross-domain	396	85.6	2,325	M+S
M3DocVQA (2025)	MP	BM	CVPR	Wikipedia Articles	3368	12.2	2,441	M+S
MM-NIAH (2024)	MP	BM	NeurIPS	Synthetic	–	–	16,486	M+S
MMDocIR (2025)	MP	BM	ACL	Cross-domain	313	65.1	1,658	M+S
MMLongBench (2025)	MP	BM	NeurIPS	Cross-domain	–	–	13,331	M+S
MMLongBench-Doc (2024)	MP	BM	NeurIPS	Cross-domain	135	47.5	1,082	M+S
PaperTab (2024)	MP	BM	NeurIPS	Academic Paper	307	~11	393	M+S
PaperText (2024)	MP	BM	NeurIPS	Academic Paper	1087	~11	2,804	M+S
ViDoSeek (2025)	MP	BM	EMNLP	Cross-domain	–	–	1,200	M+S
VisDoMBench (2025)	MP	BM	NAACL	Cross-domain	1277	19.11	2,271	M+S
VisR-Bench (2025)	MP	BM	ICLR	Cross-domain	226	–	471	M+S
DocStruct4M (2024)	MP+SP	PT+SFT	EMNLP	Cross-domain	–	–	4,000,000	M+S
TAT-DQA (2022)	MP+SP	PT+BM	ACM	Financial reports	2758	1.33	16,558	M
InstructDoc (2024)	MP+SP	SFT	AAAI	Cross-domain	–	–	–	M+S
Leopard-Instruct (2025)	MP+SP	SFT	TMLR	Cross-domain	–	–	925,000	M+S

Table 2.2 Datasets referenced in the MP-VRDU survey. **Pages:** document scope (MP = Multi-Page, SP = Single-Page, MP+SP = Mixed). **Type:** dataset role – PT (Pretraining), SFT (Supervised Fine-Tuning), BM (Benchmark), and combinations. **Hop:** question-hop type (Single, Multi, or both). ‘–’ indicates not reported.

2.7.2 Supervised Fine-Tuning

Because per-page human annotation does not scale, supervised fine-tuning has converged on two strategies, often combined within one system. The first aggregates publicly released question-answer collections into broad instruction mixes spanning heterogeneous question and answer types [24, 29, 32, 43]. The second curates a bespoke corpus for the system under study, typically synthesised by a strong vision-language model that writes question-answer pairs, reasoning traces, or agent trajectories over scraped PDFs [46, 51, 55, 62, 71]. Synthetic generation has become the dominant scaling path for fine-tuning data, and its underlying PDF pools are typically drawn from the same scientific archives, slide repositories, and benchmark sources that also supply evaluation. Layered above this, a growing body of work applies reinforcement or preference learning with rewards that credit correct evidence-page citation, faithful chain-of-thought structure, or appropriate tool use rather than final-answer accuracy alone [35, 42, 46]. These reinforcement pools remain small relative to the supervised mixes that precede them, indicating that reinforcement signals shape behaviour, instilling evidence grounding, output

formatting, and multi-step reasoning, rather than scaling the data itself.

2.7.3 Benchmarking

Benchmarks have evolved through several tiers that accumulated rather than replaced one another. Single-page extractive evaluation on DocVQA [4], InfographicVQA [18], and ChartQA [19] persists as a per-page baseline, and mid-length corpora such as MP-DocVQA [13], DUDE [103], and SlideVQA [100] extend the same extractive formats into a few-page setting. Long-document multimodal benchmarks then introduced documents whose total length routinely exceeds backbone context windows and whose evidence requires cross-page integration over mixed text, tables, charts, and figures, including MMLongBench-Doc [110], LongDocURL [105], DocBench [102], and M-LongDoc [96]. Retrieval-oriented benchmarks such as UDA-Benchmark [104] and DocLens [69] additionally score evidence-page selection and layout-level localisation alongside answer accuracy, and the most recent additions push into cross-document settings, including M3DocRAG [33], ViDoRAG [74], and VDocRAG [55]. Despite this proliferation, evaluation in practice has consolidated around a small set of long-document and mid-length benchmarks that recur in most reported comparisons. That the same scarce pool of source PDFs supplies both synthetic fine-tuning data and evaluation is a contamination risk we return to in §4.2.4.

2.8 Conclusion

This chapter established MP-VRDU as a distinct document-understanding problem organised around evidence management. It introduced a taxonomy of four architectural families and surveyed the multimodal representations, inference-time strategies, training approaches, datasets, and evaluation practices used across the field. Across these dimensions, the literature consistently shows that long-document performance depends not only on backbone capability, but on how evidence is preserved, selected, compressed, and integrated across pages. The broader challenges and future research directions that emerge from this survey are discussed in Chapter 4.

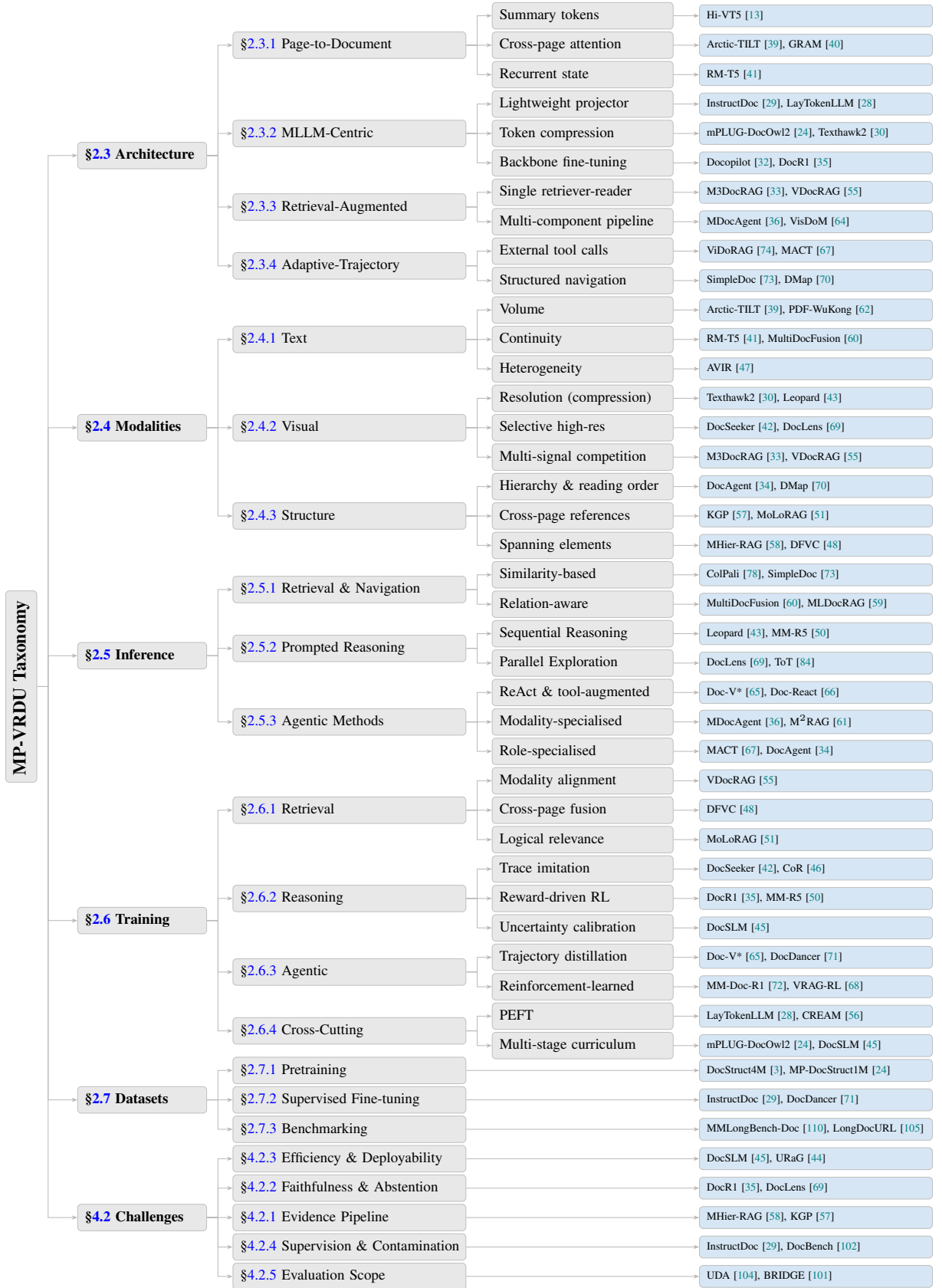


Figure 2.6 MP-VRDU Taxonomy

Chapter 3

Empirical Attribution of Failure Modes

This chapter is adapted from **Locating Failure in Multi-Page Visually Rich Document Understanding: An Empirical Attribution** [114], by Lewei Xu, Yihao Ding, Zihan Xu, Daniel Yitian Su, Daochang Liu, Siwen Luo, Yifan Peng, and Wei Liu, *submitted to EACL and currently under review*. The first author was responsible for formulating the failure taxonomy and attribution framework; designing the experimental methodology; implementing and running all experiments; analysing and interpreting the results; and preparing the full manuscript, including the main text, methodological details, empirical analysis, figures and tables, and appendix material.

3.1 Introduction

Modern MP-VRDU systems increasingly build document-specific machinery around general-domain MLLM backbones (as discussed in §1.3.2). At the frontier of this model class, however, the most capable systems are both closed-weight and deployed at a scale that generally requires substantial datacentre infrastructure, so access is typically provided through hosted services. Document workloads involving confidential or regulated material therefore favour local processing because of privacy and data-residency requirements, while high usage volumes can make commercial API costs another constraint [115, 116]. In industry, this has motivated deployments built around smaller open-weight MLLMs that can run on local consumer or enterprise hardware within fixed memory and compute budgets. Representation fidelity, evidence retrieval, and reasoner scale therefore compete for the same resources, making it important to identify where accuracy is lost before deciding where additional compute should be spent.

The literature in Chapter 2 reflects substantial disagreement about what limits MP-VRDU systems. OCR-free methods argue that operating directly on visual document inputs avoids information loss from text extraction and preserves visual and layout cues [33, 55, 65], while other studies find that combining textual and visual representations can outperform vision alone [36, 45, 64]; some instead report stronger vision-only pipelines [53]. Retrieval findings are similarly mixed: insufficient retrieval can omit required evidence [42, 47], whereas broader retrieval can introduce irrelevant context and attention degradation [5, 63, 117]. Reasoning failures have likewise been attributed to difficulty combining evidence across pages [65], interpreting evidence within individual pages [46], and answering when the available evidence is insufficient [49, 103, 110]. These competing accounts correspond to three implementation-agnostic failure loci: **representation**, where encoding fails to preserve required evidence; **selection**, where relevant evidence is absent or diluted; and **reasoning**, where adequate evidence is miscombined or judged insufficiently. We use this decomposition for controlled attribution by intervening on one locus while holding the others fixed, testing claims about modality, retrieval, and reasoning within a common MP-VRDU pipeline.

Our contributions are as follows: (1) We introduce an attribution framework that formalises MP-VRDU failure through three system-level loci: representation, selection, and reasoning (§3.3). (2) We conduct a controlled empirical study native to MP-VRDU that intervenes on one locus at a time and tests competing claims about modality, evidence selection, and reasoning under a unified experimental setting (§3.4-§3.6). (3) We analyse the resulting error patterns and derive practical implications for designing and deploying MP-VRDU systems under constrained local compute budgets (§3.7, §3.8).

3.2 Related Work

Representation quality has been studied most directly through OCR and parsing errors. OHRBench shows that OCR corruption propagates through retrieval and generation in text-based RAG pipelines, placing an upstream ceiling on downstream performance [8]. More recently, Sun et al. find that conventional character-level OCR metrics can obscure failures in tables and document structure that disproportionately affect RAG accuracy [118]. These results establish representation as a genuine source of downstream error, but do so within text-centric pipelines where the representation is fixed before retrieval and no visual channel can compensate for information lost during parsing.

Selection and reasoning have largely been studied separately. Cuconasu et al. show that retrieved distractors are not uniformly harmful: semantically related irrelevant passages can sharply reduce RAG accuracy, while their effect depends on composition and position [119]. Jin et al. similarly find that increasing retrieval depth initially improves long-context RAG before performance declines as hard negatives accumulate [120]. These findings qualify any general claim that distractors are harmless and provide a useful comparison for the limited distractor depths we test. Downstream of retrieval, context-faithful prompting can materially change whether an LLM follows supplied evidence or abstains when it is insufficient [121], showing that response behaviour is also sensitive to the reasoning procedure. These studies, however, operate primarily over text passages, where document representation is already given rather than itself being a competing source of failure.

Document-native benchmarks expose several of these difficulties together. MMLongBench-Doc reports weaker performance on cross-page questions, large differences between OCR-fed and vision-language models, and substantial variation in abstention behaviour [110]; LongDocURL separately evaluates understanding, locating, and reasoning over long multimodal documents [105]. DocScope goes further by evaluating page localisation, region grounding, fact extraction, and answer verification under progressively supplied evidence, providing a trajectory-level diagnostic view of long-document reasoning [122]. These works already establish several effects that our experiments recover. Our study instead provides a unified attribution view within a single MP-VRDU pipeline, intervening on representation, selection, and reasoning under a common setup, thereby filling the gaps between representation-quality studies, retrieval analyses, reasoning diagnostics, and document-level benchmarks.

3.3 Attribution Framework

The three loci are hierarchical: evidence lost in representation cannot be selected, and evidence never selected cannot be reasoned over. This hierarchy makes controlled attribution possible, since a locus can be tested only after those preceding it have been neutralised. Figure 3.1 summarises this decomposition and the mechanisms through which each locus can fail.

Because the loci are defined by system function rather than architecture, the techniques surveyed in Chapter 2 reduce failure at one or more of them rather than forming categories of their own. Some act

primarily on a single locus. Retrieval and reranking change what evidence reaches the reasoner without altering how the document is represented or how the reasoner integrates it. Others act on several at once. An agentic controller mainly addresses selection by gathering evidence over several rounds, but can also act on representation when it re-renders a region at higher resolution, and on reasoning when it revises an answer through self-refinement.

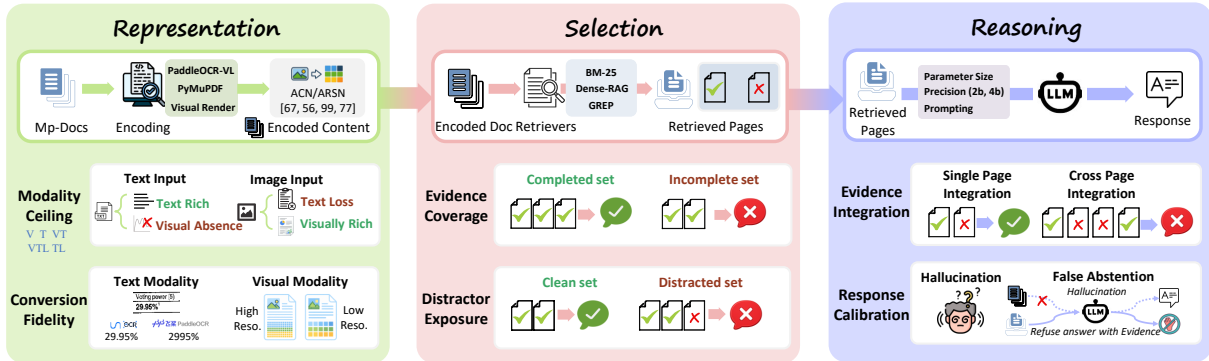


Figure 3.1 The three failure loci and their mechanisms.

3.3.1 Representation

Representation converts the document into the artifact consumed by later stages. This may be an embedded text layer, parser markdown, a cropped visual region, a rendered page, or a combination of these. Operations performed by the backbone after this artifact is tokenised, such as projection, resampling, or compression, belong to the reasoning locus because they are properties of how the backbone processes its input rather than of document conversion. The chosen representation determines both which document signals can be preserved and the cost of producing them. Representation failure occurs through two mechanisms: **(1) Modality Ceiling**. Each modality can preserve only the signals it is capable of expressing, at the granularity permitted by its representation. Text tokens, for example, cannot capture non-linguistic visual information, while visual tokens preserve fine-grained detail only when sufficient spatial resolution and token budget are available. **(2) Conversion Fidelity**. Even when a modality can represent a signal, the conversion process may degrade or corrupt it. Conversion fidelity measures how accurately the resulting artifact preserves information that the modality could otherwise carry. A document parser may, for example, misread a numeric value and leave the downstream reasoner operating on an incorrect but seemingly reliable representation [8].

3.3.2 Selection

Selection determines which parts of the encoded document are placed before the reasoner. This operation becomes necessary in the multi-page setting when the full document cannot be processed within the available context. Similarity retrieval is one implementation, but structural navigation, adaptive reranking, and agentic inspection perform the same underlying function, potentially over several iterations (§2.5.1, §2.5.3). Selection may also operate at different granularities. Most systems select whole pages, matching common benchmark annotations and late-interaction retrievers, although sections, tables, or cropped figures may instead serve as the selected unit. Selection failure occurs through two mechanisms: **(1) Evidence Coverage**. Evidence coverage measures how much of the evidence required to answer a question is present in the selected context, *i.e.*, the recall of the required evidence units. Omit-

ting required evidence leaves the reasoner with an incomplete evidence set and therefore places an upper bound on downstream accuracy regardless of reasoning capability. **(2) Distractor Exposure.** Selection may also introduce content that is irrelevant to the question. Distractor exposure measures the extent to which such content accompanies the required evidence. Unlike evidence omission, distractors do not make the evidence intrinsically insufficient, but they increase the burden on the reasoner to identify and use the relevant content.

3.3.3 Reasoning

Reasoning produces an answer from the evidence delivered by the preceding stages. Once the representation preserves the evidence a question requires and selection places that evidence before the reasoner, remaining failures are attributable to how the reasoner processes and acts on it. Such failures may be addressed through instruction, model scale, architecture, or training (§2.5.2, §2.6.2). Reasoning can fail in many ways, including arithmetic errors over correctly read values or answering a different question from the one asked. We focus on two mechanisms that are particularly important in the multi-page setting: **(1) Evidence Integration.** Multi-page questions may require complementary information distributed across several pages, including content whose meaning spans page boundaries. A reasoner may handle each piece of evidence correctly in isolation yet fail to connect them into the answer the question requires. This difficulty may be amplified in long contexts, where evidence at different positions is not used equally reliably [5]. **(2) Response Calibration.** Response calibration concerns whether the model’s answer reflects the level of support provided by the available evidence. Unlike integration, it can be tested both when sufficient evidence is present and when it is absent by construction. Failure occurs in two directions: *hallucination*, where the model answers without sufficient supporting evidence [123], and *false abstention*, where it refuses despite having sufficient evidence. We treat these as opposing outcomes of the same decision because interventions that make a model more willing to answer can reduce false abstention while increasing unsupported answers, and vice versa [124].

3.4 Experimental Setup

We instantiate the framework in a **single-pass retriever-generator** document question answering system, evaluated over two MP-VRDU benchmarks whose annotations make the controlled experiments possible. In this section, we first describe the system and the attribution experiments that isolate each mechanism (§3.4.1), followed by the experimental configurations of representation, selection, and reasoning (§3.4.2–§3.4.4), including additional experiments comparing practical pipeline choices (§3.4.5).

3.4.1 Attribution by Construction

The document question answering system consists of three stages that correspond to the three failure loci described in §3.3. For each question, the system performs the following:

1. **Encoding Stage:** each page is converted into the representation consumed by later stages, using parser markdown, a rendered page image, or both.
2. **Retrieval Stage:** a retriever scores every encoded page against the question and returns the highest-ranked k pages. Retrieval operates at page granularity, matching the evidence annotations used by both benchmarks.

3. EMPIRICAL ATTRIBUTION OF FAILURE MODES

3. **Inference Stage:** a multimodal LLM receives the selected pages, the question, and a task instruction, then produces an answer in a single pass. The model has access only to the evidence delivered by the retrieval stage.

The system’s stages act once and in order, allowing failures to be attributed by constructing conditions that isolate one locus while holding the others fixed. Across the six mechanisms defined in §3.3, we vary the relevant representation, evidence, or reasoning condition and compare relevant metrics against an appropriate controlled baseline. Table 3.1 summarises these experiments.

Locus	Mechanism	Constructed condition	Held fixed
Representation (Encoding stage: §3.4.2)	Modality ceiling	Encoding rung compared across T, TL, TLV, V by evidence source	Gold pages, reasoner, prompt
	Conversion fidelity	Text and image fidelity across three parsers and three render resolutions	Encoding rung, gold pages, reasoner, prompt
Selection (Retrieval stage: §3.4.3)	Evidence coverage	Gold pages withheld from multi-page questions	Encoding, reasoner, prompt
	Distractor exposure	Non-gold pages added at a fixed gold-page count	Gold set, encoding, reasoner, prompt
Reasoning (Inference stage: §3.4.4)	Evidence integration	Evidence conditions requiring cross-page integration	Documents, gold pages, encoding
	Response calibration	Answerable and unsupported evidence conditions	Gold pages, encoding, reasoner

Table 3.1 Attribution by construction: conditions used to isolate failures.

Default Configuration. Unless otherwise stated, pages are encoded at **TLV** using PaddleOCR-VL parser output and the `med` image preset, retrieved by ColQwen3, and passed to Qwen3-VL-8B-Instruct in `bf16`. Inference uses the prompt template in Figure 3.2, with selected pages ordered by their original document position. Generation is greedy (`do_sample=False`) with no temperature, top- p , top- k , or repetition penalty. The default output limit is 256 tokens. Each experiment below changes only the representation, evidence set, or reasoning instruction required for the mechanism being tested.

```
You are answering a question about a document.
{instruction}
Question:
{question}
Document evidence:
{context}
Answer:
```

Figure 3.2 Reasoner prompt template. `{instruction}` is empty in the default configuration. `{context}` is the selected pages in document order, as parser markdown, page images, or both according to the rung.

3.4.2 Representation Experiments

Each page is encoded from two signals: a text channel and a visual channel. The representation experiments run only on questions answerable from their annotated evidence, and modality comparisons are analysed according to the type of evidence source the question requires.

Modality ceiling. We vary which signals reach the reasoner at all, across the four rungs of Table 3.2: embedded text (**T**), layout-aware parser text (**TL**), parser text combined with the page image (**TLV**), and the page image alone (**V**). The comparison tests whether the required evidence remains recoverable when textual, structural, or visual signals are unavailable, rather than assuming that one representation is uniformly preferable across question types.

Rung	Text channel	Image	Cost	Role
T	Embedded text (PyMuPDF)	–	Extraction only	Acquisition floor; empty on scanned pages
TL	Parser markdown	–	Parser pass	Text-only ceiling; recovers reading order and table structure
TLV	Parser markdown	Page image	Parser pass + vision tokens	Combined encoding; the only full option on scanned pages
V	–	Page image	Vision tokens	Vision-only

Table 3.2 Representation ladder. Ordered by fidelity and, as a consequence, by cost.

Conversion fidelity. We vary the quality of each channel while holding the representation rung fixed, through the parser for text and the render resolution for the image (Table 3.3). These experiments test whether information that a modality can in principle represent is degraded before reaching the reasoner, and whether improving the fidelity of that conversion improves downstream performance.

Parser (text channel)			Resolution (image channel)			
Parser	Params	Rungs	Preset	Tok./page	Pixel cap	Rungs
PaddleOCR-VL	0.9B	TL, TLV	low	285	313,600	TLV, V
MinerU 2.5	1.2B	TL, TLV	med	456	501,760	TLV, V
Unlimited-OCR	3.3B	TL, TLV	high	744	752,640	TLV, V

Table 3.3 Representation fidelity variation. PaddleOCR-VL and med are the defaults; the two halves vary independently. Tokens are Qwen3-VL’s median visual tokens per page (one token per 32×32 px).

Configuration. All representation experiments use the dataset’s annotated gold pages as an oracle evidence set, removing retrieval error from the comparison. The **T** rung supplies the PDF’s embedded text layer directly through PyMuPDF, while **TL** supplies markdown produced by a layout-aware parser. **TLV** combines this markdown with the rendered page image, and **V** supplies the image alone; no bounding boxes or other explicit layout signals are passed to the reasoner. A fifth encoding, **TV**, combines embedded text with the page image without parsing and is used as a direct contrast against

3. EMPIRICAL ATTRIBUTION OF FAILURE MODES

TLV. The default parser is PaddleOCR-VL (0.9B) [125], with MinerU 2.5 (1.2B) [126] and Unlimited-OCR (3.3B) [127] used in the parser comparison. Pages are rasterised at 200 DPI and downsampled to the active image preset before inference. All experiments use `med` except the resolution sweep, which evaluates all three presets.

3.4.3 Selection Experiments

Selection determines which evidence reaches the reasoner. We investigate failures caused by incomplete evidence coverage and by exposure to irrelevant pages.

Evidence coverage. We vary how much of the required evidence reaches the reasoner by withholding gold pages from questions that cite more than one. These conditions run on multi-page questions only and are evaluated against the corresponding complete gold-page oracle. Pages are ranked by ColQwen3 within the document, allowing us to remove either the highest- or lowest-ranked gold page, or retain only one of them. A condition that a question cannot satisfy is excluded and recorded rather than silently satisfied with a different page set.

Distractor exposure. We vary how much irrelevant evidence accompanies the required evidence by adding top-ranked non-gold pages to a complete gold set. Conditions are blocked on the exact gold-page count, so questions requiring different amounts of evidence are not pooled at the same retrieval depth. The range of k narrows as the gold count rises, keeping the supplied context at six pages or fewer and preventing distractor count from being confounded with arbitrary increases in context length.

Rule	Pages supplied	Blocking	Range
<code>drop_top</code>	All gold except the highest-ranked	≥ 2 gold	–
<code>drop_bottom</code>	All gold except the lowest-ranked	≥ 2 gold	–
<code>keep_top</code>	The highest-ranked gold page only	≥ 2 gold	–
<code>keep_bottom</code>	The lowest-ranked gold page only	≥ 2 gold	–
$+k$ distractors	All gold plus the k top-ranked non-gold pages	Exactly 1 gold	$k \in \{1, 2, 3, 4, 5\}$
$+k$ distractors	All gold plus the k top-ranked non-gold pages	Exactly 2 gold	$k \in \{1, 2, 3, 4\}$

Table 3.4 Gold set perturbations for selection experiments. Read against the oracle condition of all gold pages with no distractors. The distractor rules hold the supplied set at six pages or fewer regardless of gold count.

Configuration. ColQwen3 is used only to order pages within the constructed selection conditions. For evidence coverage, it ranks the annotated gold pages so that the highest- or lowest-ranked evidence can be removed or retained. For distractor exposure, non-gold pages are ranked separately and the highest-scoring distractors are appended to the complete gold set. Each constructed page set is then evaluated under **TLV**, and **V** representations. Verdict transitions are additionally measured against the corresponding complete-evidence **TLV** condition.

3.4.4 Reasoning Experiments

Reasoning determines how the available evidence is used to produce a response. We investigate whether the reasoner can integrate complementary evidence across the supplied context and whether it can judge

when that evidence is sufficient to support an answer.

Evidence integration. We test evidence integration on answerable questions by supplying the annotated evidence pages directly and separating questions according to the structure of the evidence required to answer them. We first compare single-hop and multi-hop questions under the default `none` prompt to measure the integration deficit that remains once representation and selection are controlled. We then compare `none` against `cot` on the same evidence and question pools, testing whether explicit intermediate reasoning can recover performance specifically when evidence must be combined across pages.

Response calibration. We test response calibration by varying both whether the supplied evidence supports an answer and whether the reasoner is explicitly instructed to abstain. The comparison uses the `none` and `abstain` prompts on answerable and unanswerable questions. Answerable questions receive their annotated gold pages, while unanswerable questions receive the top three pages ranked by BM25, providing plausible but non-supporting evidence. This allows us to measure unsupported answers when evidence is insufficient and false abstentions when it is sufficient.

Mode	Instruction
<code>none</code>	(empty)
<code>grounded</code>	Use only the provided document evidence and keep the answer concise.
<code>abstain</code>	<code>grounded</code> , plus: If the evidence does not contain the answer, answer exactly: Not answerable.
<code>abstain_balanced</code>	<code>abstain</code> , plus: If the evidence does contain the answer, give it: do not decline a question the evidence supports.
<code>cot</code>	<code>grounded</code> , plus: Think step by step before answering. Then write your final answer on a new line beginning with ‘Answer:’.
<code>extract_cot</code>	<code>grounded</code> , plus a verbatim-evidence extraction step, with visual evidence described in enough detail to answer from the description alone, then reasoning constrained to what was extracted, then the <code>Answer:</code> line.

Table 3.5 Reasoning prompt modes for the `{instruction}` field. Each is a preamble only, and every mode runs one generation per condition.

Configuration. The three reasoning conditions differ only in the instruction inserted into Figure 3.2. Under `none`, the instruction field is empty. The `cot` condition instructs the model to reason through the supplied evidence before producing its answer, while `abstain` constrains the answer to the supplied evidence and permits *Not answerable* when the required support is absent. Evidence pages are presented in document order under the active representation. The default 256-token output budget is retained for `none` and `abstain`, while `cot` allows up to 2048 tokens for intermediate reasoning.

3.4.5 Additional Experiments

The attribution experiments above isolate the mechanisms that constrain MP-VRDU performance. We additionally examine practical implementation choices that affect how these findings translate into system design and deployment, focusing on retrieval strategy, prompting, and model configuration.

Retrievers. We compare six page-level retrievers spanning lexical, dense-text, and late-interaction visual retrieval. Text retrievers consist of BM25, BGE-M3, and Qwen3-Embedding, while the vision retrievers consist of ColModernVBERT, ColQwen2.5, and ColQwen3. We additionally evaluate joint

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retrieval by taking the deduplicated union of matched text and vision retrievers. Retrieval is evaluated independently of answer generation using page-level precision, recall, and F1 against the annotated evidence pages across increasing retrieval depths. This characterises the practical trade-off between retrieval precision and evidence coverage when choosing how much and what kind of retrieval machinery to deploy.

Arm	Retriever	Params	Index and cost
Text	BM25	–	Term postings per document; no encoding pass
Text	BGE-M3	568M	One vector per page
Text	Qwen3-Embedding-4B	4B	One vector per page; sequence cap 4096
Vision	ColModernVBERT	250M	One vector per image patch; requires rendering
Vision	ColQwen2.5	3B	One vector per image patch; requires rendering
Vision	ColQwen3	4B	One vector per image patch; requires rendering

Table 3.6 Selection: retriever experiments. Ordered by cost within each arm. Late-interaction scoring stores many vectors per page rather than one, so the vision arm carries a larger index and a more expensive scoring pass than its parameter count alone suggests.

Prompt modes. Beyond the `none`, `cot`, and `abstain` conditions used in the main reasoning experiments, we evaluate the remaining prompt interventions in Table 3.5. These vary how strongly the reasoner is grounded in the supplied context, how abstention is framed, and whether evidence is explicitly extracted before answering, while holding the supplied evidence fixed. This tests how far reasoning behaviour can be improved through inexpensive inference-time changes before altering the model or evidence pipeline.

Axis	Checkpoints
Scale (primary)	Qwen3-VL-Instruct at 2B, 4B, 8B, 32B
Scale (secondary)	Gemma-3-IT at 4B, 12B, 27B
Family	InternVL3-8B, GLM-4.6V-Flash, MiniCPM-V-4.5, Llama-3.2-11B-Vision
Precision	Qwen3-VL-8B quantized at bfloat16, 8-bit, 4-bit
Reasoning variant	Qwen3-VL-8B-Thinking

Table 3.7 Reasoner variation experiments. All share the same prompt template decoding configuration.

Model variations. We vary the reasoner across model scale, weight precision, and model family. Within Qwen3-VL, we compare models from 2B to 32B parameters, alongside 16-bit, 8-bit, and 4-bit variants of the 8B model. We additionally compare a quantised larger model against the full-precision 8B configuration at a similar memory budget, and compare Qwen3-VL-8B with alternative model families at approximately the same scale. These experiments directly examine the accuracy–resource trade-offs involved in selecting a reasoner for deployment.

Configuration. The additional experiments inherit the default pipeline unless the component under study is explicitly varied. For retrieval, text models index both embedded PDF text and default-parser

output, while visual retrievers operate on pages rendered at 200 DPI. Prompt and model comparisons use the same supplied evidence for each paired condition. Prompt modes requiring intermediate reasoning, and thinking-model variants, receive a 2048-token generation budget and mark the final response with `Answer:`, from which the answer is extracted before evaluation.

3.5 Datasets and Evaluation

In this section we describe the two datasets used for our experiments (§3.5.1, §3.5.2) and conclude with the evaluation protocol (§3.5.3).

3.5.1 MMLongBench-Doc Dataset

MMLongBench-Doc [110] is the primary dataset. It is the only multi-page benchmark we are aware of that annotates, per question, the evidence pages, the kind of evidence the question draws on, and the document’s domain, while also supplying a pool of plausible questions unsupported by the document. Evidence pages make coverage measurable and define the oracle retriever, evidence source resolves representation findings by what the question actually requires, and the unanswerable pool makes calibration testable rather than inferred. The benchmark contains 135 documents averaging 47.5 pages and 847 answerable questions. Its questions are manually created or revised by expert annotators and subsequently quality-controlled. Multi-page questions are constructed to require information collection and reasoning over the annotated evidence pages, so we generally treat each gold page as necessary evidence rather than as another occurrence of the same answer. As with any human annotation, however, the page sets should not be assumed perfectly minimal in every case. Tables 3.8 and 3.9 gives the composition.

Property	Value	Domain	Q	Docs	Digital	Scanned
Questions	1,091	Research report / Intro	293	34	23	11
Answerable	847	Academic paper	204	26	26	0
Unanswerable	244	Guidebook	156	22	20	2
Documents	135	Tutorial / Workshop	139	17	3	14
Mean pages/doc	48.3	Financial report	117	11	11	0
Median pages/doc	28	Brochure	101	15	13	2
Pages min–max	9–468	Admin / Industry	81	10	8	2
Total			1,091	135	104	31

Table 3.8 MMLongBench-Doc composition. Digital and scanned counts are documents, not questions.

Evidence source	Q	Gold pages	Q	Answer format	Q
Pure-text (plain)	298	1 page	485	Int	290
Figure	293	2 pages	247	Str	250
Table	218	3+ pages	113	None	244
Chart	178	0 pages	246	Float	160
Generalized-text (layout)	119			List	147
(none)	254				

Table 3.9 MMLongBench-Doc question labels. Evidence source is multi-label, so its column sums past the question count; the remaining columns are exclusive. Questions with three or more gold pages reach 24 at the tail.

Operational definitions. Four labels are derived rather than read from the annotation, and each is fixed for every experiment.

1. **Unanswerable** is a gold answer of exactly *Not answerable*, giving 244 questions. A further two questions carry an answer with no annotated evidence page; they are excluded from the oracle conditions, which have no pages to supply.
2. **Evidence pages** are converted from the dataset’s 1-based numbering to 0-based indices, de-duplicated, and read as physical page positions, which is the dataset’s own convention.
3. **Scan status** is auto-detected per document. A page counts as carrying text when at least 20 characters are extractable, and a document is scanned when none of its sampled pages does. This gives 104 born-digital and 31 scanned documents.
4. **Gold-page count** is the length of the de-duplicated evidence-page list, and is the hop label throughout: single-page questions cite one page, multi-page questions cite two or more.

3.5.2 LongDocURL Dataset

We additionally use LongDocURL [105] as a secondary dataset mainly for replication purposes. It contains 2,317 answerable questions over documents spanning 51–149 pages and provides task-family, question-type, answer-format, evidence-source, and evidence-page annotations. Its QA pairs and supporting evidence are produced through a semi-automated pipeline in which LLMs and LVLMs generate candidate questions and evidence before automated and human verification against the source document. Unlike MMLongBench-Doc, the annotated evidence pages are not a minimal set that must all be present to answer the question. Multiple annotated pages may contain redundant or independently sufficient evidence. We account for this distinction when interpreting LongDocURL coverage and integration results. Tables 3.10 and 3.11 gives the composition.

Property	Value	Evidence source	Q	Answer format	Q
Questions	2,325	Text	994	String	941
Documents	396	Table	871	List	757
Mean pages/doc	85.6	Layout	779	Integer	431
Median pages/doc	78	Figure	548	Float	185
Pages min-max	51–149	Others	8	None	11

Table 3.10 LongDocURL composition. Evidence source is multi-label and sums past the question count; answer format is exclusive.

LongDocURL is less challenging than MMLongBench-Doc in several ways. Its documents are born-digital, it contains almost no unanswerable questions, and many questions can be answered directly from the document text, even when the annotated evidence is visual. We therefore use LongDocURL as a secondary dataset to test whether the main findings also hold under a less demanding setting.

Question types. LongDocURL nests nine question types inside its three task tags [105]. Understanding and Reasoning are single types and four types respectively; the four Locating types are each defined by the pair of element types a question must move between.

- **Understanding.** `extract` asks for information stated directly in the document, found by identifying keywords or reading a table.

Task tag	Question type	Single-page	Multi-page	Total
Understanding	extract	612	631	1,243
Locating	extract_fig2tab	231	0	231
	topic2title	60	139	201
	summary2title	24	113	137
	summary2tab	8	118	126
Reasoning	calculate	89	56	145
	count	26	91	117
	compare	36	76	112
	summarize	7	6	13
All		1,093	1,230	2,325

Table 3.11 LongDocURL question type by task tag and hop. Every type belongs to exactly one tag.

- **Reasoning.** `calculate` requires arithmetic over extracted values; `count` requires enumerating instances; `compare` requires ordering or contrasting two or more values; `summarize` requires condensing content into a single statement.
- **Locating.** `topic2title` gives a description and asks which section title it matches; `summary2title` gives a paragraph summary and asks for the title of the section it came from; `summary2tab` gives a description and asks which named tables it refers to; `extract_fig2tab` requires reading a figure and a table together to answer.

3.5.3 Evaluation Protocol

Scoring and Judge Inputs. Answers are scored by an LLM judge rather than exact match because correct responses may differ from the gold answer in units, rounding, or formatting. Each response is reduced to a binary *correct/incorrect* verdict based on semantic equivalence, and accuracy is the proportion judged correct. Gemini-2.5-Flash is the primary judge, run at temperature 0 with JSON-only output. Every judge receives the same four fields: the question, gold answer, an explicit unanswerable flag derived from the gold answer, and the model answer. The judge is not shown document content, evidence annotations, representation or retrieval settings, model configuration, prompt condition, or cost telemetry. This keeps the scoring criterion fixed across experimental arms and prevents the judge from conditioning on the manipulation being evaluated.

```
You judge answers to document questions.
Return only JSON with keys:
- verdict: one of correct, incorrect
- extracted_answer: the answer extracted from the model response, or
empty string
- rationale: a short reason

Mark correct when the model answer is semantically equivalent to the
gold answer.
For unanswerable questions, mark correct only when the model abstains.
```

Figure 3.3 Judge accuracy rubric.

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Rubrics and Judge Validation. Two judge rubrics are used over the same input payload. The *accuracy rubric* in Figure 3.3 is used where only correctness is required, while the *analysis rubric* in Figure 3.4 additionally supports abstention and error categorisation. Across 19,983 responses scored under both rubrics, the analysis rubric is 0.8–2.9 points stricter, but none of the 36 pairwise rung orderings reverses and key ladder deltas change by at most 0.8 points. We further validate the primary judge against GPT-4o, an independent model family, on 500 stratified responses, obtaining 96.0% agreement and Cohen’s $\kappa = 0.918$ (95% CI [0.881, 0.951]). Against a human annotator on a 120-response subset, agreement is 98.3% with $\kappa = 0.966$ (95% CI [0.915, 1.000]). Agreement is stable across representation rungs (0.938–0.969), and the near-balanced validation pool yields a prevalence index of 0.156.

```
<accuracy only judge prompt, plus "abstained" in verdict choices>
ABSTENTION
Return the verdict "abstained" whenever the response declines to
answer because it judges the evidence insufficient, however it is
worded. "Not answerable", "the document does not provide this",
"there is no document evidence to determine this" and "it is not
possible to determine this from the given information" are all
abstentions. A response that notes a limitation and then commits to
an answer anyway is NOT an abstention: judge the answer it commits
to.

ERROR CATEGORY
Decide the verdict FIRST, using only the two rules above, and do not
revisit it. The error category explains a verdict you have already
reached; it never changes one. In particular, a response that states
the gold answer in its own words, or surrounds it with explanation, is
still correct.

<the eleven categories, in priority order>
```

Figure 3.4 Judge analysis rubric.

Category	Assigned when
insufficient_evidence	The response declines: the evidence does not support an answer.
no_answer_reached	It reasons but never commits, trailing off or stopping mid-calculation.
answered_unanswerable	The question is unanswerable and a substantive answer is given.
equivocal	The gold answer is present but not committed to.
wrong_count	A "how many" question is answered with a different count.
wrong_value	A non-count numeric answer differs from the gold value.
wrong_entity	A non-numeric answer names a different thing.
incomplete_list	The gold answer is a set and the response gives a subset.
over_inclusive_list	The response includes items the gold answer does not.
unit_or_format_mismatch	The same quantity in a non-matching unit, scale or granularity.
other	None of the above.

Table 3.12 The error taxonomy in judge priority order.

Error categories. Each incorrect response is assigned exactly one category from Table 3.12, using the listed priority order so that the first applicable category is selected. Categories are determined only from

the question, gold answer, and model response; the judge is not shown the document pages, representation rung, or retriever, avoiding attribution to pipeline stages it cannot observe. Correct responses receive no category. The taxonomy is nearly exhaustive in practice, with `other` assigned to only one of the 1,853 incorrect responses on the representation ladder.

3.6 Empirical Findings

We organise the findings by where failures arise in the MP-VRDU pipeline. §3.6.1 examines representation, showing when information is lost through modality limits or imperfect conversion. §3.6.2 studies selection, where incomplete evidence coverage is substantially more damaging than moderate distractor exposure. §3.6.3 then examines reasoning, where the remaining bottlenecks are evidence integration and response calibration.

3.6.1 Representation

Representation determines what evidence remains available for downstream reasoning. OCR-free approaches preserve visual and spatial signals that text extraction may discard [33, 55], while text-based and joint representations can expose linguistic content more directly [36, 45, 64]. Our results separate these effects into two mechanisms: *modality ceiling*, which determines what information a representation can expose at all, and *conversion fidelity*, which determines how much of that information survives encoding.

3.6.1.1 Modality Ceiling

Evidence source	Accuracy				Abstention				Incorrect			
	T	TL	TLV	V	T	TL	TLV	V	T	TL	TLV	V
Text	36.9	44.4	51.0	42.3	23.0	11.5	7.3	8.0	40.1	44.1	41.6	49.7
Layout	26.3	33.9	49.2	43.2	20.3	10.2	0.8	4.2	53.4	55.9	50.0	52.5
Table	35.6	44.7	44.7	34.1	15.9	7.2	6.7	9.1	48.6	48.1	48.6	56.7
Chart	16.3	16.9	42.7	40.4	32.6	28.1	7.3	6.2	51.1	55.1	50.0	53.4
Figure	14.5	22.1	42.4	39.0	36.6	24.8	7.6	7.9	49.0	53.1	50.0	53.1
All	29.0	35.7	49.5	42.6	26.3	17.2	6.2	7.1	44.7	47.2	44.3	50.3

Table 3.13 Answer outcomes by evidence source and representation rung on MMLongBench-Doc. Richer representations primarily reduce abstention, while the incorrect rate remains broadly stable.

Vision Is Necessary but Not Sufficient. The representation ladder shows that vision expands what the system can answer, but does not replace explicit text. At **T** and **TL**, abstention is particularly high for chart questions (32.6% and 28.1%) and figure questions (36.6% and 24.8%), but falls to 7.3% and 7.6% at **TLV**. Since the annotated gold pages are supplied throughout, these refusals expose a modality ceiling: text-only representations often do not preserve the graphical or spatial signal required by the question. Vision alone is also weaker than the joint representation, with **V** reaching 42.6% compared with 49.5% for **TLV**. This does not imply that the visual input lacks the corresponding linguistic information; rather, fine-grained text may be difficult to recover at the available resolution and visual-token budget, whereas an explicit text channel exposes it directly. This complements the original MMLongBench-Doc observation that OCR-fed LLMs often outperform vision-language models [110]. The strongest representation

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is therefore multimodal because each channel compensates for information that is unavailable or difficult to access through the other.

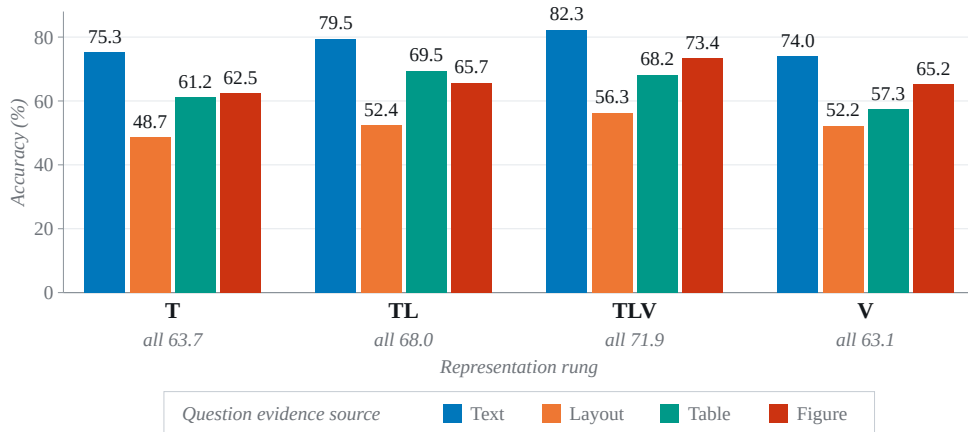


Figure 3.5 LongDocURL accuracy by representation rung and evidence source. Evidence-source differences remain substantial across representation rungs.

The Modality Ceiling Depends on the Document Distribution. LongDocURL reproduces the overall advantage of **TLV**, but with a substantially weaker modality gap (Figure 3.5). Its documents are born-digital and frequently expose answer-relevant content through the text layer even when the annotated evidence source is visual. Figure questions, for example, already reach 62.5% at **T** and 65.7% at **TL**, rising to 73.4% at **TLV**. These easier documents reveal that modality ceiling is not determined by an evidence label alone: a question annotated to a figure may still be answerable through redundant surrounding text. Vision becomes most important when the required signal is genuinely absent from the textual representation, as occurs more often for charts, complex layouts, and scanned or visually dense documents. The value of a modality therefore depends on how much unique information it contributes within the target document distribution, rather than on a universal preference for text or vision.

3.6.1.2 Conversion Fidelity

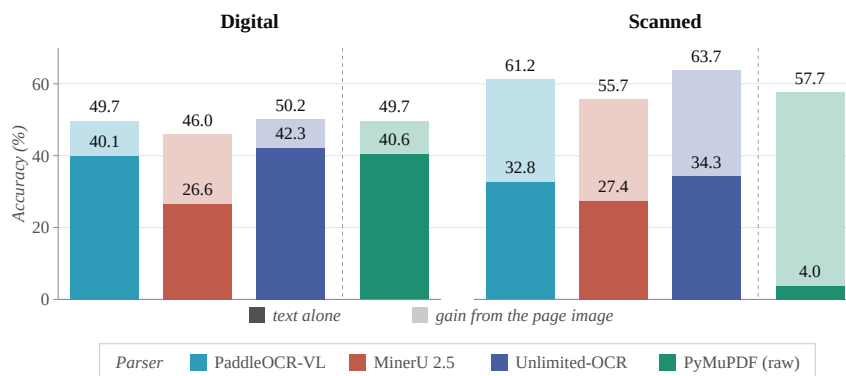


Figure 3.6 Parser accuracy by text source with and without page images. Adding vision substantially reduces sensitivity to parser choice, particularly on scanned documents.

Conversion Errors Matter Most Without a Redundant Channel. Parser choice produces substantial differences when text must carry the evidence alone, confirming that conversion is not a neutral

preprocessing step. MinerU reaches 26.6% on born-digital and 27.4% on scanned documents, compared with 40.1% and 32.8% for PaddleOCR-VL and 42.3% and 34.3% for Unlimited-OCR. The distinction between document types is important: on born-digital documents, parsing mainly restructures an existing text layer, whereas on scans it determines whether a usable text representation is recovered at all. PyMuPDF makes this extreme, falling to 4.0% on scanned documents where an embedded text layer is often absent. These results corroborate prior findings that OCR and structural parsing errors propagate into downstream RAG performance [8, 118]. Our multimodal conditions add a further qualification: once vision accompanies the parsed text, the spread between parsers contracts from 15.7 to 4.2 points on digital documents and from 30.3 to 8.0 on scans. Conversion errors are therefore most damaging when they destroy the only representation carrying the required signal; when another modality preserves the source information, downstream reasoning can often compensate.

Evidence source	TLV				V			
	low	med	high	Δ	low	med	high	Δ
Chart	38.2	44.9	48.3	+10.1	34.3	44.9	48.3	+14.0
Figure	39.7	45.5	48.3	+8.6	38.3	40.0	43.8	+5.5
Layout	46.6	50.0	54.2	+7.6	47.5	49.2	53.4	+5.9
Text	52.9	55.0	55.0	+2.1	38.8	45.4	51.5	+12.7
Table	48.8	49.8	48.8	+0.0	25.3	37.3	42.4	+17.1
All	49.2	52.5	54.7	+5.5	37.8	45.7	50.2	+12.4

Table 3.14 Render resolution by evidence source. On MMLongBench-Doc, at the TLV and V rungs. Δ indicates difference between high and low resolution.

Visual Fidelity Follows the Same Pattern. Render resolution exposes the same mechanism from the visual side. Increasing resolution produces a much larger improvement for **V** than for **TLV**, where accompanying text already preserves part of the fine-grained information that additional pixels would recover (Table 3.14). This mirrors the motivation behind high-resolution compression methods such as mPLUG-DocOwl2, where preserving visual detail across many pages must be balanced against the resulting visual-token cost [24]. Low-resolution rendering is therefore most restrictive when vision alone must carry small text, spatial detail, or other fine-grained evidence. Parser quality and render resolution expose the same underlying principle: conversion fidelity matters most when degradation removes the last usable copy of the information needed to answer.

3.6.2 Selection

Selection determines which evidence reaches the reasoner. While both incomplete retrieval and irrelevant context can limit long-document understanding, our results show that their effects are asymmetric: missing required evidence removes information that downstream stages cannot recover, whereas moderate distractor exposure is often tolerated within the context range we test.

3.6.2.1 Evidence Coverage

Evidence Coverage Is a Hard Constraint. Removing required evidence overwhelmingly converts previously correct answers into incorrect ones, while the reverse transition is rare (Figure 3.7). On MMLongBench-Doc, withholding a single gold page breaks 24.7–27.0% of previously correct answers,

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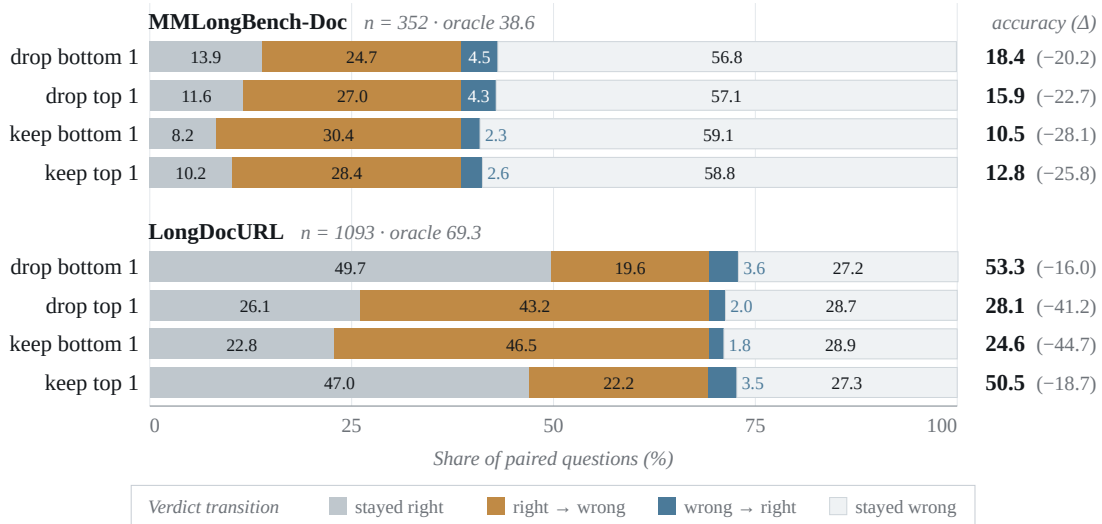


Figure 3.7 Verdict transitions after withholding or retaining gold evidence pages on MMLongBench-Doc and LongDocURL. Missing evidence predominantly converts correct answers into incorrect ones, while repairs are comparatively rare.

compared with only 4.3–4.5% being repaired. Retaining just one gold page widens this imbalance further, with 28.4–30.4% changing from correct to incorrect and only 2.3–2.6% moving in the opposite direction. The result is therefore more than the expected correlation between retrieval recall and accuracy: directly removing known evidence produces a strongly directional loss that downstream reasoning rarely reverses. This complements retrieval-depth studies showing that broader retrieval initially improves performance as evidence recall rises [120], while isolating the mechanism responsible for that early gain.

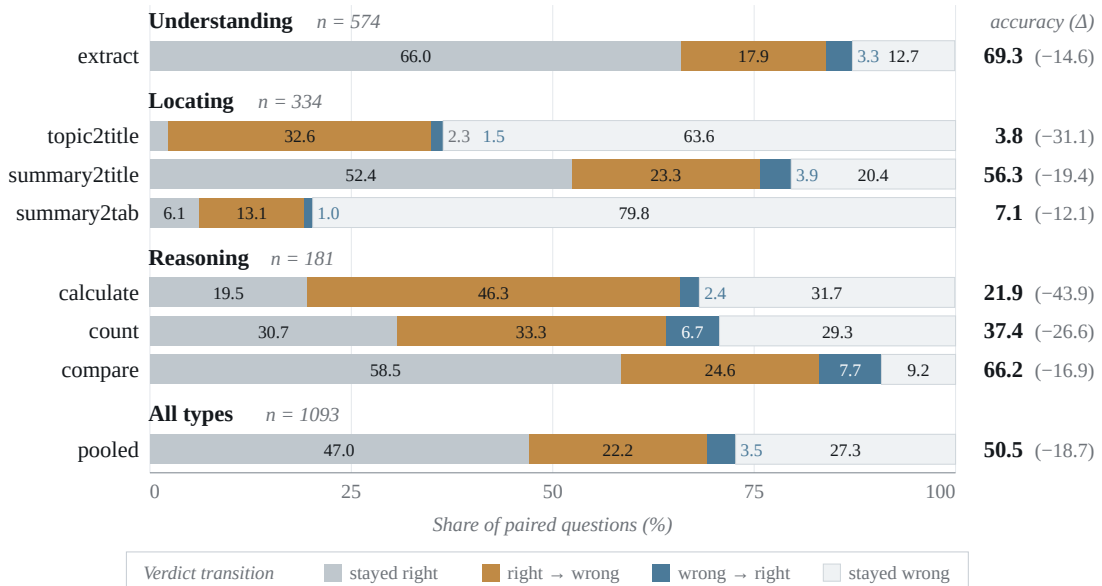


Figure 3.8 LongDocURL verdict transitions under partial evidence coverage, grouped by question type. Extractive questions are relatively robust, while question types requiring evidence combination show substantially larger losses.

Evidence Importance Shapes Coverage Loss. The effect of missing evidence also depends on which evidence survives. On LongDocURL, retaining only the highest-ranked gold page preserves substantially more performance than removing it, while dropping the lowest-ranked page is considerably less damaging. MMLongBench-Doc is less forgiving, consistent with its multi-page questions more often depending jointly on several pieces of evidence. Figure 3.8 shows the same pattern by question type: `extract` questions are comparatively robust when only part of the annotated evidence remains, whereas `calculate`, `count`, and `topic2title` suffer much larger losses. Selection quality therefore depends not only on how much evidence is retrieved, but on whether the evidence most critical to the required operation is preserved.

3.6.2.2 Distractor Exposure

Condition	MMLongBench-Doc						LongDocURL					
	T	TL	TLV	V	R→W	W→R	T	TL	TLV	V	R→W	W→R
<i>1 gold page</i>												
Oracle	37.1	44.6	63.7	55.0	—	—	65.6	74.0	74.7	64.5	—	—
+1 distractor	39.5	48.9	63.7	55.7	7.8	7.0	65.4	72.4	74.4	63.0	8.5	8.1
+2 distractors	38.8	47.9	64.3	56.1	8.2	8.0	65.1	68.7	72.0	63.9	9.3	7.5
+3 distractors	40.9	47.0	63.1	55.1	9.7	8.2	64.5	68.0	68.4	62.1	11.7	6.9
+4 distractors	41.9	45.3	62.5	52.5	9.5	6.6	63.5	68.0	72.2	62.5	10.5	8.7
+5 distractors	39.5	42.6	62.3	54.4	10.0	6.9	64.6	69.1	70.2	60.8	11.4	7.3
<i>2 gold pages</i>												
Oracle	28.9	33.3	37.8	34.6	—	—	62.2	63.4	70.4	61.5	—	—
+1 distractor	25.7	34.0	38.6	34.4	7.8	7.5	58.7	63.0	66.2	57.9	9.6	4.1
+2 distractors	25.3	33.6	35.7	32.4	9.6	6.0	58.9	62.2	66.4	57.4	10.9	4.4
+3 distractors	21.6	28.2	33.6	33.2	11.4	6.0	58.1	63.3	70.8	57.8	9.7	6.0
+4 distractors	24.6	30.7	43.7	31.3	11.8	11.8	57.8	63.2	68.3	57.1	13.3	6.1

Table 3.15 Adding distractor pages to a complete gold set. Distractors ranked by ColQwen3 and verdict transitions indicate the proportion of questions that turned incorrect when previously correct or vice versa when distractors were added at the **TLV** rung.

Distractor Exposure Produces Stable Accuracy with Mild Degradation. Across both datasets, adding non-gold pages produces only modest changes in aggregate accuracy within the tested range. On MMLongBench-Doc, performance remains broadly stable for single-page questions and declines gradually for multi-page questions as additional pages are introduced, while LongDocURL shows even less degradation under comparable distractor counts. This pattern is consistent with prior work showing that distractor effects are not monotonic or uniform: Cuconasu et al. find that their impact depends strongly on semantic relatedness and position [119], while Jin et al. report a rise-then-fall relationship in which increasing retrieval depth initially improves performance before accumulated hard negatives begin to reduce it [120]. The apparent stability in our results therefore should not be interpreted as evidence that added pages are harmless; instead, it suggests that positive and negative effects can partially cancel at the aggregate level, motivating a closer examination of the underlying verdict transitions.

Non-Gold Pages Can Both Distort and Complete the Evidence. The verdict transitions refine this picture. Cuconasu et al. show that additional context can help or hurt depending on its relevance and posi-

tion [119], while Jin et al. show that hard negatives can mislead the reasoner even when relevant evidence is already present [120]. Manual inspection of our transitions reveals a related document-specific effect: non-gold pages can contain contextual information that changes how the annotated evidence should be interpreted. In right-to-wrong cases, added pages sometimes introduce partial evidence that leads the reasoner to overcount or misinterpret the gold pages; in wrong-to-right cases, they can provide disambiguating or supplementary context that repairs an otherwise incorrect answer. Retrieval depth therefore trades not only noise against coverage, but also incomplete against more complete contextualisation, suggesting that broader evidence acquisition can be useful if the resulting context is carefully controlled.

3.6.3 Reasoning

Reasoning determines whether the supplied evidence is correctly combined and interpreted to produce an answer. With representation and selection controlled through TLV and oracle evidence, we examine two mechanisms: whether the reasoner can integrate complementary evidence, and whether it can judge when that evidence is sufficient to support an answer.

3.6.3.1 Evidence Integration

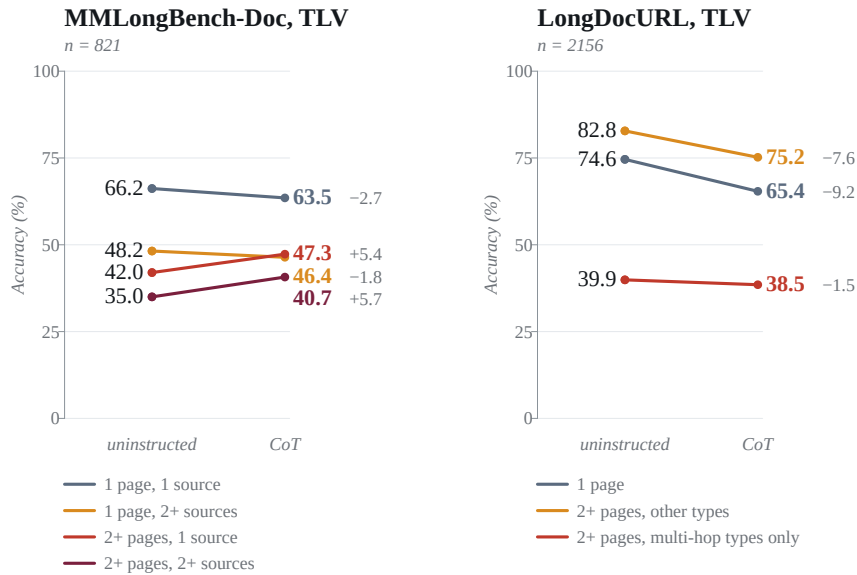


Figure 3.9 Effect of chain-of-thought prompting across evidence structures at the TLV rung with oracle evidence pages. CoT selectively improves cross-page performance on MMLongBench-Doc, but does not produce the same gains on LongDocURL.

Chain-of-Thought Selectively Improves Cross-Page Integration. Chain-of-thought does not improve reasoning uniformly across evidence structures or datasets (Figure 3.9). On MMLongBench-Doc, it slightly reduces accuracy on single-page questions, from 66.2% to 63.5% for one-source questions and from 48.2% to 46.4% when several evidence sources occur on the same page. The effect reverses once evidence is distributed across pages: accuracy rises from 42.0% to 47.3% for one-source questions and from 35.0% to 40.7% when both pages and sources must be combined. The pooled integration deficit therefore narrows from 24.6 to 18.2 points. LongDocURL shows that this benefit is not universal. CoT reduces accuracy for single-page questions from 74.6% to 65.4% and for other multi-page

questions from 82.8% to 75.2%, while its effect on multi-hop questions is much smaller, from 39.9% to 38.5%. This refines the usual motivation for chain-of-thought as a general aid to complex reasoning [80]. MP-VRDU methods similarly use explicit intermediate reasoning to combine evidence across document context [35, 43], but our results show that its value depends on the reasoning demands of the benchmark rather than on document length alone. On MMLongBench-Doc, part of the cross-page deficit identified in the original benchmark [110] is recoverable through a reasoning-only intervention; on LongDocURL, the same intervention provides no corresponding gain.

Evidence source	n	None	CoT	Δ	R→W	W→R
Text	208	44.2	46.6	+2.4	6.7	9.1
Layout	98	42.9	44.9	+2.0	8.2	10.2
Table	137	36.5	47.4	+10.9	5.8	16.8
Chart	103	35.9	38.8	+2.9	5.8	8.7
Figure	173	43.4	42.2	-1.2	7.5	6.4
<i>All</i>	457	41.6	45.3	+3.7	6.8	10.5

Table 3.16 Effect of chain-of-thought across evidence sources for multi-hop MMLongBench-Doc questions at TLV with oracle pages.

The Benefit Depends on the Evidence Being Integrated. The improvement from chain-of-thought also varies substantially with evidence type (Table 3.16). Among multi-hop questions, the largest gain occurs for tables, where accuracy increases from 36.5% to 47.4%, a 10.9-point improvement. Gains are much smaller for charts (+2.9 points), pure text (+2.4), and generalised text (+2.0), while figure questions decline slightly by 1.2 points. Chain-of-thought therefore appears most useful when cross-page integration requires explicit operations over structured evidence rather than simply more computation over any difficult input. This is consistent with the broader MP-VRDU literature, where multi-step reasoning is commonly introduced to decompose document-level questions and combine intermediate evidence [35, 43, 64]. At the same time, the weak or negative effect on several source types shows that additional reasoning is not intrinsically beneficial. Together with the structural breakdown above, the results suggest that chain-of-thought primarily mitigates a specific integration burden: relating complementary, and especially structured, evidence distributed across pages.

3.6.3.2 Response Calibration

The Reasoner Does Not Reliably Calibrate Evidence Sufficiency. Targeted abstention prompting makes the reasoner substantially more conservative, but does not improve this decision selectively (Figure 3.10). On unanswerable questions, refusal rises from 45.5% to 74.6%, substantially reducing unsupported answers. The same intervention raises refusal on answerable questions from 6.2% to 27.7%, while correct answers fall from 49.5% to 40.0%. Prior work similarly shows that prompting can shift context faithfulness and abstention behaviour [121], while MMLongBench-Doc reports large differences between aggressive and conservative answering across models [110]. Our intervention shows that this operating point can also be shifted substantially within the same model. The reduction in unsupported answers therefore comes partly from a general increase in reluctance to answer, rather than a better distinction between sufficient and insufficient evidence.

3. EMPIRICAL ATTRIBUTION OF FAILURE MODES

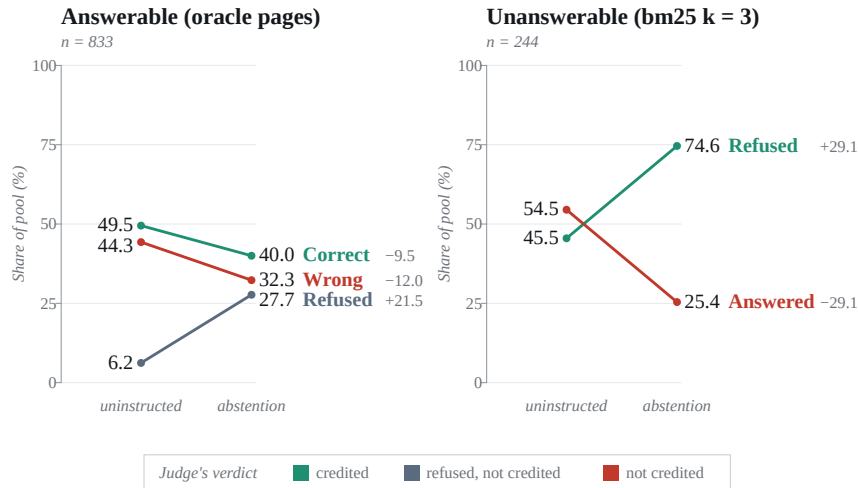


Figure 3.10 Effect of targeted abstention prompting at the TLV rung. The instruction reduces unsupported answering when evidence is insufficient, but also increases refusal on answerable questions.

Evidence source	n	No instruction			Abstention instruction		
		Acc.	Abst.	Inc.	Acc.	Abst.	Inc.
Text	286	51.0	7.3	41.6	43.7	24.1	32.2
Layout	118	49.2	0.8	50.0	41.5	19.5	39.0
Table	208	44.7	6.7	48.6	38.5	30.3	31.2
Chart	178	42.7	7.3	50.0	26.4	25.3	48.3
Figure	290	42.4	7.6	50.0	34.1	35.5	30.3
All	833	49.5	6.2	44.3	40.0	27.7	32.3

Table 3.17 Refusal outcomes by evidence source under targeted abstention prompting at TLV. False abstention is more frequent for visually and structurally demanding evidence, suggesting that reasoning difficulty influences the decision to refuse.

Reasoning Difficulty Is Mistaken for Evidence Insufficiency. False abstention also rises with the difficulty of interpreting the supplied evidence. Under the abstention instruction, refusal reaches 35.5% for figure questions and 30.3% for table questions, compared with 25.3% for charts and 24.1% for text (Table 3.17). Since these questions receive their oracle pages and the full TLV representation, the higher refusal rates cannot be attributed to evidence absence. Difficult evidence instead appears more likely to be treated as insufficient evidence. Response calibration therefore depends not only on whether support is present, but also on the reasoner’s confidence in interpreting it, which helps explain why stronger abstention instructions trade unsupported answers for false refusals rather than cleanly separating the two.

3.7 Error Analysis

The error taxonomy provides a finer view of how the reasoner fails once representation and evidence selection are controlled. Unlike the preceding analyses, each category is reported as a share of the full question pool for that condition, allowing direct comparison across prompt modes and evidence sources.

3. EMPIRICAL ATTRIBUTION OF FAILURE MODES

Modality ceiling		mmlongbench:000014 · chart · 1 gold page
Q	According to the chart on page 14 how much time was spent with family and friends in 2010?	GOLD 21%
T	○ “The document does not provide any evidence or chart on page 14, so it is not possible to determine [...]”	
TL	○ “The document does not provide sufficient textual or numerical information to determine how much [...]”	
V, TLV	✓ “According to the chart on page 14, in 2010, 21% of weekend time was spent with family and friends.”	
NOTE The gold page reaches every rung, so nothing is missing from the context. Only the rungs carrying the page image can read the bar, and the text rungs decline rather than guess.		
Conversion fidelity		mmlongbench:000513 · plain text + table · 1 gold page
Q	How many geographic market names are listed under Los Angeles area?	GOLD 13
T (raw)	✓ “The document lists 13 geographic market names under the Los Angeles area. [...]”	
TL (Paddle)	✗ “26 The document lists 26 geographic market names under the “Los Angeles Area” section [...]”	
TL (MinerU)	✗ “26 The document lists 26 geographic market names under [...]”	
TL (unlimited)	✓ “There are 13 geographic market names listed under the Los Angeles area. [...]”	
TLV	✓ “Based on the provided document, there are 13 geographic market names listed under [...]”	
NOTE The embedded text layer already carried the count; two parsers restructure the list and return its last label instead. The reasoner cannot tell a corrupted number from a real one, and both a better parser and the page image recover it.		
Evidence coverage		mmlongbench:000082 · plain text · 2 gold pages
Q	Which method integrates knowledge into white-box models via directive fine-tuning?	GOLD PKG
oracle	✓ “Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is PKG. [...]”	
drop top	✗ “Based on the provided document, the method that integrates knowledge into white-box models via directive fine-tuning is Fine-tuning Embedding Model. [...]”	
drop bottom	✗ “The method that integrates knowledge into white-box models via directive fine-tuning is directive fine-tuning. [...]”	
NOTE Neither gold page is sufficient alone. Withholding either one does not make the reasoner abstain, it answers from the page that is left, so the missing evidence never announces itself.		
Distractor exposure		mmlongbench:000041 · chart · 1 gold page
Q	On the chart showing average exit cap, there is one point that US, Europe, and China have the same average exit cap value. What is this value in USD million? Give me an integer	GOLD 300 million
gold only, +1, +2	✗ “350”	
+3 distractor	✓ “300”	
NOTE The gold page never leaves the context; the added pages only surround it. They change the reading of the same chart rather than removing anything, and this cell is one of the repairs that offset the breaks.		
Evidence integration		mmlongbench:000820 · chart + plain text · 2 gold pages
Q	According to the survey, how many US adults think it's acceptable for the criminal justice system to use automated criminal risk scores?	GOLD 1929
TLV	✗ “According to the survey, 42% of U.S. adults think it is acceptable for the criminal justice system [...]”	
V	✗ “According to the survey, 42% of U.S. adults think it is acceptable for the criminal [...]”	
T	✗ “[...] The survey data (from May 29–June 11, 2018, among 4,594 respondents) is described in detail, including methodology and weighting [...]”	
NOTE The question asks how many, not what percentage. Both operands are read: the chart's 42% at the rungs holding the page image, the 4,594 respondents on the other gold page at T. No rung multiplies them, and 42% of 4,594 is the gold 1,929, so what fails is the join across the two pages and not the evidence on either.		
Response calibration		mmlongbench:000236 · figure · 1 gold page
Q	Among all the pictures, in which page the picture contains the most number of people? Give me the page no. as an integer.	GOLD 9
none	✓ “9”	
abstain	○ “Not answerable”	
NOTE The same page, the same model, and an answer the reasoner is able to give. The abstention instruction turns it into a refusal on a question it can answer.		
✓ judged correct ✗ judged incorrect ○ declined to answer		

Figure 3.11 Worked examples of the six failure mechanisms, one question each, from MMLongBench-Doc under Qwen3-VL-8B. Every band gives the question, its gold answer, and what the reasoner returned under each condition that mechanism is tested by, marked with the judge’s verdict; the quotations are the model’s own output, elided at [...].

3.7.1 Prompt Mode

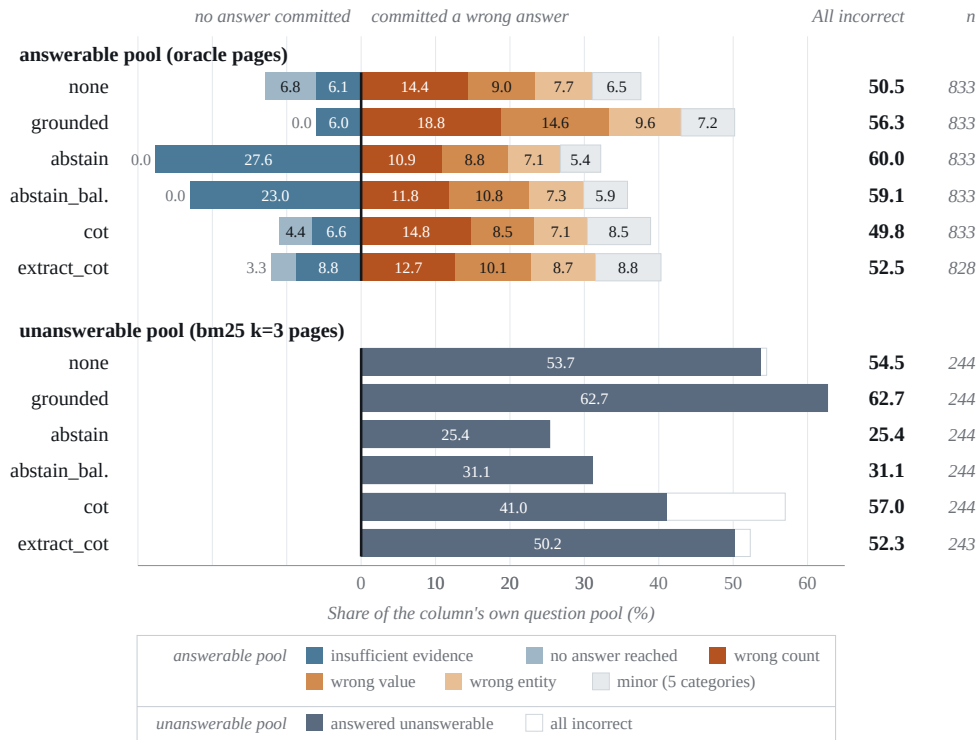


Figure 3.12 Error profile by prompt mode on answerable and unanswerable questions.

The error distribution changes substantially with the reasoning instruction, showing that prompting alters the form of failure rather than uniformly removing it. Grounding largely eliminates cases where the model fails to reach an answer, but this is offset by committed errors such as incorrect counts, values, and entities. The abstention instruction shifts the distribution more strongly toward `insufficient_evidence`, consistent with the calibration results above: it prevents some unsupported answers but also introduces refusals on answerable questions. The `abstain_balanced` variant produces a very similar profile despite explicitly encouraging the model to answer when evidence is sufficient, suggesting that this trade-off is not easily removed through wording alone. Chain-of-thought produces a different redistribution, changing the mix of substantive answer errors without the same increase in abstention, while `extract_cot` remains close to ordinary `cot` despite adding an explicit evidence-extraction step. Prompting therefore changes how the reasoner fails, but no mode removes the underlying reasoning errors across the board.

3.7.2 Evidence Source

Error type also varies substantially with the evidence a question requires. Under the uninstructed oracle-page condition, 50.5% of answerable questions are incorrect overall, but the dominant failure differs by source. Layout and figure questions are particularly prone to `wrong_count` errors, which account for 28.8% and 24.1% of their respective pools, compared with 14.4% overall. Tables and charts instead show the highest `wrong_value` rates, at 15.4% and 15.7%, suggesting that structured quantitative evidence more often fails during value extraction, comparison, or combination. Tables also stand out for `no_answer_reached`, which occurs on 14.4% of questions, more than twice the pooled rate of 6.8%. These differences indicate that providing the required multimodal evidence does not make downstream

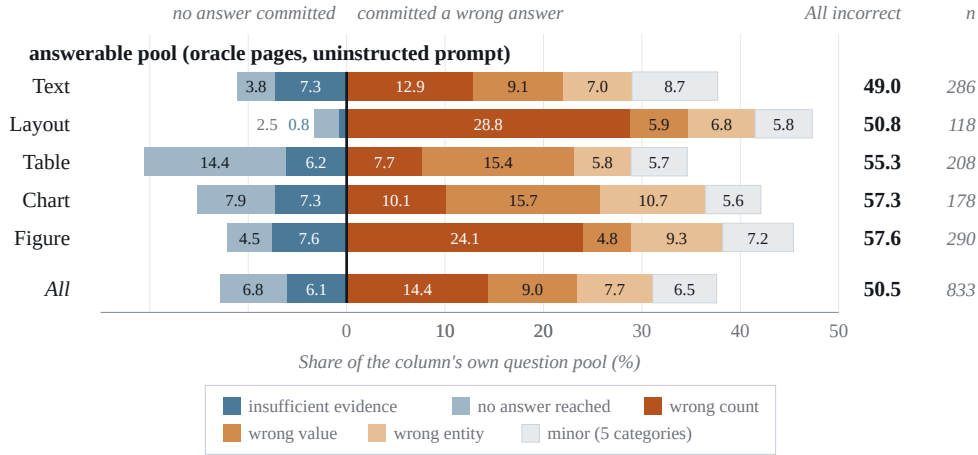


Figure 3.13 Error profile by evidence source on the answerable pool with oracle pages and no prompt instruction.

reasoning uniform: spatial and visual content more often fails during identification or aggregation, while quantitative structures produce a different mixture of extraction and comparison errors. The remaining failures are distributed across entity, list, abstention, and other categories, reinforcing that MP-VRDU difficulty depends not only on whether the evidence is available, but also on the operations required to interpret its particular form.

3.8 Practical Discussion

The preceding experiments isolate where MP-VRDU systems fail. We now consider what those findings imply for system design and deployment. Rather than treating representation, retrieval, and reasoning independently, the results suggest that compute should be allocated across the pipeline according to where information is most likely to be lost and where additional capacity can still recover it. f

3.8.1 Encoding Cost and Fidelity

Rung	Acc.	Input tokens			prefill s	Output		OOM %
		text	visual	total		tok	decode s	
T	29.0	932	0	932	1.4	115	5.7	1.2
TL	35.7	1632	0	1632	1.8	133	6.6	6.7
TV	50.1	929	851	1780	–	140	–	–
TLV	49.5	1632	857	2488	26.3	136	6.7	11.2
V	42.6	38	857	894	29.0	139	7.1	0.8

Table 3.18 Cost of each representation rung. Accuracy and token counts are means over the answerable pool under the error-taxonomy judge; the two latency columns and the OOM rate are measured separately on 2×16 GB V100s.

Table 3.18 shows that representation cost depends on both the channels retained and how they are processed. At the medium resolution, **V** averages 857 visual tokens and 894 total input tokens per question, comparable to the 932 tokens of **T**; the largest context instead comes from combining channels,

with **TLV** averaging 2,488 tokens from 1,632 text and 857 visual tokens. Visual processing nevertheless carries a substantial compute cost, with **V** and **TLV** requiring 29.0 and 26.3 s of measured prefill time compared with 1.4–1.8 s for the text-only rungs, while output length and decoding time remain comparatively stable. The richest representation also produces the highest measured OOM rate at 11.2%, compared with 0.8% for **V**. Importantly, additional representation cost does not always translate into higher accuracy: **TV** reaches 50.1% against 49.5% for **TLV** while averaging 1,780 rather than 2,488 input tokens, suggesting that embedded text paired with vision is a strong option for born-digital documents where parsing adds little once the page image is available. This does not extend to scanned documents, which lack a reliable embedded-text channel and therefore still benefit from parser-derived text. Likewise, the resolution sweep shows that additional visual fidelity is most useful for dense tables, charts, and other visually dependent evidence rather than uniformly across questions. Representation should therefore be selected around the target workload: retain text and vision where both carry complementary evidence, avoid parser overhead where embedded text is already sufficient, and spend additional visual resolution only where legibility is likely to constrain performance.

3.8.2 Retrieval Quality and Efficiency

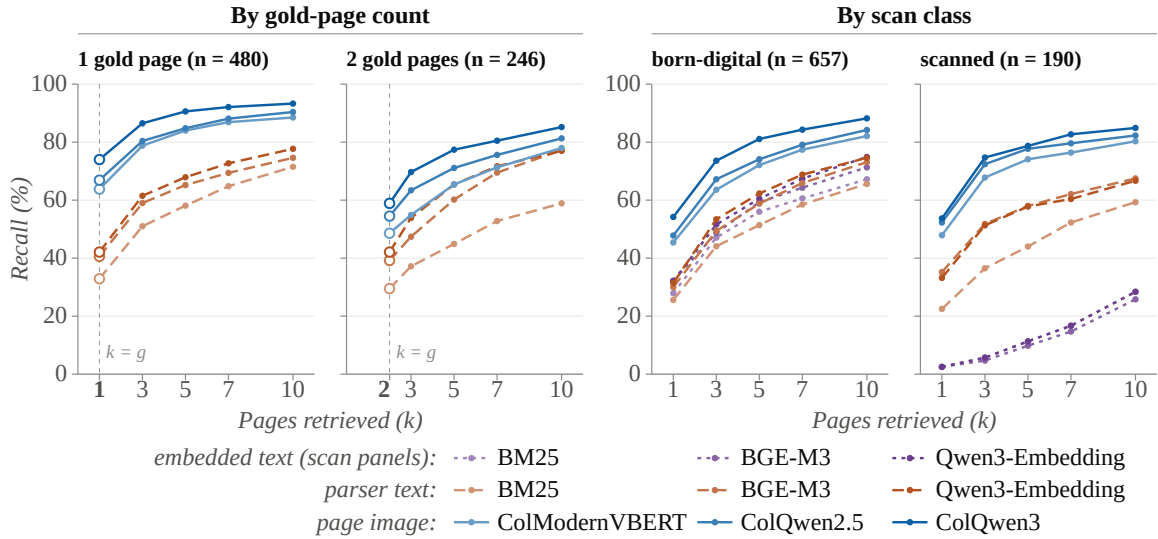


Figure 3.14 Retrieval recall on MMLongBench-Doc across retrieval depth, gold-page count, and document type. Recall increases with depth, questions requiring several evidence pages remain harder to cover, and visual retrieval is more robust across born-digital and scanned documents.

The selection findings suggest that retrieval should be configured primarily around evidence coverage, with the retriever and depth chosen according to the document distribution and the amount of downstream context the system can afford to process. Figure 3.14 shows that recall rises consistently with depth: ColQwen3 increases from 54.1% at $k = 1$ to 87.4% at $k = 10$, while the strongest joint text–vision configuration rises from 58.7% to 89.7%. This gain comes at the expected loss of precision, but the selection experiments show why that trade can be worthwhile: omitting required evidence is substantially more damaging than introducing a moderate number of additional pages, and some non-gold pages can even supply useful contextual information. Multi-page questions further strengthen this argument because complete coverage is harder when several required pages must all be recovered. Retrieval representation introduces a second deployment choice. Embedded-text retrieval avoids document parsing and is effective on born-digital documents, but its reliability falls on scans where the text layer

is absent or sparse. Parser-based text retrieval adds preprocessing cost while recovering much of this robustness, whereas visual retrievers remain comparatively consistent across both document types. A practical system should therefore favour recall where downstream evidence filtering is available, while choosing between text, parsed-text, visual, or joint retrieval according to the document mix and preprocessing budget rather than treating one retrieval representation as universally preferable.

3.8.3 Reasoner Capability and Deployment Trade-offs

Model / config	T	TL	TLV	V	GB
<i>Scale (Qwen3-VL, bf16)</i>					
2B	24.7	31.9	38.3	31.9	4.3
4B	32.3	38.6	50.4	42.4	8.9
8B	31.9	38.4	52.4	45.5	17.5
32B	35.5	43.9	59.6	56.7	66.7
<i>Precision (Qwen3-VL-8B)</i>					
16-bit	31.9	38.4	52.4	45.5	17.5
8-bit	32.5	37.9	53.6	45.7	10.3~
4-bit	31.8	39.4	51.8	45.5	6.8~
<i>Matched budget (~17–20 GB)</i>					
8B, 16-bit	31.9	38.4	52.4	45.5	17.5
32B, 4-bit	37.0	42.1	59.1	52.3	20.0~
<i>Family (8B class, bf16)</i>					
Qwen3-VL-8B	31.9	38.4	52.4	45.5	17.5
InternVL3-8B	19.3	25.3	32.6	24.1	15.9
GLM-4.6V-Flash (9B dense)	25.9	32.9	42.7	30.9	20.6
MiniCPM-V-4.5 (8B)	30.7	38.1	60.1	53.3	17.4
Llama-3.2-11B-Vision	26.5	34.3	33.3	25.3	21.3
<i>Scale (Gemma-3, bf16)</i>					
4B	22.1	30.5	39.5	33.6	8.6
12B	30.5	36.9	48.1	41.6	24.4
27B	32.8	40.0	52.0	44.6	54.9
<i>Reasoning variant (Qwen3-VL-8B)</i>					
Instruct	31.9	38.4	52.4	45.5	17.5
Thinking	34.8	42.9	62.2	52.1	17.5

Table 3.19 Reasoner accuracy across scale, precision, matched budget and family. Accuracy (%) on oracle pages over the full answerable pool; GB is the model weight footprint, not peak VRAM.

Table 3.19 shows that reasoner deployment is not simply a choice of parameter count, but a trade-off between model family, scale, numerical precision, memory, and the type of reasoning required. Within Qwen3-VL, scaling from 2B to 32B raises TLV accuracy from 38.3% to 59.6%, but the reasoning experiments show that larger models do not remove the single-to-multi-page integration deficit, so scale improves capability without directly resolving every failure mode. Models of similar size also differ substantially: Qwen3-VL-8B reaches 52.4% at TLV compared with 32.6% for InternVL3-8B, showing that nominal parameter count alone is a poor deployment criterion. Quantisation provides the clearest way to preserve model capacity under a fixed memory budget. Qwen3-VL-8B changes only from 52.4% at 16-bit to 51.8% at 4-bit while reducing its weight footprint from 17.5 GB to approximately 6.8 GB, and the 32B model at 4-bit reaches 59.1% at approximately 20 GB, outperforming the full-precision 8B model at a similar memory budget. The 4B model also remains close to the 8B model on several con-

ditions, suggesting that smaller checkpoints can still be attractive where throughput matters more than peak accuracy. Under tight VRAM constraints, preserving scale through quantisation can therefore be preferable to immediately selecting a smaller full-precision model, but the choice should still account for model family and latency because reduced weight precision does not guarantee faster decoding on all hardware.

3.9 Conclusion

This chapter examined MP-VRDU failure through controlled attribution across representation, selection, and reasoning. The results show that vision is necessary but remains complementary to text, that missing required evidence is substantially more damaging than moderate distractor exposure, and that cross-page integration remains difficult even when all required evidence is supplied. The analysis further shows that prompt interventions can shift error behaviour without necessarily improving underlying reasoning or calibration. Together, these findings show that MP-VRDU failure is distributed across the evidence pipeline and that end-to-end accuracy alone is insufficient to identify the mechanism responsible.

Chapter 4

Validation and Conclusion

This thesis treats multi-page visually rich document understanding as an evidence-management problem. Chapter 2 develops this perspective by organising MP-VRDU systems around how they preserve, select, and use document-scale evidence, while Chapter 3 tests these decisions directly through the three failure loci of representation, selection, and reasoning. This chapter brings the two together by translating their findings into a deliberately simple adaptive evidence-management pipeline before considering the broader challenges and conclusions that follow.

Across the thesis, the results show that MP-VRDU performance depends not only on backbone capability, but on how evidence is managed before and during inference. Representation must preserve complementary textual and visual information, selection must acquire sufficient evidence without unnecessarily expanding the final context, and reasoning must integrate that evidence while recognising when support is insufficient. Section 4.1 tests these principles at the system level, asking whether they can be combined into a competitive working pipeline without a specialised document backbone or complex orchestration framework. The remainder of the chapter then considers the open challenges that remain for MP-VRDU and concludes the thesis.

4.1 A Simple Adaptive-Trajectory Method

We introduce a simple adaptive evidence-management pipeline designed directly from the findings of Chapters 2 and 3. Rather than introducing a specialised document backbone or complex orchestration framework, the method combines broad evidence acquisition, selective evidence presentation, and structured reasoning within an adaptive trajectory. The aim is to test whether these principles are sufficient to produce a competitive MP-VRDU system while retaining a transparent and configurable design.

Our contributions are as follows: (1) We introduce a simple adaptive evidence-management pipeline that achieves performance competitive with leading MP-VRDU systems using the same backbone. (2) We provide a system-level validation of the evidence-management perspective and empirical attribution findings by translating them into concrete choices over representation, selection, and reasoning. (3) We characterise the resulting deployment trade-offs through configurable evidence-coverage and trajectory parameters, together with measurements of token usage, latency, memory use, and trajectories.

4.1.1 Methodology

We combines high-fidelity multimodal representation, broad evidence acquisition, selective evidence presentation, and structured reasoning. Its design follows the three failure loci identified in the attribution study. For **representation**, pages are read through complementary textual and visual channels so that

4. VALIDATION AND CONCLUSION

evidence lost or difficult to access through one remains available through the other. For **selection**, the system favours broad evidence acquisition because additional pages can either disrupt or repair an answer depending on the context they contribute; evidence gathering is therefore separated from deciding which pages should be exposed in full to the final reasoner. For **reasoning**, the final answer is produced with explicit intermediate reasoning, premise checking, and abstention so that the gathered evidence is both integrated and checked for sufficiency. The resulting pipeline uses three specialised roles implemented by the same underlying vision-language model:

1. The `reader` traverses pages in retrieval-ranked order, records answer-relevant evidence from each page, marks pages likely to contain answer-bearing information, and decides when sufficient evidence has been collected to attempt an answer.
2. The `synthesizer` receives the accumulated evidence in document order. Pages marked as answer-bearing are supplied in full through their textual and visual representations, while other useful pages remain available through the compact records produced by the `reader`. It then reasons over the assembled evidence and produces a candidate answer.
3. The `arbiter` receives the accumulated evidence and candidate answer and decides whether the current evidence is sufficient to terminate the trajectory or whether control should return to the `reader` for further evidence acquisition. It cannot replace the answer itself.

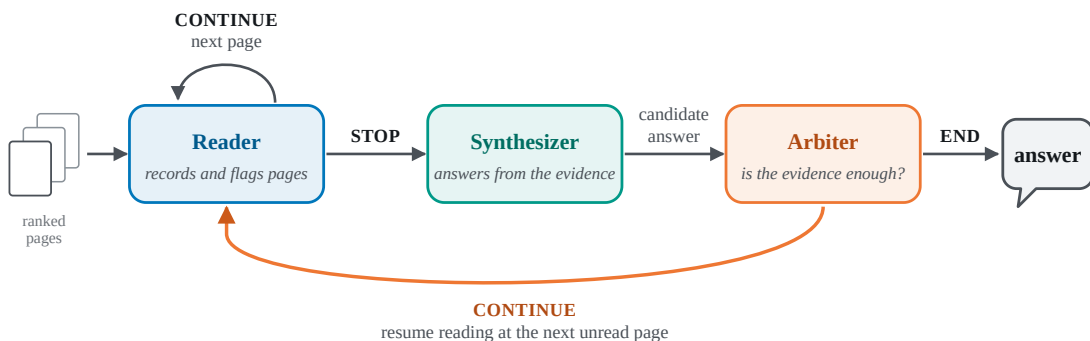


Figure 4.1 Architecture of our simple method.

The three roles form an adaptive evidence-processing loop. When the `reader` proposes `STOP`, the `synthesizer` receives the retained records and full representations of pages marked as answer-bearing, with the highest-ranked read pages used as fallback when none are marked. It then reasons over the evidence, checks question premises, and must identify the missing fact before returning *Not answerable*. The `arbiter` either accepts the answer or resumes reading from the next unread page. Full prompts and implementation details are provided in Appendix C.1.

Configuration. We instantiate all three roles with a shared 4-bit NF4 Qwen3-VL-8B-Instruct model using greedy decoding. Document pages are ranked once per question using ColQwen3.5 and visited in that order. The `reader` processes pages using the full **TLV** representation, combining Unlimited-OCR text with the high-resolution visual preset. During synthesis, every page marked as answer-bearing is reintroduced through its Unlimited-OCR text and page image at medium resolution, while unflagged pages contribute only their retained records. This preserves the redundant text and vision channels identified as beneficial in Chapter 3 while limiting the amount of full multimodal context presented.

Deployment parameters. Five parameters control the amount and depth of evidence acquisition. B sets the maximum number of pages that may be read, p controls the number of pages processed per `reader` turn, k sets the minimum initial reading depth before stopping is allowed, m sets the minimum additional reading after the `arbiter` reopens a trajectory, and C bounds the number of synthesis attempts. These parameters allow the same pipeline to scale across deployment settings: larger budgets increase evidence coverage and trajectory depth, while smaller values reduce latency, token use, and memory pressure.

4.1.2 Experiments

Setup and comparison. We reuse the experimental infrastructure of Chapter 3, including, document preprocessing, evaluation, and the Qwen3-VL model family. The deployment configuration is chosen specifically to run on a single NVIDIA V100 16 GB GPU, using 4-bit NF4 Qwen3-VL-8B-Instruct with FP16 compute and greedy decoding, with $B = 25$, $p = 1$, $k = 6$, $m = 3$, and $C = 3$. We report answer accuracy overall, by evidence source, and on unanswerable questions. We compare against M3DocRAG, MoLoRAG, MDocAgent, and SLEUTH, which represent recent MP-VRDU systems evaluated on MMLongBench-Doc with the same Qwen3-VL-8B backbone. Holding the backbone fixed makes the comparison primarily reflect differences in how each system retrieves, manages, and reasons over document evidence rather than differences in model capacity. We additionally include the gold-page **TLV** baseline from Chapter 3, which supplies the annotated evidence directly to the same reasoner and therefore provides a controlled reference for whether the proposed pipeline can recover comparable performance without access to oracle page annotations.

Efficiency and deployment. The pipeline avoids placing all retrieved pages into a single multimodal context: evidence is acquired over successive `reader` turns, while only selected pages are exposed in full during synthesis and the shared model weights remain resident throughout the trajectory. Alongside accuracy, we therefore report input and output token counts, prefill and decoding latency, peak GPU memory usage, and the numbers of `reader`, `synthesizer`, and `arbiter` turns executed per question. These measurements characterise the additional inference cost introduced by adaptive evidence acquisition and allow accuracy to be considered alongside latency, memory use, and trajectory depth.

4.1.3 Results

Performance. Table 4.1 shows that the proposed pipeline reaches 54.5% accuracy under the official MMLongBench-Doc protocol, the highest result among the open 7–8B systems reported in the table. Within the Qwen3-VL-8B group, this exceeds DocTrace at 52.9% and MARDoc at 52.7%, despite our reasoner being deployed at 4-bit precision and without task-specific training. The system also reaches 56.3% under our LLM judge, although these results should not be compared directly with published LLM-judged scores that use different judges. More importantly, the method exceeds the gold-page **TLV** baseline under both evaluation protocols, improving from 49.5% to 54.5% under the official scorer and from 52.2% to 56.3% under the LLM judge. This supports the central hypothesis that annotated evidence pages are not necessarily an upper bound when evidence acquisition and reasoning are considered jointly: a system can benefit from gathering additional document context while controlling which evidence is presented in full during synthesis. The gains are not uniform, however. Under the official scorer, the largest improvements over the gold-page baseline occur for tables (50.5% to 59.9%) and unanswerable questions (52.5% to 71.9%), while text and figure performance remain approximately unchanged

4. VALIDATION AND CONCLUSION

Method	OCR	Trained	Evidence source					Evidence pages				ACC
			TXT	LAY	CHA	TAB	FIG	SIN	MUL	UNA		
Official protocol: answer extraction + rule-based scorer												
Gold pages + VLM (TLV)	✓	×	48.1	49.6	39.4	50.5	42.5	57.7	37.4	52.5	49.5	
Qwen2.5-VL-7B												
URaG [44]	×	✓	33.6	27.7	29.3	27.5	27.2	42.7	16.9	43.5	33.8	
CoR [46]	×	✓	39.4	31.2	27.7	38.6	27.5	41.9	25.9	45.5	37.4	
MDocAgent [†] [36], via [51]	✓	×	43.1	31.2	32.6	38.7	37.9	53.5	23.8	28.3	38.5	
MoLoRAG [51]	×	×	41.7	32.4	34.2	37.9	37.6	50.1	26.1	35.0	39.3	
MoLoRAG+ [51]	×	✓ ^a	42.7	38.8	33.3	38.5	40.7	52.9	27.6	35.9	41.0	
DocSeeker [42]	×	✓	–	–	–	–	–	–	–	–	40.1	
Doc-V* [65]	×	✓	–	–	–	–	–	–	–	–	42.1	
MM-Doc-R1 [†] [72]	✓	✓	46.2	38.1	40.8	52.8	35.9	56.0	31.2	68.2	49.7	
Qwen3-VL-8B												
Whole document, via [128]	×	×	–	–	–	–	–	46.9	35.3	25.8	38.5	
DocAgent [34], via [129]	✓	×	–	–	–	–	–	–	–	–	46.6	
MARDoc [129]	✓	×	–	–	–	–	–	–	–	–	52.7	
DocTrace [128]	✓	✓	–	–	–	–	–	53.2	41.1	70.5	52.9	
Ours (4-bit)	✓	×	48.0	41.9	43.7	59.9	41.8	58.9	37.8	71.9	54.5	
LLM judge: binary correctness												
Gold pages + VLM (TLV)	✓	×	55.3	50.9	45.5	48.4	44.8	63.5	38.6	51.1	52.2	
Qwen2.5-VL-7B												
DMap [70]	✓	×	–	–	–	–	–	–	–	–	31.9	
M3DocRAG [33], via [73]	×	×	43.2	40.0	36.1	31.6	42.0	55.5	24.8	8.7	35.5	
MDocAgent [†] [36], via [73]	✓	×	47.0	39.0	47.1	41.0	39.9	59.5	28.6	33.5	43.8	
SimpleDoc [†] [73]	×	×	49.7	42.0	44.6	37.8	42.1	58.7	31.7	62.1	50.4	
Qwen3-VL-8B												
InSight-doc [130]	×	✓	–	–	–	–	–	–	–	–	58.5	
Ours (4-bit)	✓	×	52.7	42.4	46.6	56.3	45.3	62.2	39.1	71.4	56.3	

Table 4.1 MMLongBench-Doc accuracy (%) for published systems using open 7–8B vision-language backbones. Reported values are taken directly from the cited papers; “via” indicates results reproduced by another study rather than reported in the system’s original paper. **OCR**: parser or OCR text is provided as input. **Trained**: task-specific fine-tuning or RL within the pipeline. [†]A text LLM operates alongside the VLM. ^aRetriever only. ^bThe document outline includes visual-element descriptions generated by Qwen3-VL-235B.

and layout questions decline. Multi-page accuracy also remains a clear weakness at 37.8%, below DocTrace’s 41.1%, indicating that the higher overall accuracy does not eliminate the underlying difficulty of integrating evidence across pages.

Adaptive evidence acquisition. Figure 4.2 shows that the pipeline does not uniformly consume its full page budget. The largest group terminates after six pages, the earliest point permitted by the stopping floor $k = 6$, while progressively fewer questions continue deeper into the retrieval ranking and a smaller group reaches the $B = 25$ limit. Reopened trajectories account for part of this longer tail, with some questions requiring one or multiple CONTINUE decisions after an initial synthesis attempt. This differs substantially from the annotated evidence distribution, where most answerable questions contain only one or two gold pages. The gap reflects the practical setting in which the locations of those pages are unknown and additional contextual pages may also contribute useful evidence.

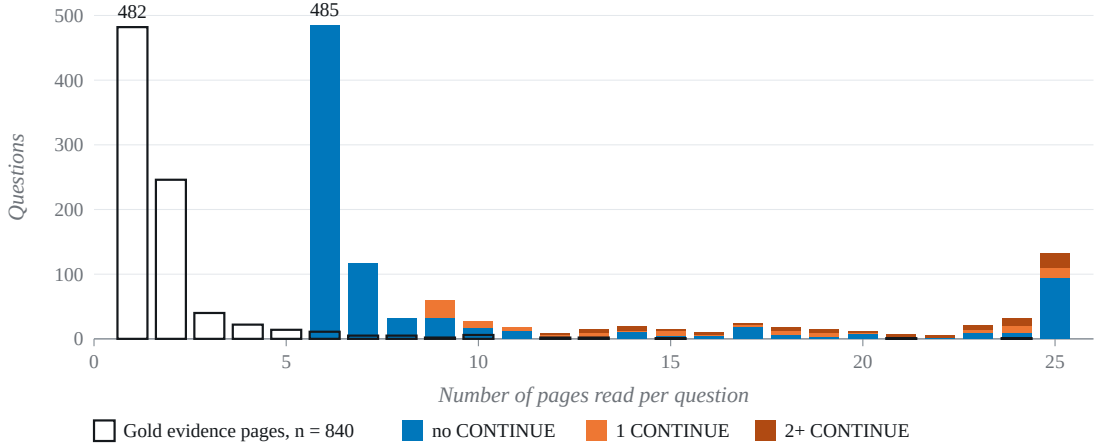


Figure 4.2 Trajectory of our method.

	Turns	Tokens			Latency (s)		Peak VRAM (GiB)
		Input text	Input visual	Output	Prefill	Decode	
reader	11.2	5,209	834	1,008	2.0	70.6	7.3
synthesizer	1.4	9,638	1,742	724	4.0	59.0	8.2
arbiter	1.3	4,988	0	1,083	1.6	72.8	7.1
Per question, mean	13.9	78,215	11,733	13,721	29.7	968.1	8.2
Per question, median	9.0	37,484	7,272	8,081	13.8	544.6	8.0

Table 4.2 Inference cost of the pipeline per agent turn and question.

Inference cost. Table 4.2 quantifies the cost of this adaptive behaviour. The pipeline uses 13.9 turns per question on average, compared with a median of 9.0, with the `reader` accounting for 11.2 turns and the `synthesizer` and `arbiter` invoked only 1.4 and 1.3 times per question respectively. Mean peak memory is 8.2 GiB, remaining comfortably within the 16 GB budget of the deployment GPU, while the average question consumes 78,215 text-input tokens, 11,733 visual tokens, and 13,721 output tokens across its trajectory. The principal cost is latency rather than memory: mean prefill time is only 29.7 s per question compared with 968.1 s of decoding, with a median total of roughly 9.3 minutes. The method therefore trades additional sequential inference for bounded memory use and broader evidence acquisition, with the deployment parameters providing a direct mechanism for reducing trajectory depth where lower latency is required.

4.2 Open Challenges and Future Work

The survey of Chapter 2 identifies five challenges that define the open MP-VRDU frontier. They arise from a consistent pattern across the surveyed systems: progress comes from moving evidence handling out of the backbone, through compression, retrieval, or iterative orchestration, but each shift relocates the bottleneck rather than removing it. Chapter 3 and the system-level validation in §4.1 address parts of the first three challenges directly, but also show that their underlying trade-offs remain unresolved. Three concern system capability, in the evidence pipeline itself, in faithfulness, and in efficiency. Two concern data and evaluation, in supervision integrity and in evaluation scope.

4.2.1 Evidence Pipeline Bottleneck

MP-VRDU systems must reconcile high-resolution per-page reading with document-scale evidence aggregation, an objective that exceeds any MLLM’s compute and context budget [110]. Every architectural family takes a lossy compromise. Compression silently drops fine-grained content and is typically fixed before the question is seen. Retrieval bounds answer accuracy from above, while broader retrieval introduces attention dilution. Iterative retrieval lifts this ceiling only partially, inheriting the recall of every tool it calls. Cross-page structure compounds the problem further, since none of the routes in §2.4.3 scales cleanly. The method in §4.1 shows that separating broad evidence acquisition from selective evidence presentation can mitigate this trade-off, but its remaining weakness on multi-page questions indicates that the bottleneck is not removed. Future work should explore query-conditional compression, retrieval that follows cross-references and logical relations, and learned cross-page structural representations.

4.2.2 Faithfulness and Abstention

Long documents amplify hallucination risk. When retrieval misses the answer-bearing region or compression discards it, current systems still produce confident answers. Few surveyed systems support abstention, calibrated uncertainty, or citation-grounded generation, leaving users without signals to distinguish supported answers from fabrications. The method in §4.1 further addresses this through premise checking and arbitration, improving the system’s ability to reject unsupported answers or acquire additional evidence, but without eliminating the underlying calibration problem. Future work should integrate evidence-grounded decoding, learned abstention against unanswerable or un retrievable questions, and trajectory-level faithfulness audits that verify the cited evidence supports the final answer.

4.2.3 Efficiency and Deployability

Agentic and retrieval-augmented pipelines routinely issue tens of LLM calls per question, with token costs scaling jointly with document length and trajectory length. The surveyed literature reports almost no inference-cost analysis: end-to-end latency, token budgets per query, and accuracy-cost Pareto curves are largely absent, despite their centrality to enterprise deployment. The method in §4.1 demonstrates that an adaptive evidence-management pipeline can remain within a 16 GB memory budget, but does so partly by trading memory pressure for repeated sequential inference and substantially higher latency. Future work should treat compute as a first-class evaluation axis, develop efficient inference techniques specific to long documents such as cross-round caching, speculative decoding, and adaptive depth, and explore compact MP-VRDU models [45] that operate under realistic on-device or privacy constraints.

4.2.4 Supervision Scarcity and Contamination

Per-page annotation does not scale to documents averaging tens of pages, so most systems rely on LVLM-synthesised QA over scraped PDFs [29, 35, 71, 73]. Compounding this, benchmark construction often reuses documents from earlier suites: MMLongBench-Doc draws documents from DUDE, SlideVQA, and ChartQA [110], while MP-DocVQA is built on DocVQA [4, 13]. Together, MLLM-generated QA data [105] and benchmark-reused fine-tuning sets [32, 35] risk contamination: models may train on benchmark-derived documents and be evaluated on overlapping data, sometimes judged by the same MLLM family. Future work should adopt held-out documents, declare cross-suite overlap, and separate generators from evaluators.

4.2.5 Evaluation Methodology and Scope

Beyond contamination, two further evaluation gaps persist. Answer-only accuracy masks retrieval failure and rewards correct answers reached through incorrect reasoning. Newer benchmarks score retrieval and page-localisation alongside VQA-style answers [69, 104], but the practice is not yet standardised, and agentic pipelines produce stochastic trajectories that further complicate reproducibility. Task, language, and domain scope also remain narrow. Real-world MP-VRDU spans summarisation and structured extraction beyond VQA, multilingual layouts beyond English, and domain-specialised settings such as medical, legal, and financial documents, as well as streaming inputs that arrive page-by-page. These dimensions are inherited from SP-VRDU [6] rather than introduced by the multi-page setting and remain underexplored in both regimes. Future work should therefore standardise trajectory-level evaluation with retrieval and grounding metrics, report stochasticity bounds, and broaden coverage along these inherited dimensions.

4.3 Concluding Remarks

The empirical analysis directly addresses two of the open challenges identified above. For faithfulness and abstention, it shows that targeted abstention can substantially reduce unsupported answers, but also introduces false refusals on difficult yet answerable questions, exposing a calibration problem rather than simply a prompting problem. For efficiency and deployability, the experiments show that practical MP-VRDU performance depends on the accuracy–cost trade-offs introduced by representation, retrieval depth, and reasoner capability, and that stronger performance does not necessarily require maximising every component simultaneously. The method introduced in §4.1 then validates these findings at the system level by combining them into a competitive adaptive evidence-management pipeline.

More broadly, this thesis argues that MP-VRDU is fundamentally an evidence-management problem. Progress therefore depends not only on stronger multimodal backbones, but on controlling what evidence is preserved, selected, and presented to the reasoner, and on understanding which stage is responsible when the system fails. Designing and evaluating MP-VRDU systems around these mechanisms provides a clearer basis for improving accuracy, faithfulness, and efficiency than treating the pipeline as a monolithic end-to-end model.

Word Count and Appendices

The main text is approximately 16,500 words, excluding the abstract, acknowledgements, tables, captions, and the appendix. The appendix contains supplementary detail including the full survey methodology, surveyed corpus details, extended result tables and full method details; these not essential for reading this thesis.

Statement of LLM Use

LLM assistants were used to correct grammatical inaccuracies and suggest phrasing revisions in the manuscript and to assist with implementation of the experimental pipeline. All generated code was read and tested by the author and the author takes full responsibility for the content of this thesis. LLM assistants did not participate in research ideation.

Additional Artefacts

- Survey code and surveyed corpus: <https://github.com/LeweiXu/MP-VRDU-Survey>
- Empirical attribution and method code: <https://github.com/LeweiXu/MP-VRDU-Analysis>
- Original project proposal: <https://www.overleaf.com/read/ntdnhjhxkgqt#386d92>

References

- [1] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, “Layoutlm: Pre-training of text and layout for document image understanding,” in *KDD '20: The 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, Virtual Event, CA, USA, August 23-27, 2020*, R. Gupta, Y. Liu, J. Tang, and B. A. Prakash, Eds. ACM, 2020, pp. 1192–1200. [Online]. Available: <https://doi.org/10.1145/3394486.3403172>
- [2] J. Ye, A. Hu, H. Xu, Q. Ye, M. Yan, G. Xu, C. Li, J. Tian, Q. Qian, J. Zhang, Q. Jin, L. He, X. Lin, and F. Huang, “Ureader: Universal ocr-free visually-situated language understanding with multimodal large language model,” in *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, ser. Findings of ACL, H. Bouamor, J. Pino, and K. Bali, Eds., vol. EMNLP 2023. Association for Computational Linguistics, 2023, pp. 2841–2858. [Online]. Available: <https://doi.org/10.18653/v1/2023.findings-emnlp.187>
- [3] A. Hu, H. Xu, J. Ye, M. Yan, L. Zhang, B. Zhang, J. Zhang, Q. Jin, F. Huang, and J. Zhou, “mplug-docowl 1.5: Unified structure learning for ocr-free document understanding,” in *Findings of the Association for Computational Linguistics: EMNLP 2024, Miami, Florida, USA, November 12-16, 2024*, ser. Findings of ACL, Y. Al-Onaizan, M. Bansal, and Y. Chen, Eds., vol. EMNLP 2024. Association for Computational Linguistics, 2024, pp. 3096–3120. [Online]. Available: <https://doi.org/10.18653/v1/2024.findings-emnlp.175>
- [4] M. Mathew, D. Karatzas, and C. V. Jawahar, “Docvqa: A dataset for VQA on document images,” in *IEEE Winter Conference on Applications of Computer Vision, WACV 2021, Waikoloa, HI, USA, January 3-8, 2021*. IEEE, 2021, pp. 2199–2208. [Online]. Available: <https://doi.org/10.1109/WACV48630.2021.00225>
- [5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” *Trans. Assoc. Comput. Linguistics*, vol. 12, pp. 157–173, 2024. [Online]. Available: https://doi.org/10.1162/tacl_a_00638
- [6] Y. Ding, S. Luo, Y. Dai, Y. Jiang, Z. Li, Q. Sun, G. Martin, W. Liu, and Y. Peng, “A survey on mllm-based visually rich document understanding: Methods, challenges, and emerging trends,” in *Findings of the Association for Computational Linguistics, ACL 2026, San Diego, California, United States, July 2-7, 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. Association for Computational Linguistics, 2026, pp. 13 319–13 340. [Online]. Available: <https://doi.org/10.18653/v1/2026.findings-acl.652>
- [7] S. Gao, S. Zhao, X. Jiang, L. Duan, Y. X. Chng, Q. Chen, W. Luo, K. Zhang, J. Bian, and M. Gong, “Scaling beyond context: A survey of multimodal retrieval-augmented generation for document understanding,” in *Proceedings of the 64th Annual Meeting of the Association*

- for Computational Linguistics (Volume 1: Long Papers)*, ACL 2026, San Diego, California, United States, July 2-7, 2026, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. Association for Computational Linguistics, 2026, pp. 4458–4489. [Online]. Available: <https://doi.org/10.18653/v1/2026.acl-long.204>
- [8] J. Zhang, Q. Zhang, B. Wang, L. Ouyang, Z. Wen, Y. Li, K. Chow, C. He, and W. Zhang, “OCR hinders RAG: evaluating the cascading impact of OCR on retrieval-augmented generation,” in *IEEE/CVF International Conference on Computer Vision, ICCV 2025, Honolulu, HI, USA, October 19-25, 2025*. IEEE, 2025, pp. 17 443–17 453. [Online]. Available: <https://doi.org/10.1109/ICCV51701.2025.01620>
- [9] S. Shaikh, “Dissecting agentic RAG: A component ablation for multi-hop QA with a local 7b model,” *CoRR*, vol. abs/2606.21553, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2606.21553>
- [10] Y. Zarrinkia, S. Venkatesh, and A. Thomo, “The reasoning bottleneck in graph-rag: Structured prompting and context compression for multi-hop QA,” *CoRR*, vol. abs/2603.14045, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2603.14045>
- [11] N. Subramani, A. Matton, M. Greaves, and A. Lam, “A survey of deep learning approaches for OCR and document understanding,” *CoRR*, vol. abs/2011.13534, 2020. [Online]. Available: <https://arxiv.org/abs/2011.13534>
- [12] G. Jaume, H. K. Ekenel, and J. Thiran, “FUNSD: A dataset for form understanding in noisy scanned documents,” in *2nd International Workshop on Open Services and Tools for Document Analysis, OST@ICDAR 2019, Sydney, Australia, September 22-25, 2019*. IEEE, 2019, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/ICDARW.2019.10029>
- [13] R. Tito, D. Karatzas, and E. Valveny, “Hierarchical multimodal transformers for multipage docvqa,” *Pattern Recognit.*, vol. 144, p. 109834, 2023. [Online]. Available: <https://doi.org/10.1016/j.patcog.2023.109834>
- [14] OpenAI, “Projects in chatgpt,” OpenAI Help Center, 2026, accessed: 2026-09-25. [Online]. Available: <https://help.openai.com/en/articles/10169521-projects-in-chatgpt>
- [15] —, “Chatgpt is now a partner for your most ambitious work,” OpenAI, Jul. 2026, accessed: 2026-09-25. [Online]. Available: <https://openai.com/index/chatgpt-for-your-most-ambitious-work/>
- [16] Anthropic, “Retrieval augmented generation (rag) for projects,” Claude Help Center, 2026, accessed: 2026-09-25. [Online]. Available: <https://support.claude.com/en/articles/11473015-retrieval-augmented-generation-rag-for-projects>
- [17] A. Lassoued, M. A. Souibgui, and Y. Kessentini, “Orca: Orchestrated reasoning with collaborative agents for document visual question answering,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.02438>
- [18] M. Mathew, V. Bagal, R. Tito, D. Karatzas, E. Valveny, and C. V. Jawahar, “Infographicvqa,” in *IEEE/CVF Winter Conference on Applications of Computer Vision, WACV 2022*,

- Waikoloa, HI, USA, January 3-8, 2022. IEEE, 2022, pp. 2582–2591. [Online]. Available: <https://doi.org/10.1109/WACV51458.2022.00264>
- [19] A. Masry, D. X. Long, J. Q. Tan, S. R. Joty, and E. Hoque, “Chartqa: A benchmark for question answering about charts with visual and logical reasoning,” in *Findings of the Association for Computational Linguistics: ACL 2022, Dublin, Ireland, May 22-27, 2022*, ser. Findings of ACL, S. Muresan, P. Nakov, and A. Villavicencio, Eds., vol. ACL 2022. Association for Computational Linguistics, 2022, pp. 2263–2279. [Online]. Available: <https://doi.org/10.18653/v1/2022.findings-acl.177>
- [20] D. Wang, N. Raman, M. Sibue, Z. Ma, P. Babkin, S. Kaur, Y. Pei, A. Nourbakhsh, and X. Liu, “Docllm: A layout-aware generative language model for multimodal document understanding,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2024, Bangkok, Thailand, August 11-16, 2024*, L. Ku, A. Martins, and V. Srikumar, Eds. Association for Computational Linguistics, 2024, pp. 8529–8548. [Online]. Available: <https://doi.org/10.18653/v1/2024.acl-long.463>
- [21] Y. Huang, T. Lv, L. Cui, Y. Lu, and F. Wei, “Layoutlmv3: Pre-training for document AI with unified text and image masking,” in *MM ’22: The 30th ACM International Conference on Multimedia, Lisboa, Portugal, October 10 - 14, 2022*, J. Magalhães, A. D. Bimbo, S. Satoh, N. Sebe, X. Alameda-Pineda, Q. Jin, V. Oria, and L. Toni, Eds. ACM, 2022, pp. 4083–4091. [Online]. Available: <https://doi.org/10.1145/3503161.3548112>
- [22] C. Luo, Y. Shen, Z. Zhu, Q. Zheng, Z. Yu, and C. Yao, “Layoutllm: Layout instruction tuning with large language models for document understanding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*. IEEE, 2024, pp. 15 630–15 640. [Online]. Available: <https://doi.org/10.1109/CVPR52733.2024.01480>
- [23] Y. Wang, W. Zhou, H. Feng, K. Zhou, and H. Li, “Towards improving document understanding: An exploration on text-grounding via mllms,” *CoRR*, vol. abs/2311.13194, 2023. [Online]. Available: <https://doi.org/10.48550/arXiv.2311.13194>
- [24] A. Hu, H. Xu, L. Zhang, J. Ye, M. Yan, J. Zhang, Q. Jin, F. Huang, and J. Zhou, “mplug-docowl2: High-resolution compressing for ocr-free multi-page document understanding,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, 2025, pp. 5817–5834. [Online]. Available: <https://doi.org/10.18653/v1/2025.acl-long.291>
- [25] Y. Zhu, Y. Zhang, D. Liu, C. Xie, Z. Xiong, B. Zheng, and S. Guo, “Enhancing document understanding with group position embedding: A novel approach to incorporate layout information,” in *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. [Online]. Available: <https://openreview.net/forum?id=Dj9a4zQsSI>
- [26] Z. Wang, T. Guan, P. Fu, C. Duan, Q. Jiang, Z. Guo, S. Guo, J. Luo, W. Shen, and X. Yang, “Marten: Visual question answering with mask generation for multi-modal document understanding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*. Computer Vision Foundation / IEEE, 2025, pp.

- 14 460–14 471. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Wang_Marten_Visual_Question_Answering_with_Mask_Generation_for_Multi-modal_Document_CVPR_2025_paper.html
- [27] V. Perot, K. Kang, F. Luisier, G. Su, X. Sun, R. S. Boppana, Z. Wang, Z. Wang, J. Mu, H. Zhang, C. Lee, and N. Hua, “LMDX: language model-based document information extraction and localization,” in *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, ser. Findings of ACL, L. Ku, A. Martins, and V. Srikumar, Eds., vol. ACL 2024. Association for Computational Linguistics, 2024, pp. 15 140–15 168. [Online]. Available: <https://doi.org/10.18653/v1/2024.findings-acl.899>
- [28] Z. Zhu, C. Luo, Z. Shao, F. Gao, H. Xing, Q. Zheng, and J. Zhang, “A simple yet effective layout token in large language models for document understanding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*. Computer Vision Foundation / IEEE, 2025, pp. 14 472–14 482. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Zhu_A_Simple_yet_Effective_Layout_Token_in_Large_Language_Models_CVPR_2025_paper.html
- [29] R. Tanaka, T. Iki, K. Nishida, K. Saito, and J. Suzuki, “Instructdoc: A dataset for zero-shot generalization of visual document understanding with instructions,” in *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*, M. J. Wooldridge, J. G. Dy, and S. Natarajan, Eds. AAAI Press, 2024, pp. 19 071–19 079. [Online]. Available: <https://doi.org/10.1609/aaai.v38i17.29874>
- [30] Y. Yu, M. Liao, J. Zhang, and J. Wu, “Texthawk2: A large vision-language model excels in bilingual OCR and grounding with 16x fewer tokens,” *CoRR*, vol. abs/2410.05261, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2410.05261>
- [31] L. Xu, Y. Ding, Z. Xu, S. Luo, D. Liu, and W. Liu, “Managing evidence at document scale: A survey of multi-page visually rich document understanding,” 2026.
- [32] Y. Duan, Z. Chen, Y. Hu, W. Wang, S. Ye, B. Shi, L. Lu, Q. Hou, T. Lu, H. Li, J. Dai, and W. Wang, “Docopilot: Improving multimodal models for document-level understanding,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*. Computer Vision Foundation / IEEE, 2025, pp. 4026–4037. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Duan_Docopilot_Improving_Multimodal_Models_for_Document-Level_Understanding_CVPR_2025_paper.html
- [33] J. Cho, D. Mahata, O. Irsoy, Y. He, and M. Bansal, “M3docrag: Multi-modal retrieval is what you need for multi-page multi-document understanding,” *CoRR*, vol. abs/2411.04952, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2411.04952>
- [34] L. Sun, L. He, S. Jia, Y. He, and C. You, “Docagent: An agentic framework for multi-modal long-context document understanding,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 17 701–17 716. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.893>

- [35] J. Xiong, Y. Wang, W. Zhao, C. Liu, B. Yin, W. Zhou, and H. Li, “Docr1: Evidence page-guided GRPO for multi-page document understanding,” in *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, S. Koenig, C. Jenkins, and M. E. Taylor, Eds. AAAI Press, 2026, pp. 11 178–11 186. [Online]. Available: <https://doi.org/10.1609/aaai.v40i13.38097>
- [36] S. Han, P. Xia, R. Zhang, T. Sun, Y. Li, H. Zhu, and H. Yao, “Mdocagent: A multi-modal multi-agent framework for document understanding,” *CoRR*, vol. abs/2503.13964, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2503.13964>
- [37] Y. Ding, S. C. Han, J. Lee, and E. H. Hovy, “Deep learning based visually rich document content understanding: a survey,” *Artif. Intell. Rev.*, vol. 59, no. 4, p. 114, 2026. [Online]. Available: <https://doi.org/10.1007/s10462-025-11477-3>
- [38] H. Gbada, K. Kalti, and M. A. Mahjoub, “Deep learning approaches for information extraction from visually rich documents: datasets, challenges and methods,” *Int. J. Document Anal. Recognit.*, vol. 28, no. 1, pp. 121–142, 2025. [Online]. Available: <https://doi.org/10.1007/s10032-024-00493-8>
- [39] L. Borchmann, M. Pietruszka, W. Jaskowski, D. Jurkiewicz, P. Halama, P. Józiak, L. Garncarek, P. Liskowski, K. Szyndler, A. Gretkowski, J. Oltusek, G. Nowakowska, A. Zawlocki, L. Duhr, P. Dyda, and M. Turski, “Arctic-tilt. business document understanding at sub-billion scale,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, G. Rehm and Y. Li, Eds. Association for Computational Linguistics, 2025, pp. 264–283. [Online]. Available: <https://doi.org/10.18653/v1/2025.acl-industry.20>
- [40] T. Blau, S. Fogel, R. Ronen, A. Golts, S. Tsiper, E. Ben-Avraham, A. Aberdam, R. Ganz, and R. Litman, “GRAM: global reasoning for multi-page VQA,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*. IEEE, 2024, pp. 15 598–15 607. [Online]. Available: <https://doi.org/10.1109/CVPR52733.2024.01477>
- [41] Q. Dong, L. Kang, and D. Karatzas, “Multi-page document VQA with recurrent memory transformer,” in *Document Analysis Systems - 16th IAPR International Workshop, DAS 2024, Athens, Greece, August 30-31, 2024, Proceedings*, ser. Lecture Notes in Computer Science, G. Sfikas and G. Retsinas, Eds., vol. 14994. Springer, 2024, pp. 57–70. [Online]. Available: https://doi.org/10.1007/978-3-031-70442-0_4
- [42] H. Yan, Y. Liu, X. Liu, Y. Zhang, M. Liao, J. Wu, W. Chen, and X. Bai, “Docseeker: Structured visual reasoning with evidence grounding for long document understanding,” *CoRR*, vol. abs/2604.12812, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2604.12812>
- [43] M. Jia, W. Yu, K. Ma, T. Fang, Z. Zhang, S. Ouyang, H. Zhang, D. Yu, and M. Jiang, “Leopard: A vision language model for text-rich multi- image tasks,” *Trans. Mach. Learn. Res.*, vol. 2025, 2025. [Online]. Available: <https://openreview.net/forum?id=R2rasAEPVi>
- [44] Y. Shi, J. Wang, Z. Shan, D. Peng, Z. Lin, and L. Jin, “Urag: Unified retrieval and generation in multimodal llms for efficient long document understanding,” in *Fortieth AAAI Conference*

- on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, S. Koenig, C. Jenkins, and M. E. Taylor, Eds. AAAI Press, 2026, pp. 25 357–25 365. [Online]. Available: <https://doi.org/10.1609/aaai.v40i30.39729>
- [45] T. Hannan, D. Mallios, P. Pathak, F. Sardari, T. Seidl, G. Bertasius, M. Fayyaz, and S. Sengupta, “Docslm: A small vision-language model for long multimodal document understanding,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 9337–9346.
- [46] J. Guo, C. Huang, H. Tao, Z. An, G. Ke, and X. Fang, “End-to-end document understanding via chain-of-reading,” 2026. [Online]. Available: <https://openreview.net/forum?id=6YXMyPrDEN>
- [47] Z. Li, Y. Li, L. Kang, D. Karatzas, and W. Ma, “AVIR: adaptive visual in-document retrieval for efficient multi-page document question answering,” *CoRR*, vol. abs/2601.11976, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2601.11976>
- [48] B. Qian, K. Deng, Y. Wu, J. Zhang, J. Sun, Y. Xue, and C. Fan, “Accurate multi-page document retrieval by effectively fusing context information across pages,” *Electronics*, vol. 15, no. 7, p. 1353, 2026.
- [49] J. Zhang, Q. Fan, Y. Yu, and Y. Zhang, “DREAM: integrating hierarchical multimodal retrieval with multi-page multimodal language model for documents VQA,” in *Proceedings of the 33rd ACM International Conference on Multimedia, MM 2025, Dublin, Ireland, October 27-31, 2025*, C. Gurrin, K. Schoeffmann, M. Zhang, L. Rossetto, S. Rudinac, D. Dang-Nguyen, W. Cheng, P. Chen, and J. Benois-Pineau, Eds. ACM, 2025, pp. 4213–4221. [Online]. Available: <https://doi.org/10.1145/3746027.3755357>
- [50] M. Xu, J. Dong, J. Hou, Z. Wang, S. Li, Z. Gao, R. Zhong, and H. Cai, “MM-R5: multimodal reasoning-enhanced reranker via reinforcement learning for document retrieval,” *CoRR*, vol. abs/2506.12364, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2506.12364>
- [51] X. Wu, Y. Tan, N. Hou, R. Zhang, and H. Cheng, “Molorag: Bootstrapping document understanding via multi-modal logic-aware retrieval,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 14 024–14 045. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.708>
- [52] L. Kang, R. Tito, E. Valveny, and D. Karatzas, “Multi-page document visual question answering using self-attention scoring mechanism,” in *Document Analysis and Recognition - ICDAR 2024 - 18th International Conference, Athens, Greece, August 30 - September 4, 2024, Proceedings, Part VI*, ser. Lecture Notes in Computer Science, E. H. B. Smith, M. Liwicki, and L. Peng, Eds., vol. 14809. Springer, 2024, pp. 219–232. [Online]. Available: https://doi.org/10.1007/978-3-031-70552-6_13
- [53] K. Liu, Z. Chen, M. Li, J. Tang, D. Yang, and L. Zhang, “Resolving evidence sparsity: Agentic context engineering for long-document understanding,” *CoRR*, vol. abs/2511.22850, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2511.22850>

- [54] J. Chen, R. Zhang, Y. Zhou, T. Yu, F. Dernoncourt, J. Gu, R. A. Rossi, C. Chen, and T. Sun, “SV-RAG: lora-contextualizing adaptation of mllms for long document understanding,” in *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. [Online]. Available: <https://openreview.net/forum?id=FDaHjwInXO>
- [55] R. Tanaka, T. Iki, T. Hasegawa, K. Nishida, K. Saito, and J. Suzuki, “Vdocrag: Retrieval-augmented generation over visually-rich documents,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2025, Nashville, TN, USA, June 11-15, 2025*. Computer Vision Foundation / IEEE, 2025, pp. 24 827–24 837. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2025/html/Tanaka_VDocRAG_Retrieval-Augmented_Generation_over_Visually-Rich_Documents_CVPR_2025_paper.html
- [56] J. Zhang, Y. Yu, and Y. Zhang, “CREAM: coarse-to-fine retrieval and multi-modal efficient tuning for document VQA,” in *Proceedings of the 32nd ACM International Conference on Multimedia, MM 2024, Melbourne, VIC, Australia, 28 October 2024 - 1 November 2024*, J. Cai, M. S. Kankanhalli, B. Prabhakaran, S. Boll, R. Subramanian, L. Zheng, V. K. Singh, P. César, L. Xie, and D. Xu, Eds. ACM, 2024, pp. 925–934. [Online]. Available: <https://doi.org/10.1145/3664647.3680750>
- [57] Y. Wang, N. Lipka, R. A. Rossi, A. F. Siu, R. Zhang, and T. Derr, “Knowledge graph prompting for multi-document question answering,” in *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*, M. J. Wooldridge, J. G. Dy, and S. Natarajan, Eds. AAAI Press, 2024, pp. 19 206–19 214. [Online]. Available: <https://doi.org/10.1609/aaai.v38i17.29889>
- [58] Z. Gong, C. Mai, and Y. Huang, “Mhier-rag: Multi-modal RAG for visual-rich document question-answering via hierarchical and multi-granularity reasoning,” in *Advanced Intelligent Computing Technology and Applications - 22nd International Conference on Intelligent Computing, ICIC 2026, Toronto, ON, Canada, July 22-26, 2026, Proceedings, Part XXVIII*, ser. Lecture Notes in Computer Science, D. Huang, B. Li, Q. Zhang, and W. Bao, Eds., vol. 16669. Springer, 2026, pp. 285–297. [Online]. Available: https://doi.org/10.1007/978-981-92-3492-9_24
- [59] Y. Zhang and Y. Wu, “Mldocrag: Multimodal long-context document retrieval augmented generation,” *CoRR*, vol. abs/2602.10271, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2602.10271>
- [60] J. Shin, C. Park, J. Park, J. Seo, and H. Lim, “Multidocfusion: Hierarchical and multimodal chunking pipeline for enhanced RAG on long industrial documents,” *CoRR*, vol. abs/2604.12352, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2604.12352>
- [61] T. Duan, M. Hu, L. Xue, C. Liu, G. Geng, W. Luo, and Z. Luo, “M2rag: A multi-agent and multimodal fusion framework for retrieval-augmented document qa,” in *International Conference on Neural Information Processing*. Springer, 2025, pp. 507–520.
- [62] X. Xie, H. Yan, L. Yin, Y. Liu, J. Ding, M. Liao, Y. Liu, W. Chen, and X. Bai, “Pdf-wukong: A large multimodal model for efficient long PDF reading with end-to-end

- sparse sampling,” *Int. J. Comput. Vis.*, vol. 134, no. 7, p. 324, 2026. [Online]. Available: <https://doi.org/10.1007/s11263-026-02860-w>
- [63] E. López, A. Llabrés, and E. Valveny, “Enhancing document VQA models via retrieval-augmented generation,” in *Document Analysis and Recognition - ICDAR 2025 Workshops - Wuhan, China, September 20-21, 2025, Proceedings, Part I*, ser. Lecture Notes in Computer Science, L. Jin, R. Zanibbi, and V. Eglin, Eds., vol. 16225. Springer, 2025, pp. 92–107. [Online]. Available: https://doi.org/10.1007/978-3-032-09368-4_6
- [64] M. Suri, P. Mathur, F. Dernoncourt, K. Goswami, R. A. Rossi, and D. Manocha, “Visdom: Multi-document QA with visually rich elements using multimodal retrieval-augmented generation,” in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2025 - Volume 1: Long Papers, Albuquerque, New Mexico, USA, April 29 - May 4, 2025*, L. Chiruzzo, A. Ritter, and L. Wang, Eds. Association for Computational Linguistics, 2025, pp. 6088–6109. [Online]. Available: <https://doi.org/10.18653/v1/2025.naacl-long.310>
- [65] Y. Zheng, P. Fu, H. Li, Z. Wang, Y. Zhang, W. Ruan, X. Zhang, Z. Wei, Z. Luo, J. Luan, W. Chen, and X. Bai, “Doc-v*: Coarse-to-fine interactive visual reasoning for multi-page document VQA,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. Association for Computational Linguistics, 2026, pp. 45 901–45 923. [Online]. Available: <https://doi.org/10.18653/v1/2026.acl-long.2129>
- [66] J. Wu, Y. Xia, T. Yu, X. Chen, S. S. Harsha, A. V. Maharaj, R. Zhang, V. S. Bursztyrn, S. Kim, R. A. Rossi, J. J. McAuley, Y. Li, and R. Sinha, “Doc-react: Multi-page heterogeneous document question-answering,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, 2025, pp. 67–78. [Online]. Available: <https://doi.org/10.18653/v1/2025.acl-short.6>
- [67] X. Yu, C. Xu, Z. Chen, Y. Zhang, S. Lu, C. Yang, J. Zhang, S. Yan, and X. Hu, “Visual document understanding and reasoning: A multi-agent collaboration framework with agent-wise adaptive test-time scaling,” *arXiv preprint arXiv:2508.03404*, 2025.
- [68] Q. Wang, R. Ding, Y. Zeng, Z. Chen, L. Chen, S. Wang, P. Xie, F. Huang, and F. Zhao, “VRAG-RL: empower vision-perception-based RAG for visually rich information understanding via iterative reasoning with reinforcement learning,” in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*, D. Belgrave, C. Zhang, L. N. Montoya, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, N. Chen, I. V. M. Ruíz, and A. Loaiza-Bonilla, Eds., 2025. [Online]. Available: http://papers.nips.cc/paper_files/paper/2025/hash/529b876b5d87f997e98ed925a3ce9f01-Abstract-Conference.html
- [69] D. Zhu, R. Meng, J. Chen, S. Li, T. Pfister, and J. Yoon, “Doclens: A tool-augmented multi-agent framework for long visual document understanding,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San*

- Diego, California, United States, July 2-7, 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. Association for Computational Linguistics, 2026, pp. 26 804–26 829. [Online]. Available: <https://doi.org/10.18653/v1/2026.acl-long.1234>
- [70] S. Fu, Y. Zhang, Y. Xiang, X. Du, and J. Tang, “DMAP: human-aligned structural document map for multimodal document understanding,” in *Proceedings of the ACM Web Conference 2026, WWW 2026, Dubai, United Arab Emirates, originally scheduled for April 13-17, 2026, rescheduled for June 29 - July 3, 2026*, H. Hacid, Y. Maarek, F. Bonchi, I. Guy, and E. Yilmaz, Eds. ACM, 2026, pp. 2037–2048. [Online]. Available: <https://doi.org/10.1145/3774904.3792229>
- [71] Q. Zhang, X. Lv, J. Wu, B. Li, Z. Tao, G. Yan, H. Zhang, B. Wang, J. Xu, H. Mi, and W. Zhang, “Docdancer: Towards agentic document-grounded information seeking,” *CoRR*, vol. abs/2601.05163, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2601.05163>
- [72] J. Lin, K. Hu, B. Wang, Y. Zhou, Z. Xi, H. Guo, S. Liu, J. Wang, S. Dou, E. Zhou, H. Yan, Z. Han, T. Gui, Q. Zhang, and X. Huang, “Mm-doc-rl: Training agents for long document visual question answering through multi-turn reinforcement learning,” in *Findings of the Association for Computational Linguistics, ACL 2026, San Diego, California, United States, July 2-7, 2026*, M. Liakata, V. P. Moreira, J. Zhang, and D. Jurgens, Eds. Association for Computational Linguistics, 2026, pp. 29 770–29 783. [Online]. Available: <https://doi.org/10.18653/v1/2026.findings-acl.1488>
- [73] C. Jain, Y. Wu, Y. Zeng, J. Liu, S. Dai, Z. Shao, Q. Wu, and H. Wang, “Simpledoc: Multi-modal document understanding with dual-cue page retrieval and iterative refinement,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 28 410–28 427. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.1443>
- [74] Q. Wang, R. Ding, Z. Chen, W. Wu, S. Wang, P. Xie, and F. Zhao, “Vidorag: Visual document retrieval-augmented generation via dynamic iterative reasoning agents,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 9113–9134. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.464>
- [75] A. Plaat, M. J. van Duijn, N. van Stein, M. Preuss, P. van der Putten, and K. J. Batenburg, “Agentic large language models, a survey,” *J. Artif. Intell. Res.*, vol. 84, 2025. [Online]. Available: <https://doi.org/10.1613/jair.1.18675>
- [76] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J. Wen, “A survey on large language model based autonomous agents,” *Frontiers Comput. Sci.*, vol. 18, no. 6, p. 186345, 2024. [Online]. Available: <https://doi.org/10.1007/s11704-024-40231-1>
- [77] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. R. Narasimhan, and Y. Cao, “React: Synergizing reasoning and acting in language models,” in *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. [Online]. Available: https://openreview.net/forum?id=WE_vluYUL-X

- [78] M. Faysse, H. Sibille, T. Wu, B. Omrani, G. Viaud, C. Hudelot, and P. Colombo, “Colpali: Efficient document retrieval with vision language models,” in *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. [Online]. Available: <https://openreview.net/forum?id=ogjBpZ8uSi>
- [79] O. Khattab and M. Zaharia, “Colbert: Efficient and effective passage search via contextualized late interaction over BERT,” in *Proceedings of the 43rd International ACM SIGIR conference on research and development in Information Retrieval, SIGIR 2020, Virtual Event, China, July 25-30, 2020*, J. X. Huang, Y. Chang, X. Cheng, J. Kamps, V. Murdock, J. Wen, and Y. Liu, Eds. ACM, 2020, pp. 39–48. [Online]. Available: <https://doi.org/10.1145/3397271.3401075>
- [80] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou, “Chain-of-thought prompting elicits reasoning in large language models,” in *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, Eds., 2022. [Online]. Available: http://papers.nips.cc/paper_files/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html
- [81] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, “Reflexion: language agents with verbal reinforcement learning,” in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html
- [82] M. Besta, F. Scheidl, L. Gianinazzi, G. Kwasniewski, S. Klaiman, J. Müller, and T. Hoefler, “Demystifying higher-order graph neural networks,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 3, pp. 2544–2565, 2026. [Online]. Available: <https://doi.org/10.1109/TPAMI.2025.3637114>
- [83] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, and D. Zhou, “Self-consistency improves chain of thought reasoning in language models,” in *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023. [Online]. Available: <https://openreview.net/forum?id=1PL1NIMMrw>
- [84] S. Yao, D. Yu, J. Zhao, I. Shafran, T. Griffiths, Y. Cao, and K. Narasimhan, “Tree of thoughts: Deliberate problem solving with large language models,” in *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., 2023. [Online]. Available: http://papers.nips.cc/paper_files/paper/2023/hash/271db9922b8d1f4dd7aaef84ed5ac703-Abstract-Conference.html
- [85] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “Lora: Low-rank adaptation of large language models,” in *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022. [Online]. Available: <https://openreview.net/forum?id=nZeVKeeFYf9>

- [86] N. Houlsby, A. Giurgiu, S. Jastrzebski, B. Morrone, Q. de Laroussilhe, A. Gesmundo, M. Attariyan, and S. Gelly, “Parameter-efficient transfer learning for NLP,” in *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, ser. Proceedings of Machine Learning Research, K. Chaudhuri and R. Salakhutdinov, Eds., vol. 97. PMLR, 2019, pp. 2790–2799. [Online]. Available: <http://proceedings.mlr.press/v97/houlsby19a.html>
- [87] L. Nan, C. Hsieh, Z. Mao, X. V. Lin, N. Verma, R. Zhang, W. Kryscinski, H. Schoelkopf, R. Kong, X. Tang, M. Mutuma, B. Rosand, I. Trindade, R. Bandaru, J. Cunningham, C. Xiong, and D. R. Radev, “Fetaqa: Free-form table question answering,” *Trans. Assoc. Comput. Linguistics*, vol. 10, pp. 35–49, 2022. [Online]. Available: https://doi.org/10.1162/tac1_a_00446
- [88] Y. Ding, S. Luo, H. Chung, and S. C. Han, “PDF-VQA: A new dataset for real-world VQA on PDF documents,” in *Machine Learning and Knowledge Discovery in Databases: Applied Data Science and Demo Track - European Conference, ECML PKDD 2023, Turin, Italy, September 18-22, 2023, Proceedings, Part VI*, ser. Lecture Notes in Computer Science, G. D. F. Morales, C. Perlich, N. Ruchansky, N. Kourtellis, E. Baralis, and F. Bonchi, Eds., vol. 14174. Springer, 2023, pp. 585–601. [Online]. Available: https://doi.org/10.1007/978-3-031-43427-3_35
- [89] W. Chen, H. Wang, J. Chen, Y. Zhang, H. Wang, S. Li, X. Zhou, and W. Y. Wang, “Tabfact: A large-scale dataset for table-based fact verification,” in *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net, 2020. [Online]. Available: <https://openreview.net/forum?id=rkeJRhNYDH>
- [90] A. Singh, V. Natarajan, M. Shah, Y. Jiang, X. Chen, D. Batra, D. Parikh, and M. Rohrbach, “Towards VQA models that can read,” in *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*. Computer Vision Foundation / IEEE, 2019, pp. 8317–8326. [Online]. Available: http://openaccess.thecvf.com/content_CVPR_2019/html/Singh_Towards_VQA_Models_That_Can_Read_CVPR_2019_paper.html
- [91] R. Tanaka, K. Nishida, and S. Yoshida, “Visualmrc: Machine reading comprehension on document images,” in *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021, Thirty-Third Conference on Innovative Applications of Artificial Intelligence, IAAI 2021, The Eleventh Symposium on Educational Advances in Artificial Intelligence, EAAI 2021, Virtual Event, February 2-9, 2021*. AAAI Press, 2021, pp. 13 878–13 888. [Online]. Available: <https://doi.org/10.1609/aaai.v35i15.17635>
- [92] P. Pasupat and P. Liang, “Compositional semantic parsing on semi-structured tables,” in *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing of the Asian Federation of Natural Language Processing, ACL 2015, July 26-31, 2015, Beijing, China, Volume 1: Long Papers*. The Association for Computer Linguistics, 2015, pp. 1470–1480. [Online]. Available: <https://doi.org/10.3115/v1/p15-1142>
- [93] R. Xia, S. Mao, X. Yan, H. Zhou, B. Zhang, H. Peng, J. Pi, D. Fu, W. Wu, H. Ye, S. Feng, B. Wang, C. Xu, C. He, P. Cai, M. Dou, B. Shi, S. Zhou, Y. Wang, B. Wang, J. Yan, F. Wu, and Y. Qiao, “Docgenome: An open large-scale scientific document benchmark for training and testing multi-modal large language models,” *CoRR*, vol. abs/2406.11633, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2406.11633>

- [94] L. Qi, S. Lv, H. Li, J. Liu, Y. Zhang, Q. She, H. Wu, H. Wang, and T. Liu, “Dureader_{vis}: A chinese dataset for open-domain document visual question answering,” in *Findings of the Association for Computational Linguistics: ACL 2022, Dublin, Ireland, May 22-27, 2022*, ser. Findings of ACL, S. Muresan, P. Nakov, and A. Villavicencio, Eds., vol. ACL 2022. Association for Computational Linguistics, 2022, pp. 1338–1351. [Online]. Available: <https://doi.org/10.18653/v1/2022.findings-acl.105>
- [95] T. Stanislawek, F. Gralinski, A. Wróblewska, D. Lipinski, A. Kaliska, P. Rosalska, B. Topolski, and P. Biecek, “Kleister: Key information extraction datasets involving long documents with complex layouts,” in *16th International Conference on Document Analysis and Recognition, ICDAR 2021, Lausanne, Switzerland, September 5-10, 2021, Proceedings, Part I*, ser. Lecture Notes in Computer Science, J. Lladós, D. Lopresti, and S. Uchida, Eds., vol. 12821. Springer, 2021, pp. 564–579. [Online]. Available: https://doi.org/10.1007/978-3-030-86549-8_36
- [96] Y. K. Chia, L. Cheng, H. P. Chan, M. Song, C. Liu, M. Aljunied, S. Poria, and L. Bing, “M-longdoc: A benchmark for multimodal super-long document understanding and A retrieval-aware tuning framework,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 9233–9250. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.469>
- [97] Y. Ding, K. Ren, J. Huang, S. Luo, and S. C. Han, “MMVQA: A comprehensive dataset for investigating multipage multimodal information retrieval in pdf-based visual question answering,” in *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI 2024, Jeju, South Korea, August 3-9, 2024*. ijcai.org, 2024, pp. 6243–6251. [Online]. Available: <https://www.ijcai.org/proceedings/2024/690>
- [98] Y. Zhao, Y. Li, C. Li, and R. Zhang, “MultihierTT: Numerical reasoning over multi hierarchical tabular and textual data,” in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2022, Dublin, Ireland, May 22-27, 2022*, S. Muresan, P. Nakov, and A. Villavicencio, Eds. Association for Computational Linguistics, 2022, pp. 6588–6600. [Online]. Available: <https://doi.org/10.18653/v1/2022.acl-long.454>
- [99] P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner, “A dataset of information-seeking questions and answers anchored in research papers,” in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2021, Online, June 6-11, 2021*, K. Toutanova, A. Rumshisky, L. Zettlemoyer, D. Hakkani-Tür, I. Beltagy, S. Bethard, R. Cotterell, T. Chakraborty, and Y. Zhou, Eds. Association for Computational Linguistics, 2021, pp. 4599–4610. [Online]. Available: <https://doi.org/10.18653/v1/2021.naacl-main.365>
- [100] R. Tanaka, K. Nishida, K. Nishida, T. Hasegawa, I. Saito, and K. Saito, “Slidevqa: A dataset for document visual question answering on multiple images,” in *Thirty-Seventh AAAI Conference on Artificial Intelligence, AAAI 2023, Thirty-Fifth Conference on Innovative Applications of Artificial Intelligence, IAAI 2023, Thirteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2023, Washington, DC, USA, February 7-14, 2023*, B. Williams,

- Y. Chen, and J. Neville, Eds. AAAI Press, 2023, pp. 13 636–13 645. [Online]. Available: <https://doi.org/10.1609/aaai.v37i11.26598>
- [101] B. Xiang, S. C. Han, and Y. Ding, “BRIDGE: benchmark for multi-hop reasoning in long multimodal documents with grounded evidence,” *CoRR*, vol. abs/2603.07931, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2603.07931>
- [102] A. Zou, W. Yu, H. Zhang, K. Ma, D. Cai, Z. Zhang, H. Zhao, and D. Yu, “DOCBENCH: A benchmark for evaluating llm-based document reading systems,” *CoRR*, vol. abs/2407.10701, 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2407.10701>
- [103] J. V. Landeghem, R. Powalski, R. Tito, D. Jurkiewicz, M. B. Blaschko, L. Borchmann, M. Coustaty, S. Moens, M. Pietruszka, B. Anckaert, T. Stanislawek, P. Józiak, and E. Valveny, “Document understanding dataset and evaluation (DUDE),” in *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*. IEEE, 2023, pp. 19 471–19 483. [Online]. Available: <https://doi.org/10.1109/ICCV51070.2023.01789>
- [104] Y. Hui, Y. Lu, and H. Zhang, “UDA: A benchmark suite for retrieval augmented generation in real-world document analysis,” in *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/7c06759d1a8567f087b02e8589454917-Abstr-act-Datasets_and_Benchmarks_Track.html
- [105] C. Deng, J. Yuan, P. Bu, P. Wang, Z. Li, J. Xu, X. Li, Y. Gao, J. Song, B. Zheng, and C. Liu, “Longdocurl: a comprehensive multimodal long document benchmark integrating understanding, reasoning, and locating,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, W. Che, J. Nabende, E. Shutova, and M. T. Pilehvar, Eds. Association for Computational Linguistics, 2025, pp. 1135–1159. [Online]. Available: <https://doi.org/10.18653/v1/2025.acl-long.57>
- [106] J. Cho, D. Mahata, O. Irsoy, Y. He, and M. Bansal, “M3docvqa: Multi-modal multi-page multi-document understanding,” in *IEEE/CVF International Conference on Computer Vision, ICCV 2025 - Workshops, Honolulu, HI, USA, October 19-20, 2025*. IEEE, 2025, pp. 6237–6247. [Online]. Available: <https://doi.org/10.1109/ICCVW69036.2025.00649>
- [107] W. Wang, S. Zhang, Y. Ren, Y. Duan, T. Li, S. Liu, M. Hu, Z. Chen, K. Zhang, L. Lu, X. Zhu, P. Luo, Y. Qiao, J. Dai, W. Shao, and W. Wang, “Needle in A multimodal haystack,” in *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/24a8968affe71ffe4067d022b9d16566-Abstract-Datasets_and_Benchmarks_Track.html
- [108] K. Dong, Y. Chang, D. Deik, D. Li, R. Tang, and Y. Liu, “Mmdocir: Benchmarking multimodal retrieval for long documents,” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China,*

- November 4-9, 2025, C. Christodoulopoulos, T. Chakraborty, C. Rose, and V. Peng, Eds. Association for Computational Linguistics, 2025, pp. 30971–31005. [Online]. Available: <https://doi.org/10.18653/v1/2025.emnlp-main.1576>
- [109] Z. Wang, W. Yu, X. Ren, J. Zhang, Y. Zhao, R. Saxena, L. Cheng, G. Y. Wong, S. See, P. Minervini, Y. Song, and M. Steedman, “Mmlongbench: Benchmarking long-context vision-language models effectively and thoroughly,” in *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*, D. Belgrave, C. Zhang, L. N. Montoya, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, N. Chen, I. V. M. Ruíz, and A. Loaiza-Bonilla, Eds., 2025. [Online]. Available: http://papers.nips.cc/paper_files/paper/2025/hash/9c8712a9e6d34d60edb8c4c980d4a0f2-Abstract-Datasets_and_Benchmarks_Track.html
- [110] Y. Ma, Y. Zang, L. Chen, M. Chen, Y. Jiao, X. Li, X. Lu, Z. Liu, Y. Ma, X. Dong, P. Zhang, L. Pan, Y. Jiang, J. Wang, Y. Cao, and A. Sun, “MMLONGBENCH-DOC: benchmarking long-context document understanding with visualizations,” in *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/ae0e43289bffa0c1fa34633fc608e92-Abstract-Datasets_and_Benchmarks_Track.html
- [111] F. Zhu, W. Lei, F. Feng, C. Wang, H. Zhang, and T. Chua, “Towards complex document understanding by discrete reasoning,” in *MM ’22: The 30th ACM International Conference on Multimedia, Lisboa, Portugal, October 10 - 14, 2022*, J. Magalhães, A. D. Bimbo, S. Satoh, N. Sebe, X. Alameda-Pineda, Q. Jin, V. Oria, and L. Toni, Eds. ACM, 2022, pp. 4857–4866. [Online]. Available: <https://doi.org/10.1145/3503161.3548422>
- [112] D. D. Lewis, G. Agam, S. Argamon, O. Frieder, D. A. Grossman, and J. Heard, “Building a test collection for complex document information processing,” in *SIGIR 2006: Proceedings of the 29th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, Seattle, Washington, USA, August 6-11, 2006*, E. N. Efthimiadis, S. T. Dumais, D. Hawking, and K. Järvelin, Eds. ACM, 2006, pp. 665–666. [Online]. Available: <https://doi.org/10.1145/1148170.1148307>
- [113] A. F. Biten, R. Tito, L. Gómez, E. Valveny, and D. Karatzas, “OCR-IDL: OCR annotations for industry document library dataset,” in *Computer Vision - ECCV 2022 Workshops - Tel Aviv, Israel, October 23-27, 2022, Proceedings, Part IV*, ser. Lecture Notes in Computer Science, L. Karlinsky, T. Michaeli, and K. Nishino, Eds., vol. 13804. Springer, 2022, pp. 241–252. [Online]. Available: https://doi.org/10.1007/978-3-031-25069-9_16
- [114] L. Xu, Y. Ding, Z. Xu, D. Y. Su, D. Liu, S. Luo, Y. Peng, and W. Liu, “Locating failure in multi-page visually rich document understanding: An empirical attribution,” *arXiv preprint arXiv:2608.07943*, 2026.
- [115] S. Liang, Z. Qiao, T. Bai, Y. He, D. Chen, and S. Fu, “A system-level framework for evaluating privacy in hybrid LLM deployments,” *Algorithms*, vol. 19, no. 6, 2026. [Online]. Available: <https://doi.org/10.3390/a19060500>

- [116] G. Pan, V. Chodnekar, A. Roy, and H. Wang, “A cost-benefit analysis of on-premise large language model deployment: Breaking even with commercial LLM services,” in *IEEE International Conference on Big Data, BigData 2025, Macau, China, December 8-11, 2025*. IEEE, 2025, pp. 2234–2239. [Online]. Available: <https://doi.org/10.1109/BigData66926.2025.11400957>
- [117] W. Wang, S. Zhang, Y. Ren, Y. Duan, T. Li, S. Liu, M. Hu, Z. Chen, K. Zhang, L. Lu, X. Zhu, P. Luo, Y. Qiao, J. Dai, W. Shao, and W. Wang, “Needle in A multimodal haystack,” in *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, A. Globersons, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. M. Tomczak, and C. Zhang, Eds., 2024. [Online]. Available: http://papers.nips.cc/paper_files/paper/2024/hash/24a8968affe71ffe4067d022b9d16566-Abstract-Datasets_and_Benchmarks_Track.html
- [118] L. Sun, D. Wang, J. Huang, L. Zhang, C. Jia, Z. Cheng, and X. Zhang, “When good OCR is not enough: Benchmarking OCR robustness for retrieval-augmented generation,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track), ACL 2026, San Diego, California, USA, July 2-7, 2026*, Y. Li, G. Rehm, and M. Tu, Eds. Association for Computational Linguistics, 2026, pp. 884–894. [Online]. Available: <https://doi.org/10.18653/v1/2026.acl-industry.60>
- [119] F. Cuconasu, G. Trappolini, F. Siciliano, S. Filice, C. Campagnano, Y. Maarek, N. Tonellotto, and F. Silvestri, “The power of noise: Redefining retrieval for RAG systems,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, G. H. Yang, H. Wang, S. Han, C. Hauff, G. Zuccon, and Y. Zhang, Eds. ACM, 2024, pp. 719–729. [Online]. Available: <https://doi.org/10.1145/3626772.3657834>
- [120] B. Jin, J. Yoon, J. Han, and S. Ö. Arik, “Long-context llms meet RAG: overcoming challenges for long inputs in RAG,” in *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025. [Online]. Available: <https://openreview.net/forum?id=oU3tpaR8fm>
- [121] W. Zhou, S. Zhang, H. Poon, and M. Chen, “Context-faithful prompting for large language models,” in *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, ser. Findings of ACL, H. Bouamor, J. Pino, and K. Bali, Eds., vol. EMNLP 2023. Association for Computational Linguistics, 2023, pp. 14 544–14 556. [Online]. Available: <https://doi.org/10.18653/v1/2023.findings-emnlp.968>
- [122] X. Feng, J. Zhou, Z. Huang, K. Wang, S. Ye, J. Hu, Z. Chen, Y. Luo, and J. Zhang, “Docscope: Benchmarking verifiable reasoning for trustworthy long-document understanding,” *CoRR*, vol. abs/2605.08888, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2605.08888>
- [123] D. Li, Z. Sun, X. Hu, Z. Liu, Z. Chen, B. Hu, A. Wu, and M. Zhang, “A survey of large language models attribution,” *CoRR*, vol. abs/2311.03731, 2023. [Online]. Available: <https://doi.org/10.48550/arXiv.2311.03731>
- [124] S. Whitehead, S. Petryk, V. Shakib, J. Gonzalez, T. Darrell, A. Rohrbach, and M. Rohrbach, “Reliable visual question answering: Abstain rather than answer incorrectly,” in *Computer Vision - ECCV 2022 - 17th European Conference, Tel Aviv, Israel, October 23-27, 2022, Proceedings*,

- Part XXXVI*, ser. Lecture Notes in Computer Science, S. Avidan, G. J. Brostow, M. Cissé, G. M. Farinella, and T. Hassner, Eds., vol. 13696. Springer, 2022, pp. 148–166. [Online]. Available: https://doi.org/10.1007/978-3-031-20059-5_9
- [125] C. Cui, T. Sun, S. Liang, T. Gao, Z. Zhang, J. Liu, X. Wang, C. Zhou, H. Liu, M. Lin, Y. Zhang, Y. Zhang, H. Zheng, J. Zhang, J. Zhang, Y. Liu, D. Yu, and Y. Ma, “Paddleocr-vl: Boosting multilingual document parsing via a 0.9b ultra-compact vision-language model,” *CoRR*, vol. abs/2510.14528, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2510.14528>
- [126] J. Niu, Z. Liu, Z. Gu, B. Wang, L. Ouyang, Z. Zhao, T. Chu, T. He, F. Wu, Q. Zhang, Z. Jin, G. Liang, R. Zhang, W. Zhang, Y. Qu, Z. Ren, Y. Sun, Z. Tang, B. Niu, Y. Zheng, D. Ma, Z. Miao, H. Dong, S. Qian, J. Zhang, F. Wang, J. Chen, X. Zhao, L. Wei, W. Li, S. Wang, R. Xu, Y. Cao, L. Chen, Q. Wu, H. Gu, L. Lu, D. Lin, Shenguanlin, X. Zhou, L. Zhang, Y. Zang, X. Dong, J. Wang, B. Zhang, L. Bai, P. Chu, W. Li, J. Wu, L. Wu, Z. Li, G. Wang, Z. Tu, C. Xu, K. Chen, B. Zhou, D. Lin, W. Zhang, and C. He, “Mineru2.5: A decoupled vision-language model for efficient high-resolution document parsing,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, ACL 2026, San Diego, California, USA, July 2-7, 2026, Y. Li, G. Rehm, and M. Tu, Eds. Association for Computational Linguistics, 2026, pp. 13–42. [Online]. Available: <https://doi.org/10.18653/v1/2026.acl-industry.3>
- [127] Y. Yin, H. Liu, Yy, Q. Xie, C. Liu, S. Yang, S. Wang, Z. Liu, H. Zou, J. Chen, S. Wei, J. Wu, M. Huang, Z. Wu, G. Wang, T. Du, and L. Jia, “Unlimited OCR works,” *CoRR*, vol. abs/2606.23050, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2606.23050>
- [128] L. Xiang, Z. Guan, H. Chen, X. Lin, Z. Lei, T. Hu, B. He, and L. Zeng, “Doctrace: Towards traceable long document VQA via hierarchical evidence graph reasoning,” *CoRR*, vol. abs/2608.03292, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2608.03292>
- [129] K. Chen, H. Liu, Q. Peng, J. Yang, Y. Liu, X. Zhang, and Q. Yang, “Mardoc: A memory-aware refinement agent framework for multimodal long document QA,” *CoRR*, vol. abs/2606.05749, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2606.05749>
- [130] K. Li, W. Xie, L. Yao, J. Wu, L. Hong, Y. Huang, and N. L. Zhang, “Insight-doc: Agentic visual perception for long-document understanding,” *CoRR*, vol. abs/2608.10628, 2026. [Online]. Available: <https://doi.org/10.48550/arXiv.2608.10628>

Appendix A

Additional Survey Details

A.1 Comparison with Prior Surveys

Survey	Venue	Scope	Multi-Page Treatment	Organising Framing
A Survey of Deep Learning Approaches for OCR and Document Understanding Subramani et al.	arXiv 2020	Pre-MLLM OCR pipelines	Not covered	Pipeline stage (detection, recognition, layout)
Deep Learning Approaches for Information Extraction from Visually Rich Documents: Datasets, Challenges and Methods Gbada et al.	IJDAR 2025	Information extraction over VRDs	Not covered	Extraction task and method
Deep Learning based Visually Rich Document Content Understanding: A Survey Ding et al.	AI Review 2026	Content understanding over VRDs	Not covered	Task formulation, architecture, pretraining
A Survey on MLLM-based Visually Rich Document Understanding: Methods, Challenges, and Emerging Trends Ding et al.	ACL 2026	MLLM-era VRDU	In passing	OCR dependency; architecture, training, benchmarks
Scaling Beyond Context: A Survey of Multimodal Retrieval-Augmented Generation for Document Understanding Gao et al.	ACL 2026	Multimodal RAG for long documents	Partial (RAG techniques only)	Retrieval-augmented generation
This survey	EMNLP 2026	Multi-page VRDU	Distinct problem	Evidence management

Table A.1 Comparison of this survey with prior VRDU surveys. Prior surveys either predate the multi-page literature or address it only within a broader single-page or extraction-focused scope. This survey is the first to treat multi-page VRDU as a distinct problem organised around evidence management.

Table A.1 contrasts this survey with the five prior VRDU surveys closest to our scope. The comparison is along three axes. **Scope** records the era and problem each survey targets, from pre-MLLM OCR pipelines to the recent MLLM and retrieval-augmented literature. **Multi-page treatment** records whether multi-page understanding is treated as a distinct problem with its own design axes, addressed only in passing within a broader scope, or left out. **Organising framing** records the lens each survey uses to structure the field. Prior surveys either predate the multi-page literature or treat it as one topic within a single-page or extraction-focused scope. This survey is the first to define MP-VRDU as a distinct problem, to organise it around evidence management, and to cover the four architectural families, modality pressures, inference-time and training strategies, and datasets specific to the multi-page setting.

A.2 Full Survey Methodology

A.2.1 Search keywords.

Queries combined a multi-page cluster with a task or method cluster. Multi-page terms included “multi-page document”, “long document”, “long-context document”, “cross-page reasoning”, “page-level retrieval”, “document-scale evidence”, and “hundred-page”. Task and method terms included “document visual question answering”, “visually rich document understanding”, “document VQA”, “document agent”, “visual document retrieval”, “document RAG”, “hierarchical document transformer”, and “long-context vision-language model”. Benchmark-name probes (“MP-DocVQA”, “MMLongBench-Doc”, “LongDocURL”, “DUDE”, “SlideVQA”, “M3DocVQA”, “DocBench”, “M-LongDoc”) were used as anchors for forward-citation sweeps to capture systems that report against multi-page evaluation suites.

A.2.2 Inclusion and exclusion criteria.

A paper was retained if it (i) proposed a multi-page model, framework, system, or pipeline rather than a benchmark or dataset alone, and (ii) reported quantitative results on at least one multi-page benchmark, with cross-page evidence handling as an explicit design contribution. Excluded categories were single-page-only systems benchmarked solely on DocVQA, InfographicVQA, ChartQA, FUNSD, or CORD; survey, position, and opinion papers; generic OCR or table-extraction tools without VQA or reasoning evaluation; long-context LLM benchmarks where documents are one input type among many; and dataset-only contributions, which are catalogued separately in §2.7. Peer-reviewed venue publications and arXiv preprints were assessed against the same substantive criteria, since recent MP-VRDU progress has appeared predominantly in preprint form; venue status is recorded per entry in Table 2.1. Table A.2 lists the surveyed systems grouped by architectural family, with the multi-page contribution on which each was included.

A.2.3 Knowledge extraction and verification.

Each retained paper was converted to markdown and processed through LLM-assisted key-information extraction with evidence-span location, populating per-model fields covering architectural family, OCR dependency, vision encoder, LLM backbone, adaptors and projectors, page-rendering resolution, training stages, datasets used per stage, and reported scores on each benchmark. Every extracted field was manually verified against the source paper before being added.

A.2.4 Dataset catalogue.

The dataset catalogue in §2.7 was built primarily from the pretraining, fine-tuning, and benchmarking datasets named by every paper in the surveyed corpus, supplemented with a small set of datasets the corpus does not yet adopt but that cover MP-VRDU-relevant settings such as long scientific documents, multi-page tables, and cross-page reasoning suites, included to surface datasets whose adoption may strengthen future cross-system comparison.

Model	Venue	Multi-Page Contribution
Page-to-Document Encoding Models		
Arctic-TILT (2025)	ACL	Layer-wise tensor-product text-image fusion with chunked sparse attention, serving up to 389k tokens on a single GPU
GRAM (2024)	CVPR	Interleaves page-level and document-level attention sub-layers with learnable global Doc tokens, avoiding quadratic concatenation cost
Hi-VT5 (2023)	PR	Learnable [PAGE] tokens summarise each page; decoder consumes the concatenated page tokens, with answer-page prediction for explainability
RM-T5 (2024)	DAS	Recurrent memory tokens propagate context page-by-page; per-page memories concatenated for decoding
MLLM-Centric Adaptation Models		
CoR (2026)	preprint	Four-stage Chain-of-Reading paradigm with a caption-reconstruction objective for figure and table grounding
Docopilot (2025)	CVPR	Native multi-page MLLM trained on Doc-750K cross-page proxy tasks with context-extension engineering
DocR1 (2026)	AAAI	EviGRPO adds an evidence-page F1 reward over structured think/evidence/answer outputs
DocSeeker (2026)	CVPR	Analysis-Localization-Reasoning paradigm with page-identifier grounding and evidence-guided resolution allocation
DocSLM (2026)	CVPR	Hierarchical compressor fixes 576 tokens per page; streaming segments with entropy-based abstention for edge deployment
InstructDoc (2024)	AAAI	Document-former fuses OCR, layout, and visual features into learnable tokens, with pages encoded in parallel
LayTokenLLM (2025)	CVPR	Per-segment layout token shares its segment’s position ID, adding layout without consuming context
Leopard (2025)	TMLR	Adaptive high-resolution multi-image encoding under a global token budget with pixel-shuffle compression
mPLUG-DocOwl2 (2025)	ACL	DocCompressor cross-attends global low-resolution queries to sub-image features, compressing each page to 324 tokens
Texthawk2 (2024)	preprint	Two-step resampling and rearrangement for $16\times$ visual-token compression with a coordinate-token detection head
URaG (2026)	AAAI	Cross-modal retrieval module at an early Transformer layer discards non-retained pages before deeper layers
Retrieval-Augmented Pipelines		
AVIR (2026)	MMAAsia	Adaptive page selector uses score-distribution clustering to set per-query retrieval depth
CREAM (2024)	ACM	Coarse-to-fine retrieval with iterative reranking, paired with attention-pooled multi-page visual features
DFVC (2026)	MDPI	Adapter predicts gating weights over neighbouring-page embeddings to fuse cross-page context into page representations
DREAM (2025)	MM	Learning-to-rank fusion of embedding similarity and keyword-guided MLLM scores, with cross-page mixture-of-experts routing
KGP (2024)	AAAI	Knowledge graph over passage and structural nodes traversed by an LLM agent to gather supporting facts
M2RAG (2025)	NeurIPS	Dual-tower sparse and dense retrieval with text, visual, and fusion agents performing consistency checking
M3DocRAG (2024)	preprint	ColPali late-interaction page retrieval with approximate indexing, scaling to 40k+ pages
MDocAgent (2025)	preprint	Five-stage pipeline with parallel text and image retrieval and specialised modality agents
MHier-RAG (2026)	preprint	Combines flattened in-page indexing with a recursively summarised cross-page tree for multi-granularity retrieval
MLDocRAG (2026)	preprint	Chunk-Query Graph of LVLM-generated queries links chunks for retrieval by query-to-query similarity
MM-R5 (2025)	AAAI	Two-stage reranker training with structured reasoning chains and rank-discounted GRPO rewards
MoLoRAG (2025)	EMNLP	Traverses a page graph combining semantic and VLM-judged logical relevance to reach weakly-similar evidence
MultiDocFusion (2026)	ACL	LLM-constructed document hierarchy tree drives depth-first hierarchical chunking
PDF-WuKong (2026)	IJCV	End-to-end sparse sampler jointly trained with the LLM selects top-K text and image evidence
RAG-DocVQA (2025)	ICDAR	Systematic comparison of textual and visual RAG variants for multi-page DocVQA
Self-Attention Scoring (2024)	ICDAR	Self-attention scoring module ranks independently encoded pages, extending evaluation to 793-page documents
SLEUTH (2025)	CVPR	Training-free clue discovery and page screening keep effective context fixed as retrieval depth grows
SV-RAG (2025)	ICLR	Dual LoRA adapters over one MLLM backbone serve both late-interaction page retrieval and answering
VDocRAG (2025)	CVPR	Compresses each page image to a single dense embedding via self-supervised compression pretraining tasks
VisDoMRAG (2025)	NAACL	Parallel textual and visual RAG chains reconciled by consistency-constrained modality fusion
Adaptive-Trajectory Pipelines		
DMAP (2026)	WWW	Hierarchical document map traversed by tri-path retrieval with a reflective loop for insufficient evidence
Doc-V* (2026)	ACL	Thumbnail overview then alternating retrieval and high-resolution page fetch, trained by SFT then GRPO
DocAgent (2025)	EMNLP	Hierarchical XML outline navigated by an actor agent with five tools, plus a reviewer and persistent memory bank
DocDancer (2026)	preprint	End-to-end trained agent with only search and read tools, learned from synthesised exploration trajectories
DocLens (2026)	ACL	Page navigator and element localiser zoom to fine-grained evidence; sampling-adjudication selects among candidates
DocReact (2025)	ACL	Frames iterative retrieval as information-gain maximisation, halting when residual entropy stops decreasing
MACT (2025)	preprint	Four specialised agents with independent judgment routing and agent-wise adaptive test-time scaling
MM-Doc-R1 (2026)	ACL	Planner-seeker-answer workflow trained by similarity-weighted policy optimisation for multi-turn RL
SimpleDoc (2025)	EMNLP	Dual-cue retrieval over embeddings and page summaries, with a memory-carrying reasoner that reissues refined queries
ViDoRAG (2025)	EMNLP	Gaussian-mixture adaptive top-K retrieval with seeker, inspector, and answer agents over multi-scale views
VRAG-RL (2025)	NeurIPS	Visual perception action space lets the agent crop and zoom regions, trained with retrieval-efficiency rewards

Table A.2 The 46 surveyed MP-VRDU systems. Grouped by architectural family, with the multi-page contribution motivating each system’s inclusion. Venue indicates the publication venue at time of writing.

A. ADDITIONAL SURVEY DETAILS

		MP-DocVQA	DUDE	LongDocURL	MMLongBench-Doc	SlideVQA		
Model	Variant	ANLS	ANLS	Accuracy	Accuracy	F1	EM	F1
General Domain MLLM								
Claude-3-Opus	Claude-3-Opus	–	–	–	17.4	18.1	–	–
Gemini-1.5-Pro	Gemini-1.5-Pro	–	–	50.9	28.2	20.6	–	–
GPT-4o	GPT-4o (LVLM mode)	–	–	64.5	42.8	44.9	–	–
GPT-4V	GPT-4V(ision)	–	–	–	32.4	31.2	–	–
InternVL-Chat-v1.5	InternVL-Chat-v1.5-26B	–	–	–	14.6	13.0	–	–
LLaVA-OneVision-7B	LLaVA-OneVision-Chat-7B	–	–	25.0	–	–	–	–
MiniCPM-Llama3-V2.5-8B	MiniCPM-Llama3-V2.5	–	–	–	8.5	8.6	–	–
Qwen2-VL-7B	Qwen2-VL-7B	–	–	30.6	–	–	–	–
Page-to-Document Encoding Models								
Arctic-TILT	Arctic-TILT	81.2	58.1	–	25.8	–	55.1	–
GRAM	GRAM-859M	80.32	51.15	–	–	–	–	–
Hi-VT5	Hi-VT5 (multipage)	62.01	–	–	–	–	–	–
RM-T5	RM-T5	64.01	–	–	–	–	–	–
MLLM-Centric Adaptation Models								
DocR1	DocR1	87.45	54.39	–	–	–	–	71.96
DocSeeker	DocSeeker (full SFT + EviGRPO)	86.2	57.4	51.7	40.1	–	–	77.1
Leopard	Leopard-LLaVA-Pro	70.6	43.82	–	–	–	–	37.51
mPLUG-DocOwl2	DocOwl2-8B	69.42	46.77	–	–	–	–	–
URaG	URaG-7B	88.2	57.6	52.2	33.8	32.8	72.1	–
Docopilot	Docopilot-8B	81.3	–	–	28.8	23	–	–
Texthawk2	Texthawk2	–	–	–	–	–	–	–
DocSLM	DocSLM-2B	70	47.6	–	–	–	–	–
InstructDoc	InstructDoc (finetuned)	–	46.8	–	–	–	37.7	47.3
LayTokenLLM	LayTokenLLM-8B	74.31	52	–	–	–	–	–
CoR	Qwen2.5-VL-CoR-7B	–	–	51.5	37.4	36	–	–
Retrieval-Augmented Pipelines								
AVIR	AVIR	84.58	–	–	–	–	60.3	68.9
DFVC	DFVC (ColQwen backbone)	–	–	–	–	–	–	–
DREAM	DREAM† (InternVL2-4B + retrieved text)	85.72	52.14	–	27.3	28.6	–	–
M3DocRAG	M3DOC RAG (ColPali + Qwen2-VL 7B)	84.44	–	–	21	22.6	–	–
MM-R5	MM-R5 (Qwen2.5-VL-7B SFT+GRPO)	–	–	–	–	–	–	–
MoLoRAG	MoLoRAG+ (Qwen2.5-VL-7B)	–	–	51.85	41.01	–	–	–
Self-Attention Scoring	Pix2Struct-SA proposed	61.99	–	–	–	–	–	–
SLEUTH	SLEUTH (Qwen3-VL-8B + ColPali Top-5)	–	–	59.96	52.77	–	–	–
SV-RAG	SV-RAG-InternVL2† R5 (4B)	71.0	45.0	–	23.0	24.2	31.98	–
VDocRAG	VDocRAG	–	44	–	–	–	–	42
CREAM	CREAM	65.32	52.46	–	–	–	–	–
MHier-RAG	MHier-RAG (Qwen-turbo)	–	–	55.7	52.3	46	–	–
MLDocRAG	MLDocRAG	–	–	50.8	47.9	–	–	–
MultiDocFusion	MultiDocFusion	–	–	–	–	–	–	–
M2RAG	M[2]RAG top-5	–	–	57.68	39.9	–	–	–
MDocAgent	MDocAgent (top-4)	–	–	57.8	31.5	–	–	–
PDF-WuKong	PDF-WuKong	76.9	56.1	–	–	–	–	–
RAG-DocVQA	RAG-Qwen2.5-VL	73.7	39.9	–	–	–	–	–
VisDoMRAG	VisDoMRAG (ChatGPT-4o + ColPali + BGE)	–	–	–	–	–	–	67.22
KGP	KGP	–	–	–	–	–	–	–
Adaptive-Trajectory Pipelines								
Doc-V*	Doc-V* (GRPO)	86.2	64.5	56.3	42.1	–	–	77.2
DocReact	DocReact (w/ ColPali)	–	–	–	38.29	38.07	54.87	55.04
MACT	MACT-MiMo-VL-Series-28B	–	70.8	–	–	47.4	–	83.9
VRAG-RL	VRAG-RL (Qwen2.5-VL-7B-Instruct)	–	–	–	–	–	–	–
DocLens	DocLens (Gemini-2.5-Pro)	–	–	–	67.6	–	–	–
DMAP	DMAP Ours-Top4 (GPT-4o backbone)	–	–	60.7	43.2	–	–	–
DocAgent	DocAgent (Claude-3.5-Sonnet)	–	–	–	57.3	54.1	–	–
DocDancer	DocDancer (GPT-5.2)	–	–	–	57	56.8	–	–
MM-Doc-R1	MM-Doc-R1 (Qwen3-8B + SPO)	–	–	–	49.7	46.1	–	–
SimpleDoc	SimpleDoc (3.5 pages)	–	–	72.3	60.58	–	–	–
ViDoRAG	ViDoRAG	–	–	–	–	–	–	–

Table A.3 Performance of MP-VRDU models grouped by architecture category. Reported metrics follow the source papers and are not always directly comparable. ‘–’ indicates not reported.

A.3 Additional Resources

A.3.1 Reported Performance by Architecture

Table A.3 summarises reported scores on the five most-adopted multi-page benchmarks, grouped by architectural family. It is included for coverage rather than ranking. The matrix is sparse and metric conventions diverge across MP-DocVQA, DUDE, LongDocURL, MMLongBench-Doc, and SlideVQA, so scores are not comparable across columns; for the same benchmark, differences in backbone scale,

proprietary-LLM dependency, evidence budget, and the training-versus-zero-shot distinction mean that a higher score does not by itself indicate a better architecture. Several of the strongest systems orchestrate closed-source frontier backbones of undisclosed scale, so their scores cannot be attributed to architecture alone. Read with these caveats, the table supports one robust cross-family observation: coverage tracks cost structure, with page-to-document and MLLM-centric systems confined to mid-length benchmarks and selective-evidence families dominating the long-document frontier.

A.3.2 General Domain MLLM for MP-VRDU

Frontier general-domain MLLMs handle short multi-page inputs well when the document fits in context, but the multi-page setting introduces three pressures absent in single-page VRDU: context-window overrun, attention dilution over sparsely distributed evidence [5], and a per-page reading fidelity versus whole-document coverage trade-off under a fixed token budget. Table A.3 shows that even the strongest frontier MLLMs trail specialised MP-VRDU systems by a wide margin on MMLongBench-Doc and LongDocURL, motivating the specialised architectures examined in §2.3.

A.3.3 Model Components Details

Table A.4 lists the concrete components of each surveyed system, organised by architectural family. The four families assemble these parts in characteristically different ways. Page-to-document transformers pair smaller LLM backbones (T5 variants, DocFormerv2) with custom layout-aware projectors. MLLM-centric adaptation models reuse standard vision-language stacks such as Qwen2.5-VL, mPLUG-Owl2, and SigLIP variants, and update only light projection or compression modules. Retriever-augmented pipelines decouple a frozen retriever (ColPali, ColQwen, VisRAG) from a frozen LVLM and train only the scorer, fuser, or reranker. Adaptive-trajectory pipelines mostly orchestrate frozen frontier VLMs, with parameter updates restricted to a few systems that fine-tune the controller via SFT or GRPO.

Table A.4 Per-model components of the surveyed MP-VRDU systems. Resolution lists the input image resolution or page-rendering setting; *dynamic* denotes encoders that accept variable-resolution input. ‘–’ indicates not reported not applicable.

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
Page-to-Document Encoding Models				
Arctic-TILT	U-Net	400k tokens / 500p	T5-Large	TILT fusion
GRAM	DocFormerv2	800 tok/page train, 8000 test	DocFormerv2	–
Hi-VT5	DiT	224x224 (DiT default), 20480 max tokens	T5	–
RM-T5	DiT	512 tok/page max	T5-base	–
MLLM-Centric Adaptation Models				
DocR1	Qwen2.5-VL	dynamic, max 1024 patches	Qwen2.5-VL-7B	Frozen Qwen MLP projector
DocSeeker	Qwen2.5-VL ViT	1024x784 eval, EGRA training	Qwen2.5-VL-7B-Instruct	–
Leopard	SigLIP-SO400M	364x364 / 980x980, max 50 sub-imgs	LLaMa-3.1-8B / Mistral-7B	MLP / Idefics2 resampler

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A. ADDITIONAL SURVEY DETAILS

Table A.4 – continued from previous page

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
mPLUG-DocOwl2	ViT-L/14	504x504, max 12 crops	mPLUG-Owl2	H-Reducer + DocCompressor
URaG	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B / Qwen2.5-VL-7B	Cross-modal retrieval module (2 linear + GELU) + LoRA
Docopilot	InternViT-300M	448 tile, max 24 tiles	InternLM2-1.8B / InternLM2.5-7B	2-layer MLP
Texthawk2	SigLIP-SO400M	224x224 base, max 72 sub-imgs	Qwen2-7B-Instruct	QPN + ReSA
DocSLM	SigLIP2	dynamic 384-1152, 576 tok/page	Qwen2.5-1.5B	OCR/Vision compressor + MLP
InstructDoc	CLIP	224x224	FlanT5 / BLIP2	Doc-former + proj
LayTokenLLM	–	–	LLaMa3-8B / Qwen1.5-7B	Layout tokenizer + LoRA
CoR	Qwen2.5-VL	dynamic, 32k max	Qwen2.5-VL-7B	Frozen Qwen MLP projector + LoRA
Retrieval-Augmented Pipelines				
AVIR	Pix2Struct	dynamic, 768-5120 patches	Qwen2.5-VL-3B-AWQ	Frozen Qwen MLP projector
DFVC	VisRAG, ColQwen	dynamic	Qwen2-VL-2B-Instruct	Cross-page MLP
DREAM	InternVL2 ViT	448 tile, max 24 tiles	InternVL2-4B	Cross-page MoE attention connector + LoRA
M3DocRAG	ColPali / ColQwen	144 DPI, ~1700px wide	Qwen2-VL-7B / InternVL2-8B / Idefics2-8B / Idefics3-8B	–
MM-R5	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-7B	–
MoLoRAG	ColPali	448x448	Qwen2.5-VL 3B/7B / Deepseek-VL-16B / LLaVa-Next-7B	Frozen base projector
Self-Attention Scoring	Pix2Struct	2048 patches	Pix2Struct	–
SLEUTH	ColPali, Qwen3-VL ViT	448x448	Qwen3-VL-8B / GLM-4.1V-Thinking-8B / Gemini-2.5-Flash	–
SV-RAG	InternVL2 / PaliGemma / Phi-3-vision	dynamic (InternVL2 default), 448x448	InternVL2-4B / PaliGemma-3B / Phi-3-vision-4B	Dual LoRA adaptors (retrieval + QA) + Col-projection
VDocRAG	Phi3V-4.2B	336x336 dynamic, up to 1344x1344	Phi3V-4.2B	LoRA + MLP
CREAM	Pix2Struct	2048 patches	LLaMA2-7B	Attn pooling + MLP + LoRA
KGP	–	–	T5-large, LLaMa-7B, MDR (RoBERTa), ChatGPT	LoRA
MHier-RAG	Qwen-VL-Plus	dynamic	Qwen-VL-Plus / GPT-3-turbo / Qwen-turbo / GPT-4o / DeepSeek-chat / ERNIE-turbo	–
MLDocRAG	LVLm Qwen2.5-VL	dynamic	Qwen2.5-VL-32B-Instruct, Qwen2.5-VL-7B, Qwen2.5-72B	Frozen Qwen MLP projector

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Table A.4 – continued from previous page

Model	Vision Backbone	Resolution	LLM Backbone	Adaptors and Projectors
MultiDocFusion	–	–	Mistral-8B / Llama-3.2-3B / Qwen-2.5-3B / Qwen-2.5-7B	LoRA
M2RAG	VisRAG-Ret	dynamic	Qwen2.5-7B-Instruct, Qwen2.5-VL-7B-Instruct, Qwen2.5-VL-32B	Frozen Qwen MLP projector + LoRA + modal fuser
MDocAgent	ColPali / ColQwen2	144 DPI	LLaMA-3.1-8B-Instruct, Qwen2-VL-7B-Instruct	–
PDF-WuKong	IXC2-VL-4KHD	dynamic 336px-4K, max 16 tiles	IXC2-VL-4KHD	Sparse sampler
RAG-DocVQA	DiT / Pix2Struct	2048 patches	VT5 / Qwen2.5-VL-7B-Instruct	LoRA
VisDoMRAG	ColPali / ColQwen2	448x448 (ColPali), dynamic	Qwen2-VL-7B-Instruct / Gemini-1.5-Flash / GPT-4o	–
Adaptive-Trajectory Pipelines				
Doc-V*	Qwen2.5-VL ViT	1024x768 / 256x256 thumb	Qwen2.5-VL-7B-Instruct	Frozen MLP
DocReact	ColPali, VisRAG-Ret	448x448 (ColPali default) / dynamic	GPT-4o	–
MACT	Qwen2.5-VL-7B-Instruct / MiMo-VL-7B-SFT / InternVL3-9B	dynamic	Qwen2.5-VL-7B / MiMo-VL-7B-SFT / InternVL3-9B, Qwen2.5-7B/3B / MiMo-7B-SFT / InternVL3-8B/2B	–
VRAG-RL	Qwen2.5-VL ViT	dynamic	Qwen2.5-VL-3B-Instruct / Qwen2.5-VL-7B-Instruct	–
DocLens	frontier VLMs	dynamic	Gemini-2.5-Pro / Gemini-2.5-Flash / Claude-4-Sonnet	–
DMAP	ColPali	dynamic	GPT-4o / Qwen-plus	–
DocAgent	–	144 DPI	Gemini 2.0 Flash / GPT-4o / Claude Sonnet 3.5	–
DocDancer	Qwen3-VL-235B-A22B-Instruct	144 DPI	Qwen3-4B/30B-A3B	–
MM-Doc-R1	Qwen2.5-VL ViT	dynamic	Qwen3-4B / Qwen3-8B	–
SimpleDoc	ColQwen-2.5	dynamic	Qwen2.5-VL-32B-Instruct, Qwen3-30B-A3B	–
ViDoRAG	Nv-embed-V2, ColQwen2	dynamic	Qwen2.5-VL / LLaMA3.2-Vision / GPT-4o	–

Appendix B

Additional Attribution Details

B.1 Extended Result Tables

B.1.1 Representation Ladder by Document Domain

Table B.1 breaks the representation ladder of Table 3.13 down by document domain, and sweeps the two image-carrying rungs across render resolution. Domain rows partition the pool, so each question is counted exactly once. The ordering of rungs is stable within a row, with TLV winning in six of the seven domains, but variation across domains is considerably wider than variation across rungs: at TLV accuracy ranges from 42.9% on academic papers to 73.2% on tutorial and workshop material. Resolution matters most where the image is the only channel, with pooled accuracy at V rising 12.4 points from low to high against 5.5 points at TLV, and the largest gains falling on the domains dominated by dense tables and figures.

Domain	Ladder @ med (%)				TLV		V		ΔV
	T	TL	TLV	V	low	high	low	high	
Academic paper	36.4	33.1	42.9	33.8	37.0	46.8	19.5	38.3	−9.1
Admin / Industry	50.0	56.2	67.2	54.7	64.1	70.3	42.2	64.1	−12.5
Brochure	28.6	32.5	45.5	40.3	40.3	48.1	39.0	49.4	−5.2
Financial report	49.1	52.8	51.9	38.9	54.6	53.7	21.3	46.3	−13.0
Guidebook	35.8	40.0	51.7	45.0	48.3	51.7	38.3	47.5	−6.7
Research / Intro	22.6	27.4	47.6	45.3	42.9	50.0	38.2	47.6	−2.3
Tutorial / Workshop	14.3	48.2	73.2	67.9	71.4	74.1	74.1	70.5	−5.3
All	31.9	38.8	52.5	45.6	49.2	54.7	37.8	50.2	−6.9

Table B.1 Representation ladder by document domain and render resolution. Teal deepens with accuracy and bold marks each row’s winning rung; the warm strip deepens with how far vision alone falls below the combined rung. Resolution lifts the domains whose evidence is pixel-limited and leaves the text-heavy ones where they were.

B.1.2 Parser Fidelity by Evidence Source

Table B.2 splits each parser’s text and text-plus-image accuracy across the five evidence sources, and adds the paired within-question verdict changes across that step. Adding the page image raises accuracy on almost every parser and source combination, the one consistent exception being table evidence on born-digital pages, where three of the four parsers lose ground: a well-parsed table is already a faithful representation, and the image mainly adds an opportunity to misread it. The image also compresses the

differences between parsers, with pooled born-digital accuracy spanning 26.6 to 42.3 across parsers on the text channel alone and only 46.0 to 50.2 once the image is added. Regression is otherwise rare, so the image seldom overturns an answer the text channel had already reached.

Parser	Source	Accuracy (%)				Verdict transition (%)			
		Digital		Scanned		Digital		Scanned	
		Text	+Vis	Text	+Vis	R→W	W→R	R→W	W→R
PaddleOCR-VL	Chart	23.8	40.8	4.2	54.2	3.8	20.8	0.0	50.0
	Figure	17.4	37.0	31.1	57.5	1.6	21.2	0.9	27.4
	Generalized-text	43.8	47.5	26.3	57.9	3.8	7.5	0.0	31.6
	Pure-text	50.9	54.9	35.4	55.4	3.5	7.5	0.0	20.0
	Table	47.6	46.5	59.4	71.9	5.9	4.9	3.1	15.6
	All	40.1	49.7	32.8	61.2	3.7	13.3	1.0	29.4
MinerU 2.5	Chart	17.7	38.5	14.6	47.9	3.1	23.8	2.1	35.4
	Figure	12.5	38.6	25.5	50.9	1.6	27.7	1.9	27.4
	Generalized-text	20.0	43.8	18.4	50.0	2.5	26.2	2.6	34.2
	Pure-text	30.1	49.1	30.8	53.8	3.1	22.1	0.0	23.1
	Table	36.8	42.7	56.2	65.6	5.9	11.9	6.2	15.6
	All	26.6	46.0	27.4	55.7	3.4	22.8	2.0	30.3
Unlimited-OCR	Chart	23.8	42.3	2.1	58.3	3.8	22.3	0.0	56.2
	Figure	20.7	38.0	34.0	58.5	2.7	20.1	0.9	25.5
	Generalized-text	43.8	52.5	36.8	60.5	2.5	11.2	0.0	23.7
	Pure-text	50.4	53.1	36.9	56.9	4.4	7.1	0.0	20.0
	Table	49.7	48.1	59.4	71.9	6.5	4.9	6.2	18.8
	All	42.3	50.2	34.3	63.7	4.6	12.5	1.5	30.8
PyMuPDF	Chart	24.6	37.7	0.0	54.2	3.8	16.9	0.0	54.2
	Figure	19.6	41.3	7.5	52.8	3.3	25.0	0.9	46.2
	Generalized-text	40.0	47.5	2.6	52.6	3.8	11.2	2.6	52.6
	Pure-text	50.0	54.4	3.1	50.8	4.4	8.8	0.0	47.7
	Table	48.1	44.9	0.0	59.4	10.3	7.0	0.0	59.4
	All	40.6	49.7	4.0	57.7	4.8	13.9	0.5	54.2

Table B.2 Parser fidelity by evidence source and scan status. Teal deepens with accuracy and green with the vision-step recovery rate; red deepens with the vision-step regression rate. Adding the page image lifts every parser on every source, and it lifts them furthest exactly where the text channel had least to give.

B.1.3 Retrieval Across Depth

Table B.3 reports page-level precision, recall, and F1 for all six retrievers at five depths on both corpora, which is the complete sweep behind the operating-point comparison in the main text. Vision retrieval leads text retrieval at every depth on both datasets, and ColQwen3 is the strongest arm on each. F1 peaks at $k=1$ for every retriever, since precision falls faster than recall climbs, so a system optimising a single retrieval score should read shallowly while one optimising evidence coverage should read deeply. The gap between the modalities is not an artefact of a weak text arm, since Qwen3-Embedding leads the text arm at every depth beyond the first and still trails ColQwen3 by 22.9 recall points at $k=10$ on MMLongBench-Doc. The two corpora are separate pools, so the table is read for the ordering of retrievers within a corpus and for the direction of the depth trend across both.

B. ADDITIONAL ATTRIBUTION DETAILS

Retriever	Precision					Recall					F1				
	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
MMLongBench-Doc ($n=847$)															
<i>Text</i>															
BM25	29.5	18.5	14.0	11.4	9.5	22.2	37.8	46.0	50.8	58.5	24.2	23.4	20.1	17.5	15.4
BGE-M3	33.8	18.7	14.7	12.0	10.0	25.5	38.8	48.2	53.2	61.1	27.9	23.9	21.2	18.4	16.3
Qwen3-Embedding	33.5	19.9	15.0	12.6	10.5	25.5	41.3	49.4	56.0	64.5	27.9	25.4	21.7	19.4	17.1
<i>Vision</i>															
ColModernVBERT	58.8	31.5	22.7	17.9	14.0	46.0	64.6	72.5	77.2	81.7	49.6	39.6	32.2	27.1	22.2
ColQwen2.5	63.3	33.8	23.6	18.5	14.4	48.8	68.3	74.9	79.2	83.8	52.8	42.5	33.5	27.9	22.9
ColQwen3	70.2	36.8	25.5	19.9	15.2	54.1	73.9	80.6	83.9	87.4	58.6	46.0	36.1	29.9	24.1
LongDocURL ($n=2,317$)															
<i>Text</i>															
BM25	49.6	27.0	19.0	14.8	11.3	35.6	53.9	61.1	65.6	69.5	39.7	34.3	27.7	23.3	18.7
BGE-M3	47.6	27.8	19.8	15.5	11.9	33.4	54.2	62.3	67.3	72.6	37.6	35.0	28.7	24.1	19.8
Qwen3-Embedding	48.9	28.8	20.6	16.4	12.5	34.1	56.0	64.8	70.6	75.5	38.5	36.3	29.9	25.5	20.7
<i>Vision</i>															
ColModernVBERT	64.7	33.5	23.2	17.9	13.6	47.5	67.3	74.5	78.6	83.1	52.7	42.7	33.9	28.0	22.5
ColQwen2.5	68.1	35.8	24.8	18.9	14.2	49.3	70.6	78.0	81.5	85.2	54.9	45.2	35.9	29.4	23.4
ColQwen3	73.3	38.2	25.9	19.8	14.8	53.6	74.5	81.1	84.6	88.1	59.5	48.1	37.5	30.7	24.4

Table B.3 Page-level retrieval across depth on both corpora. Amber deepens with the value on a per-metric scale and bold marks each retriever’s best-F1 depth. Vision retrieval leads text at every depth on both corpora, and recall climbs with k as precision falls.

B.1.4 Prompting by Evidence Source

Table B.4 opens the prompting sweep up by evidence source, reporting answerable accuracy, abstention on the answerable pool, and abstention on the unanswerable pool for six prompt modes at four rungs. The abstention instruction is the only prompt that materially raises correct refusal, lifting unanswerable abstention from roughly 13% uninstructed to between 75 and 82% across the four rungs, but it does so indiscriminately: abstention on the answerable pool rises with it at every rung, and the per-source rows show that the coupling is not confined to the harder evidence types. The two chain-of-thought modes leave both abstention rates near the uninstructed baseline, and `cot` is the only mode that matches `none` on answerable accuracy at TLV. Evidence-source ordering is stable across prompt modes, with pure-text and table questions strongest and chart and figure questions weakest, and adding the page image closes most of that spread. The unanswerable block carries no per-source rows, an unanswerable question having no gold evidence pages from which a source could be read, and it is scored under the original judge rather than the error-taxonomy judge used for the two answerable blocks, so levels are compared within a block rather than across the table.

B.1.5 Reasoner Scale by Evidence-Page Count

Table B.5 reports accuracy by the number of gold evidence pages a question cites, for four Qwen3-VL sizes at the two rungs that carry the page image. Scale buys multi-page accuracy steadily, and buys most where the image carries the evidence, with the 32B gaining 7.8 points over the 8B at TLV and 12.5 at V. What scale does not buy is integration, since single-page accuracy rises just as fast and the deficit between multi-page and single-page questions is essentially unchanged across the size ladder. Read alongside Table 3.19, this supports treating cross-page integration as a distinct failure mode rather than a capability that additional parameters resolve. The 3+ bucket is thin at every size and is read for trend rather than for level.

B. ADDITIONAL ATTRIBUTION DETAILS

Prompt	Source	Answerable acc.				Abstention (ans.)				Abstention (unans.)			
		T	TL	TLV	V	T	TL	TLV	V	T	TL	TLV	V
none	Chart	16.3	16.9	42.7	40.4	32.6	28.1	7.3	6.2				
	Figure	14.5	22.1	42.4	39.0	36.6	24.8	7.6	7.9				
	Generalized-text	26.3	33.9	49.2	43.2	20.3	10.2	0.8	4.2				
	Pure-text	36.9	44.4	51.0	42.3	23.0	11.5	7.3	8.0				
	Table	35.6	44.7	44.7	34.1	15.9	7.2	6.7	9.1				
	All	29.0	35.7	49.5	42.6	26.3	17.2	6.2	7.1	13.1	13.1	14.3	13.5
grounded	Chart	15.2	14.6	31.5	28.1	28.1	23.0	5.1	5.6				
	Figure	11.4	21.4	40.7	37.9	34.1	21.4	7.2	7.9				
	Generalized-text	19.5	29.7	43.2	44.9	17.8	6.8	0.0	0.8				
	Pure-text	35.2	42.3	49.0	40.6	19.9	10.1	7.0	6.3				
	Table	27.9	37.5	38.5	28.4	17.3	13.0	7.2	8.2				
	All	24.9	32.3	43.7	37.9	24.1	16.2	6.0	6.5	22.1	6.6	8.2	5.3
abstain	Chart	13.5	11.8	26.4	22.5	52.2	59.0	25.3	27.5				
	Figure	6.9	13.8	34.1	30.3	76.2	65.2	35.5	38.3				
	Generalized-text	19.5	25.4	41.5	37.3	55.9	44.1	19.5	25.4				
	Pure-text	31.7	37.8	43.7	34.3	42.5	36.0	24.1	31.1				
	Table	24.5	37.0	38.5	26.4	43.3	37.0	30.3	38.9				
	All	21.7	27.9	40.0	31.9	52.8	47.7	27.7	32.3	81.1	79.9	74.6	81.6
abstain_bal.	Chart	12.4	11.8	28.1	23.0	44.9	51.7	18.5	24.2				
	Figure	6.9	14.5	35.5	32.8	71.7	62.4	31.4	33.4				
	Generalized-text	19.5	28.8	41.5	39.8	52.5	41.5	18.6	21.2				
	Pure-text	31.0	37.8	45.5	34.3	39.7	33.2	19.2	25.5				
	Table	24.0	36.1	35.6	28.4	38.5	30.3	25.5	34.6				
	All	21.7	28.6	40.9	33.5	48.2	43.5	23.2	27.9	75.4	77.0	68.9	76.6
cot	Chart	19.7	16.3	43.3	38.2	16.3	42.1	9.6	5.6				
	Figure	11.0	17.9	43.8	43.8	15.9	26.2	6.9	5.9				
	Generalized-text	25.4	34.7	49.2	54.2	3.4	11.0	3.4	3.4				
	Pure-text	32.4	42.3	51.0	46.2	13.9	15.7	7.7	5.6				
	Table	37.5	48.6	51.4	44.0	10.6	12.0	7.2	5.3				
	All	26.9	34.1	50.2	46.9	13.3	21.7	6.6	5.3	8.6	13.9	11.9	5.7
extract_cot	Chart	13.1	16.3	36.6	35.6	19.3	37.6	6.3	7.3				
	Figure	10.3	20.1	43.3	39.4	27.6	36.5	11.1	11.8				
	Generalized-text	21.2	34.2	51.3	51.7	7.6	18.8	2.6	4.2				
	Pure-text	34.5	40.4	48.2	41.2	17.1	19.3	9.9	11.6				
	Table	37.7	46.2	49.0	36.9	13.0	14.4	9.6	17.0				
	All	26.1	33.5	47.5	41.6	19.4	26.5	8.8	11.6	8.6	18.0	18.4	18.4

Table B.4 Faithfulness across six prompt modes by evidence source. Teal deepens with the good-direction rate and warm with abstention on the answerable pool; the abstention instruction couples correct refusal to false refusal on every evidence source, and no prompt separates the two.

Size	TLV					V				
	1	2	3+	M	ΔM	1	2	3+	M	ΔM
2B	48.1	27.4	25.2	26.7	-11.9	40.7	20.7	22.5	21.3	-12.8
4B	60.5	42.3	32.4	39.2	+0.6	50.4	34.9	31.5	33.8	-0.3
8B	64.6	38.6	38.7	38.6	-	55.7	35.3	31.5	34.1	-
32B	71.5	49.0	40.9	46.4	+7.8	66.0	48.5	42.3	46.6	+12.5

Table B.5 Accuracy by gold evidence-page count for each reasoner size. **M** pools every multi-page question and ΔM reads it against the 8B at the same rung. Scale buys multi-page accuracy, most of it on the image encodings, but it raises single-page accuracy just as fast, so the multi-page deficit does not close.

Appendix C

Additional Method Details

C.1 Implementation Details

This appendix provides the main implementation details for the evidence-management pipeline introduced in §4.1. We report the role-specific prompts and the configuration used for the MMLongBench-Doc experiments. Lower-level implementation details, including caching, batching, and output recovery, are omitted.

C.1.1 Role Prompts

The three roles share the same underlying vision-language model but receive different instructions and evidence. The `reader` extracts persistent evidence from each visited page and decides whether enough evidence has been collected to attempt an answer. The `synthesizer` reasons over the assembled evidence and produces the answer, while the `arbiter` determines whether the available evidence is sufficient or whether reading should continue. Figures C.1–C.3 show the principal instructions; repetitive formatting rules and error-recovery instructions are omitted for brevity.

C.1.2 Evidence Assembly

The `reader` maintains an append-only evidence record containing only information relevant to the question. Each record is associated with its source page, and pages considered likely to contain answer-bearing evidence are additionally marked for full presentation during synthesis. When the `synthesizer` is invoked, evidence is reordered by document position rather than retrieval rank. A marked page contributes the `reader`’s record, Unlimited-OCR text, and the rendered page image, while an unmarked page contributes only its retained record. Pages with an empty record are omitted. If no page is marked, the two highest-ranked pages already inspected are supplied in full as a fallback. This preserves broad evidence coverage while preventing every inspected page from entering the final multimodal context.

C.1.3 Configuration

Table C.1 details specific pipeline configurations. The `reader` traverses the fixed retrieval ranking until it proposes stopping or exhausts the page budget. A stopping proposal is ignored before k pages have been inspected. If the `arbiter` requests further evidence, at least m additional pages are read before another stopping proposal is accepted. At most C synthesis attempts are permitted. These controls bound evidence coverage and trajectory depth independently of the underlying model.

READER prompt (abridged)

You are reading a long document to answer a question.
You see one page this turn. Pages are ordered by retrieval relevance rather than document order. Previous pages are available only through NOTES.

WHAT TO RECORD

Record what is on the CURRENT PAGE, and only evidence that bears on the QUESTION.

- Preserve exact numbers, names, dates, identifiers, units, and relevant table structure.
- Describe relevant charts, figures, diagrams, and images in sufficient detail for later reasoning.
- Do not repeat evidence already present in NOTES.
- If the page adds nothing relevant, write none.
- If the QUESTION asks for a count or list, enumerate each matching item on this page and finish with:

COUNT ON THIS PAGE: <number> (<what was counted>)

Only items matching the QUESTION's exact definition should be counted.

OUTPUT

REASONING: <reasoning>

RECORD: <page evidence or none>

ANSWER PAGE: <true/false>

DECISION: <CONTINUE/STOP>

Write STOP only when the accumulated evidence appears sufficient to answer the QUESTION completely. Otherwise continue reading.

Figure C.1 Abridged prompt for the reader role. The production prompt additionally specifies formatting and output-recovery rules.

SYNTHESIZER prompt (abridged)

You are answering a question about a long document. You cannot see all of it. A reader went through it page by page.

The EVIDENCE is organised by document page. A page may contain:

- **READER'S RECORD:** a summary that may be incomplete;
- **PARSED TEXT:** the extracted text, preferred for exact words, numbers, names, and identifiers;
- **PAGE IMAGE:** the original visual evidence, preferred for charts, figures, layout, colour, and other visual information.

Use only the supplied EVIDENCE. Where sources disagree, prefer parsed text for textual content and the page image for visual content.

Think step by step before answering.

CHECK THE QUESTION

Verify the specific premises assumed by the QUESTION, including names, dates, pages, figures, tables, groups, metrics, and whether the requested object exists.

If a required premise is contradicted or absent where it would have to appear, do not substitute the nearest matching fact.

ABSTENTION

Answer Not answerable only when the supplied evidence genuinely does not contain what is required. Before abstaining, inspect the full parsed text and page images and write:

MISSING: <specific unsupported fact>

COUNTS AND LISTS

For counting or list questions, construct a page-wise tally. Verify items against the full page evidence where available, exclude non-matching and duplicate items, and then aggregate across pages.

OUTPUT

REASONING: <step-by-step reasoning>

ANSWER: <concise final answer>

Figure C.2 Abridged prompt for the synthesizer role. The prompt combines explicit reasoning, premise verification, calibrated abstention, and structured aggregation.

ARBITER prompt (abridged)

A reader inspected pages from a long document and a synthesizer produced a candidate answer.

Your job is not to answer the QUESTION. Decide whether the accumulated evidence is sufficient to support an answer or whether additional pages should be read.

Think step by step:

- What facts does the QUESTION require?
- Which required facts are present in the accumulated evidence?
- Does the proposed answer rely on information not supported by that evidence?

If a specific required fact is missing and unread pages remain, request further reading. If the evidence is complete, terminate even if the proposed answer itself is incorrect; additional pages cannot repair a reasoning error over already sufficient evidence.

A proposed Not answerable while unread pages remain is evidence that further reading may be required.

OUTPUT

REASONING: <reasoning>

MISSING: <specific missing fact or none>

DECISION: <CONTINUE/END>

Figure C.3 Abridged prompt for the arbiter role. The arbiter evaluates evidence sufficiency but cannot replace the candidate answer.

Component	Configuration
Reasoner	Qwen3-VL-8B-Instruct
Quantisation	4-bit NF4
Compute dtype	FP16
Decoding	Greedy
Retriever	ColQwen3.5
Parser	Unlimited-OCR
Reader representation	TLV , high resolution
Synthesizer full pages	TLV , medium resolution
Page budget B	25
Pages per reader turn p	1
Initial stopping floor k	6
Post-reopen floor m	3
Maximum synthesis attempts C	3
Reader decode budget	4096 tokens
Synthesizer decode budget	4096 tokens
Arbiter decode budget	4096 tokens
Completion budget	1024 tokens
Flagged synthesis pages	All that fit in context
Hardware	1 × NVIDIA V100 16 GB

Table C.1 Configuration of our system used evaluation